

CYBER ELIAS ACADEMY

TEACHING MANUAL

Microsoft Office

Session Roadmap · Lesson Notes · Concept Diagrams · Answer Keys

READY TO TEACH

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OVERVIEW

Course Roadmap

What we cover in every session

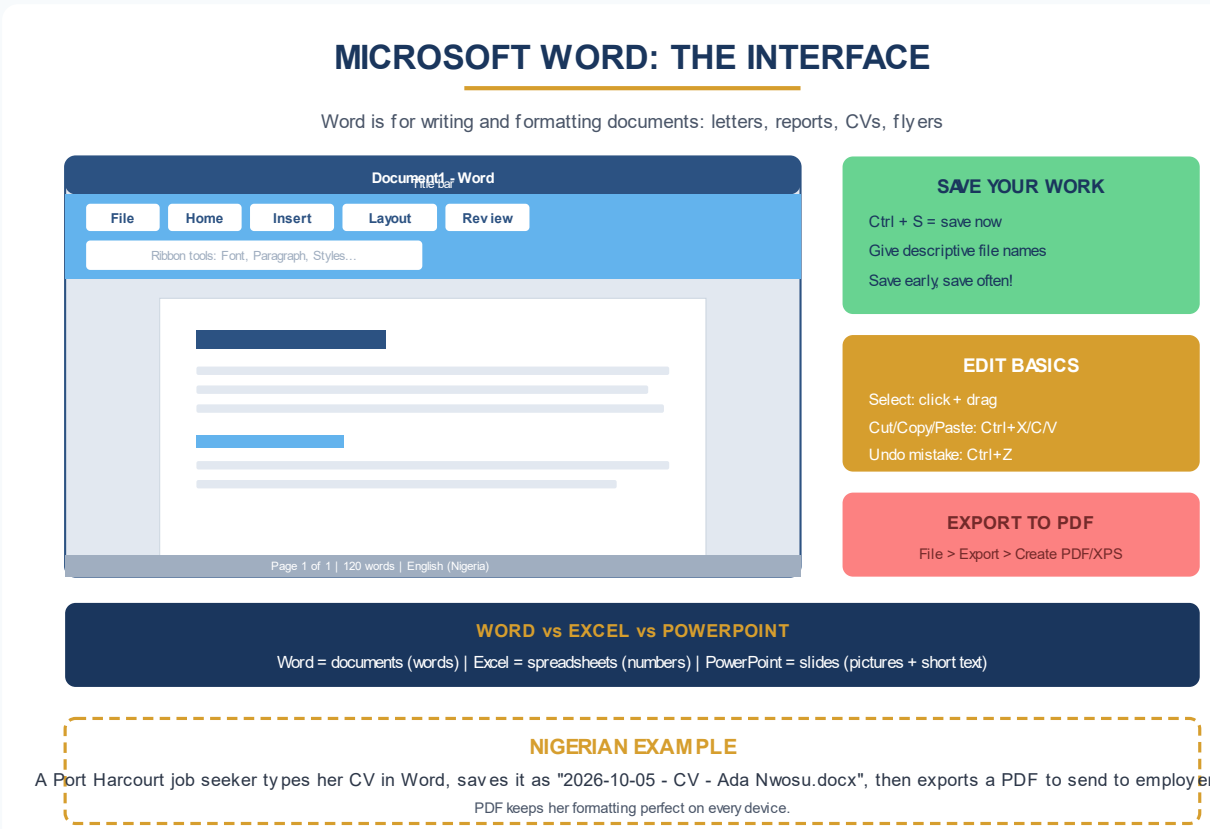
#	SESSION TOPIC	WEEK / MODULE	DURATION
01	Word Fundamentals		105 minutes
02	Document Formatting		105 minutes
03	Professional Documents		105 minutes
04	Excel Fundamentals		105 minutes
05	Formulas Functions		105 minutes
06	Data Management		105 minutes
07	PowerPoint Fundamentals		105 minutes
08	Professional Presentations		105 minutes
09	Final Project Capstone		105 minutes

How to use this book: Each session below is a complete, ready-to-teach lesson note. Section A tells you what to prepare, B lists the outcomes, D–F are the teaching content, G–H are your demos and practicals, and the knowledge check with answer key is at the end. The concept diagram for each session is embedded as a visual aid.

SESSION 01 OF 9

Word Fundamentals

CONCEPT DIAGRAM — SESSION 01



SESSION 1: WORD FUNDAMENTALS

Week: 1 **Module:** Microsoft Word **Duration:** 105 minutes **Difficulty:** Absolute Beginner **Prerequisite Knowledge:** Basic computer operation (mouse, keyboard, file management) — taught in Typing & Computer Basics **Connection to Previous Session:** First session of Microsoft Office **Preparation Required Before Class:**

- Ensure Microsoft Word is installed and activated on all academy machines
- Open Word on all machines to a blank document
- Have a projector connected for instructor demonstrations

A. SESSION IDENTIFICATION

Field	Detail
Session Number	1
Title	Word Fundamentals
Week	1

Field	Detail
Module	Microsoft Word
Duration	105 minutes
Difficulty	Absolute Beginner
Prerequisite Knowledge	Basic computer operation
Connection to Previous Session	First session
Preparation	Word installed and open on all machines

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Explain what Microsoft Word is** and what it is used for.
2. **Open Microsoft Word** from the Start menu or desktop shortcut.
3. **Identify the main parts of the Word interface:** the Ribbon, tabs, document area, status bar, and Quick Access Toolbar.
4. **Type and edit text** in a Word document, including inserting, deleting, and replacing text.
5. **Select text** using the mouse (click and drag) and keyboard (Shift + arrow keys).
6. **Cut, copy, and paste** text using right-click menu and keyboard shortcuts.
7. **Save a document** to a specific location with a descriptive file name.
8. **Open a saved document** from the File menu or recent documents list.
9. **Export a document to PDF** using the Save As or Export function.
10. **Explain why** saving work regularly and using descriptive file names matters.

C. REQUIRED MATERIALS

Hardware:

- One desktop computer per learner
- One keyboard per learner
- One mouse per learner
- One monitor per learner
- Projector for instructor demonstration

Software:

- Windows 10 or Windows 11
- Microsoft Word 2019 or Microsoft 365 (pre-installed)

Printed Materials:

- "Word Interface Diagram" handout (one per learner) — labelled screenshot of the Word interface
- "Keyboard Shortcuts" handout (one per learner)

Sample Files:

- A sample text document (provided by instructor) for editing practice

Internet: Not required for this session.

Alternative for Learners Without Word: Use Microsoft Word Online (free at office.com) or Google Docs.

D. OPENING ACTIVITY (10 minutes)

"What Do You Use Word For?"

Instructor Setup: Open Word on the projector. Show a blank document.

Instructor Script: "Good morning, everyone. Welcome to the Microsoft Office course. Over the next three weeks, you will learn to use the three most important productivity tools in the world: Microsoft Word, Excel and PowerPoint.

Today we start with Word. Before we open anything, I want to know: who has used Microsoft Word before? [Pause for hands.] What did you use it for? [Pause for answers — common answers: writing letters, assignments, CVs.]

Word is the most common document creation tool in the world. It is used for:

- Writing letters and business correspondence
- Creating CVs and cover letters
- Writing reports and proposals
- Creating flyers and invitations
- Writing meeting minutes
- And much more

By the end of this course, you will be able to create professional documents that look like they were made by a professional. Let us begin."

Expected Learner Response: Some learners will have used Word before. Most will be complete beginners. The activity sets the context and activates prior knowledge.

E. PREREQUISITE REVIEW (5 minutes)

Instructor Script: "Before we start, let us remember the basic computer skills you need for this course. Quick review."

Review Questions and Expected Answers:

1. "How do you open a program?"
    - Click the Start button, find the program, and click it. Or double-click a desktop shortcut.
2. "How do you create a new folder?"
    - Right-click in an empty space, click "New," click "Folder," type a name, and press Enter.
3. "How do you save a file?"
    - Click "File," click "Save" or "Save As," choose a location, type a name, and click "Save." Or press Ctrl+S.
4. "How do you select text with the mouse?"
    - Click and drag over the text.
5. "What does Ctrl+Z do?"
    - It undoes the last action.

How to Address Common Gaps:

- If a learner cannot open a program, sit with them and guide them through the Start menu.
- If a learner forgot how to save, remind them: Ctrl+S.
- If a learner is still struggling with the mouse, pair them with a more confident learner for the first practical exercise.

F. FULL LECTURE NOTES (35 minutes)

Topic 1: What Is Microsoft Word? (5 minutes)

Instructor Script: "Microsoft Word is a word processing program. A word processing program is software that lets you create, edit, format and print text documents. It is like a typewriter, but much more powerful. You can undo mistakes, change fonts, insert images, create tables and much more.

Word is used in almost every office in the world. When a job advert says "proficient in Microsoft Word," they mean you should be able to create clean, professional documents without help. That is exactly what you will learn in this course.

What Word Is Used For:

- Business letters and correspondence
- CVs and cover letters
- Reports and proposals
- Meeting minutes
- Flyers, invitations and certificates
- Newsletters and brochures
- And much more

Word vs Other Tools: You might have used Google Docs or Notepad. Google Docs is similar to Word but works in a web browser. Notepad is a very basic text editor with no formatting. Word is the professional standard. It has more features than both and is what most employers expect."

Topic 2: Opening Word and the Interface (10 minutes)

Instructor Script: "Let us open Word and explore the interface. The interface is everything you see on the screen — the buttons, menus and tools.

Opening Word:

1. Click the Start button (Windows logo in the bottom-left corner).
2. Type "Word" in the search box.
3. Click on "Microsoft Word" in the results.
4. Word opens. You will see a start screen with templates. Click "Blank document" to create a new document.

The Word Interface — Main Parts:

1. The Title Bar (top): This shows the name of the document and the program. By default, a new document is called "Document1." When you save it, you will give it a real name.

2. The Quick Access Toolbar (top-left): This is a small toolbar with buttons for common actions: Save, Undo and Redo. You can customise it, but we will leave it as it is for now.

3. The Ribbon (below the title bar): The Ribbon is the most important part of the Word interface. It contains all the tools you need to create and format documents. The Ribbon is organised into tabs:

- **File:** Open, save, print, export
- **Home:** Font, paragraph, styles, editing
- **Insert:** Tables, images, shapes, headers, footers
- **Design:** Themes, colours, fonts
- **Layout:** Margins, orientation, size
- **References:** Table of contents, citations
- **Mailings:** Mail merge
- **Review:** Spelling, comments, track changes
- **View:** Zoom, layout views

We will use the Home, Insert and Layout tabs most in this course.

4. The Document Area (centre): This is the large white area where you type your document. It looks like a sheet of paper. The blinking vertical line is called the cursor. It shows where the text will appear when you type.

5. The Status Bar (bottom): This shows information about your document: the page number, word count, language and zoom level.

6. The Scroll Bars (right and bottom): These let you scroll up and down or side to side to see different parts of your document."

Topic 3: Typing and Editing Text (10 minutes)

Instructor Script: "Now let us type some text and learn how to edit it.

Typing:

1. Click in the document area. The cursor will appear where you click.
2. Start typing. The text appears at the cursor position.
3. Press Enter to create a new paragraph (a new block of text).
4. Press the space bar to create a space between words.
5. Press Backspace to delete the character to the left of the cursor.
6. Press Delete to delete the character to the right of the cursor.

Selecting Text: To format or edit text, you first need to select it. Selecting means highlighting the text so you know which part you are working with.

- **To select a word:** Double-click on it.
- **To select a sentence:** Hold Ctrl and click anywhere in the sentence.
- **To select a paragraph:** Triple-click anywhere in the paragraph.
- **To select all text:** Press Ctrl+A.
- **To select with the mouse:** Click and drag over the text.
- **To select with the keyboard:** Hold Shift and use the arrow keys.

Cut, Copy and Paste:

- **Cut (Ctrl+X):** Removes the selected text and places it on the clipboard (temporary storage).
- **Copy (Ctrl+C):** Copies the selected text to the clipboard. The original stays where it is.
- **Paste (Ctrl+V):** Inserts the contents of the clipboard at the cursor position.

Think of it like this: Cut is like cutting out a piece of paper and moving it. Copy is like photocopying a piece of paper. Paste is like sticking the piece of paper somewhere.

Undo and Redo:

- **Undo (Ctrl+Z):** Reverses your last action. If you make a mistake, press Ctrl+Z.
- **Redo (Ctrl+Y):** Repeats your last action. If you undo something by mistake, press Ctrl+Y.

These are your best friends in Word. Use them often."

Topic 4: Saving and Opening Documents (5 minutes)

Instructor Script: "Saving your work is one of the most important habits you can develop. If you do not save, you lose your work when the computer crashes or the power goes out.

Saving a Document for the First Time:

1. Click "File" in the top-left corner (or press Ctrl+S).
2. Click "Save As."
3. Choose where to save: Desktop, Documents, or a specific folder.
4. Type a file name in the "File name" box. Use a descriptive name, like "My First Document" or "Business Letter."
5. Click "Save."

Saving an Existing Document: If you have already saved the document, just press Ctrl+S or click the Save icon (floppy disk) in the Quick Access Toolbar. Word will save the changes without asking for a location.

Opening a Saved Document:

1. Click "File."
2. Click "Open."
3. Navigate to where you saved the document.
4. Click on the document name.
5. Click "Open."

Or, if the document is recent:

1. Click "File."
2. Click "Open."
3. Click on the document name in the "Recent" list.

Good File Names:

- Good: "Business Letter to GTBank," "CV Adaeze Okafor," "Meeting Notes October 2026"
- Bad: "Document1," "Letter," "Stuff"

A good file name tells you what the document is without opening it. This saves you time when you are looking for a document later."

Topic 5: Exporting to PDF (5 minutes)

Instructor Script: "PDF stands for Portable Document Format. It is a file format that preserves the exact layout and formatting of your document. When you send someone a PDF, they see the document exactly as you designed it — regardless of what software they have on their computer.

PDF is the standard format for sharing finished documents. When you apply for a job and the advert says "send your CV as a PDF," they mean export your Word document to PDF.

How to Export to PDF:

1. Click "File."
2. Click "Export" (or "Save As" in older versions).
3. Click "Create PDF/XPS Document."
4. Choose where to save the PDF.
5. Type a file name (or keep the same name as the Word document).
6. Click "Publish" or "Save."

You now have two versions of your document: the Word version (which you can edit) and the PDF version (which you share). Always keep both."

G. INSTRUCTOR DEMONSTRATION (15 minutes)

Demonstration 1: Opening Word and Typing Text (5 minutes)

What the Instructor Prepares:

- Word open on the projector
- A blank document

What the Instructor Says and Does: "Watch carefully. I am going to open Word and type a short paragraph."

1. "I click the Start button. I type 'Word.' I click on Microsoft Word."
2. "The start screen appears. I click 'Blank document.'"
3. "A new document opens. I click in the document area. The cursor appears."
4. "I type: 'My name is Instructor Name. I am a learner at Cyber Elias Academy. This is my first Word document.'"
5. "I press Enter to create a new paragraph. I type: 'I am learning how to use Microsoft Word to create professional documents.'"

What Learners Should Observe: The instructor opening Word, creating a blank document, and typing text.

Expected Result: Learners can describe the steps to open Word and type text.

Demonstration 2: Selecting, Cutting, Copying and Pasting (5 minutes)

What the Instructor Prepares:

- The document from Demonstration 1 still open

What the Instructor Says and Does: "Now I am going to show you how to select, cut, copy and paste text."

1. "I want to select the first sentence. I double-click on the word 'name.' The word is selected. I want to select the whole sentence. I triple-click on the sentence. The whole sentence is selected."
2. "I press Ctrl+C to copy it. Nothing visible happens, but it is copied."
3. "I click at the end of the document. I press Ctrl+V. The sentence appears again. That is paste."
4. "Now I want to move the second paragraph. I triple-click on it to select it. I press Ctrl+X to cut it. The paragraph disappears."
5. "I click at the beginning of the document. I press Ctrl+V. The paragraph moves to the top."
6. "I made a mistake. I press Ctrl+Z to undo. The paragraph moves back. I press Ctrl+Y to redo. It moves again."

What Learners Should Observe: The instructor selecting text with the mouse, copying, pasting, cutting, undoing and redoing.

Expected Result: Learners can explain the difference between cut, copy and paste, and use undo/redo.

Questions the Instructor Can Ask:

- "What is the difference between cut and copy?" (Answer: Cut removes the text. Copy creates a duplicate.)
- "What does Ctrl+Z do?" (Answer: It undoes the last action.)

Demonstration 3: Saving and Exporting to PDF (5 minutes)

What the Instructor Prepares:

- The document from Demonstrations 1 and 2 still open

What the Instructor Says and Does: "Now I am going to save this document and export it to PDF."

1. "I click 'File.' I click 'Save As.' I navigate to the Desktop. I type 'My First Document' in the File name box. I click 'Save.'"
2. "Now I will export it to PDF. I click 'File.' I click 'Export.' I click 'Create PDF/XPS Document.' I navigate to the Desktop. I type 'My First Document' in the File name box. I click 'Publish.'"
3. "I now have two files on the Desktop: 'My First Document.docx' (the Word version) and 'My First Document.pdf' (the PDF version)."

What Learners Should Observe: The instructor saving a document and exporting it to PDF.

Expected Result: Learners can describe the steps to save and export to PDF.

H. GUIDED PRACTICAL EXERCISE (20 minutes)

Exercise: "My First Word Document"

Practical Objective: The learner will open Word, type a short paragraph, select and edit text, cut/copy/paste, save the document, and export it to PDF.

Starting Materials:

- Word open on each machine to a blank document

Instructions:

Part 1: Open Word and Type Text (5 minutes)

1. "Click the Start button. Type 'Word.' Click on Microsoft Word."
2. "Click 'Blank document' to create a new document."
3. "Click in the document area. Type your full name."
4. "Press Enter. Type: 'I am a learner at Cyber Elias Academy.'"
5. "Press Enter. Type: 'I am learning how to use Microsoft Word to create professional documents.'"

Instructor Prompts:

- "Raise your hand if you can see your text on the screen."
- "What happens when you press Enter?" (Answer: A new paragraph is created.)

Part 2: Select and Edit Text (5 minutes)

6. "Triple-click on your name to select it."
7. "Press Ctrl+C to copy it."
8. "Click at the end of the last sentence. Press Enter to create a new line. Press Ctrl+V to paste your name."
9. "Double-click on the word 'professional' in the last sentence. Press Delete to remove it. Type 'excellent' to replace it."
10. "Press Ctrl+Z to undo. Press Ctrl+Y to redo."

Instructor Prompts:

- "What happened when you pressed Delete?" (Answer: The selected word was deleted.)
- "What happened when you pressed Ctrl+Z?" (Answer: The word came back.)

Part 3: Save the Document (5 minutes)

11. "Click 'File.' Click 'Save As.' Navigate to the Desktop."
12. "Type 'My First Document' in the File name box. Click 'Save.'"
13. "Press Ctrl+S to save again. Nothing happens — it just saves silently."

Instructor Prompts:

- "Where did you save the document?" (Answer: On the Desktop.)
- "What is the file name?" (Answer: My First Document.)

Part 4: Export to PDF (5 minutes)

14. "Click 'File.' Click 'Export.' Click 'Create PDF/XPS Document.'"
15. "Navigate to the Desktop. Type 'My First Document' in the File name box. Click 'Publish.'"
16. "Look on your Desktop. You should see two files: 'My First Document.docx' and 'My First Document.pdf.'"

Instructor Prompts:

- "What is the difference between the .docx file and the .pdf file?" (Answer: The .docx file is the Word version you can edit. The .pdf file is the finished version you share.)

Expected Result: Every learner has created a Word document, typed text, edited text, saved it to the Desktop, and exported it to PDF.

Troubleshooting Guidance:

- If the learner cannot find Word: "Click the Start button and type 'Word.' It should appear in the search results."
- If the learner cannot save: "Click 'File,' then 'Save As,' then navigate to the Desktop. Make sure you click 'Save' in the dialog box."
- If the learner cannot export to PDF: "In some versions of Word, the option is called 'Save As' and you choose PDF from the 'Save as type' dropdown."

Completion Checklist:

- ☐ Opened Word and created a blank document
- ☐ Typed three lines of text
- ☐ Selected text using triple-click
- ☐ Used Ctrl+C, Ctrl+V, Ctrl+X, Ctrl+Z, Ctrl+Y
- ☐ Saved the document to the Desktop as "My First Document"
- ☐ Exported the document to PDF

I. INDEPENDENT CLASS EXERCISE (10 minutes)

Exercise: "Word Scavenger Hunt"

Scenario: "You are helping your friend who is new to Word. They need to find and use several tools. Your task is to find each tool in the Word interface and use it."

Task: Without looking at your notes or your neighbour's screen, complete the following tasks in Word:

1. Find the "Home" tab on the Ribbon. Click it. What do you see?

2. Find the "Insert" tab. Click it. What do you see?
3. Find the "File" tab. Click it. What do you see?
4. Type your full name. Select it. Make it bold. (Hint: Look for the Bold button on the Home tab.)
5. Type the name of your school or workplace. Select it. Make it italic. (Hint: Look for the Italic button.)
6. Type today's date. Select it. Underline it. (Hint: Look for the Underline button.)
7. Select all the text. Change the font size to 14. (Hint: Look for the font size box.)
8. Save the document as "Scavenger Hunt" on the Desktop.

Requirements:

- Complete all 8 tasks independently
- Do not ask the instructor for help unless you are truly stuck
- You may look at your notes from the lecture

Time Allowance: 10 minutes

Success Criteria:

- All 8 tasks completed correctly
- The learner can explain what each tool does

Hints Available If the Learner Gets Stuck:

- Hint 1: "The Home tab has the most common tools: font, paragraph and styles."
- Hint 2: "The Bold button looks like a 'B.' The Italic button looks like an 'I.' The Underline button looks like a 'U' with a line under it."
- Hint 3: "The font size box is next to the font name box on the Home tab. It usually shows a number like '11' or '12.'"

Complete Answer Key for the Instructor:

1. The Home tab contains tools for font formatting (bold, italic, underline, font size, font colour), paragraph formatting (alignment, spacing, bullets) and styles.
2. The Insert tab contains tools for inserting tables, images, shapes, headers, footers, page numbers and other elements.
3. The File tab contains options for opening, saving, printing and exporting documents.
4. The learner types their name, selects it, and clicks the Bold button (B) on the Home tab.
5. The learner types their school name, selects it, and clicks the Italic button (I) on the Home tab.
6. The learner types today's date, selects it, and clicks the Underline button (U) on the Home tab.
7. The learner selects all text (Ctrl+A) and changes the font size to 14 using the font size box on the Home tab.
8. The learner saves the document to the Desktop with the name "Scavenger Hunt."

J. REAL-WORLD CASE STUDY (5 minutes)

Case: "Adaeze's Job Application"

Situation: Adaeze is a fresh graduate. She sees a job advert for an administrative assistant at a company in Port Harcourt. The advert says to send a CV and cover letter to `hr@companyname.com` as PDF attachments. Adaeze has a CV, but it is a plain Word document with no formatting. She opens Word and stares at the screen. She does not know how to make it look professional.

The Problem: Adaeze's CV looks like a school assignment, not a professional document. If she sends it as it is, she will not get an interview.

Why It Matters: Your CV is often the first impression you make on a potential employer. A poorly formatted CV suggests you are careless or unskilled. A well-formatted CV suggests you are professional and competent. In a competitive job market, the difference between a good CV and a bad CV can be the difference between getting an interview and being rejected.

Relevant Concept: Word formatting. If Adaeze knew how to use fonts, sizes, alignment and spacing, she could transform her plain CV into a professional document.

Worked Solution: If Adaeze had taken this course, here is what she would do:

1. She would open her CV in Word.

2. She would select the headings (name, education, experience) and make them bold and slightly larger.
3. She would use a clean, professional font like Calibri or Arial.
4. She would align the text consistently (left-aligned for body text, centred for the name).
5. She would add spacing between sections.
6. She would save the document and export it to PDF.
7. She would send the PDF to the employer.

Reasoning: Each formatting decision makes the CV easier to read and more professional. A clean, well-organised CV shows that the candidate cares about details and takes pride in their work.

Possible Mistake: Adaeze might use too many fonts and colours. She might think that more formatting makes it look better. In reality, too much formatting makes a document look cluttered and unprofessional.

Correction: Use a maximum of two fonts and two colours. Keep it simple and clean. Professional does not mean fancy. Professional means clear, consistent and easy to read.

Transfer Exercise: "Your friend Chinedu has a CV with five different fonts and three different colours. It looks messy. What advice would you give him?"

Professional Connection: Every job application requires a CV. Every CV should be professional. This is not optional — it is expected. The skills you learn in this course will serve you every time you apply for a job, write a report or create any professional document.

K. COMMON MISTAKES AND TROUBLESHOOTING (5 minutes)

Mistake 1: Not Saving Regularly

Symptoms: The learner types for 20 minutes without saving. The computer crashes and all work is lost. **Likely Cause:** The learner forgets to save or does not realise the importance. **How to Investigate:** Ask the learner, "When did you last save?" **Safe Corrective Steps:** Teach the learner to press Ctrl+S every few minutes. Make it a habit. **How to Confirm the Problem Is Resolved:** The learner saves regularly. **How to Prevent Recurrence:** Remind the learner at the start of every session: "Save early, save often."

Mistake 2: Typing Over Existing Text

Symptoms: The learner clicks in the middle of a sentence and starts typing, replacing the existing text. **Likely Cause:** The learner does not understand the difference between insert mode and overtype mode. **How to Investigate:** Ask the learner, "What happened to the text after the cursor?" **Safe Corrective Steps:** Explain that Word is usually in insert mode (new text pushes existing text forward). If the learner accidentally pressed the Insert key, it switches to overtype mode (new text replaces existing text). Press the Insert key again to return to insert mode. **How to Confirm the Problem Is Resolved:** The learner types in insert mode. **How to Prevent Recurrence:** Show the learner where the Insert key is and explain what it does.

Mistake 3: Using the Space Bar Instead of Tab or Indent

Symptoms: The learner presses the space bar five times to indent a paragraph. **Likely Cause:** The learner does not know about the Tab key or the indent button. **How to Investigate:** Ask the learner, "How did you indent the paragraph?" **Safe Corrective Steps:** Teach the learner to use the Tab key to indent the first line of a paragraph. Or use the Increase Indent button on the Home tab. **How to Confirm the Problem Is Resolved:** The learner uses Tab or the indent button. **How to Prevent Recurrence:** Show the learner the Tab key and the indent button. Explain that using the space bar for indentation is unreliable.

Mistake 4: Not Exporting to PDF

Symptoms: The learner saves the Word document but does not export it to PDF. **Likely Cause:** The learner does not know how or does not understand why PDF is important. **How to Investigate:** Ask the learner, "Did you export to PDF?" **Safe Corrective Steps:** Show the learner how to export to PDF. Explain that PDF is the standard format for sharing finished documents. **How to Confirm the Problem Is Resolved:** The learner exports to PDF. **How to Prevent Recurrence:** Make PDF export a standard step in every assignment.

Mistake 5: Using Descriptive File Names

Symptoms: The learner saves files as "Document1," "Document2," "Document3." **Likely Cause:** The learner does not change the default file name. **How to Investigate:** Ask the learner, "What did you name the file?" **Safe Corrective Steps:** Teach the learner to always use a descriptive file name. Show them examples of good and bad file names. **How to Confirm the Problem Is Resolved:** The learner uses descriptive file names. **How to Prevent Recurrence:** Remind the learner at the start of every save: "What is this document about? Put that in the name."

L. SESSION ASSIGNMENT (5 minutes)

Assignment Title: "My First Professional Document"

Purpose: To reinforce Word fundamentals by having the learner create a simple professional document.

Realistic Scenario: "You need to write a short letter to your landlord about a maintenance issue in your apartment. You want the letter to look professional. Create this letter in Microsoft Word."

Detailed Instructions:

1. Open Microsoft Word. Create a new blank document.
2. Type the following letter (you can change the details to match your own situation):
    - Your name and address (top-right corner)
    - Today's date
    - Landlord's name and address
    - Subject: "Maintenance Request — [Your Apartment Number]"
    - Body: A short paragraph explaining a maintenance issue (e.g., a leaking tap, a broken window)
    - Polite closing: "Thank you for your prompt attention to this matter."
    - Sign-off: "Yours sincerely," followed by your name
3. Save the document as "Letter to Landlord" on the Desktop.
4. Export the document to PDF.
5. Show both files to the instructor before leaving.

Required Deliverables:

- A Word document named "Letter to Landlord" saved on the Desktop
- A PDF version of the same document on the Desktop

Submission Format:

- Both files must be visible on the Desktop of the academy machine
- The learner must show the instructor both files before leaving

Deadline Recommendation:

- Before leaving the session (in-class assignment)

Marking Rubric:

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Content completeness	All parts of the letter included	Most parts included	Several parts missing
Formatting	Name, address, date, subject, body, sign-off all present and properly placed	Most elements present and placed correctly	Elements missing or misplaced
File saved correctly	Saved on Desktop with correct name	Saved but wrong location or name	Not saved
PDF exported	PDF exported correctly	PDF exported but wrong location	Not exported
Professional appearance	Clean, well-organised, easy to read	Mostly clean	Cluttered or hard to read

Total: 25 marks

Common Mistakes to Avoid:

- Forgetting to include the date
- Not including a subject line
- Saving the file with a vague name (e.g., "Document1")
- Not exporting to PDF

Instructor Reference Solution:

```
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Port Harcourt, Rivers State

8 October 2026

Mr. Chinedu Okafor
456 Aba Road
Port Harcourt, Rivers State

Subject: Maintenance Request – Apartment 5B

Dear Mr. Okafor,

I am writing to bring to your attention a maintenance issue in my apartment. The kitchen tap has been leaking for the past three day

I would appreciate it if you could send a plumber to fix the tap as soon as possible. Please let me know a convenient time for the r

Thank you for your prompt attention to this matter.

Yours sincerely,
Adaeze Okafor
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M. KNOWLEDGE CHECK (5 minutes)

Questions

1. Multiple Choice: What is Microsoft Word? a) A spreadsheet program b) A word processing program c) A presentation program d) A web browser

Answer: b) A word processing program

2. Short Answer: What are the main parts of the Word interface?

Answer: Title bar, Quick Access Toolbar, Ribbon (with tabs), document area, status bar, and scroll bars.

3. Multiple Choice: What does Ctrl+S do? a) Select all text b) Save the document c) Copy text d) Print the document

Answer: b) Save the document

4. Explain in Your Own Words: Why is it important to export a document to PDF?

Answer: PDF preserves the exact layout and formatting of the document. When you send someone a PDF, they see the document exactly as you designed it, regardless of what software they have. PDF is the standard format for sharing finished documents.

5. Practical Identification: Open Word. Identify the Ribbon, the Home tab, the document area, and the status bar.

Answer: The learner correctly identifies all four elements.

6. Scenario-Based: Your friend says, "I typed a whole page but then the computer crashed and I lost everything. What did I do wrong?" What advice would you give?

Answer: You should have saved your work regularly. Press Ctrl+S every few minutes. Saving takes two seconds and protects your work from crashes and power outages.

7. Error-Detecting: A learner writes in their assignment: "I saved my file as 'Document1' on the Desktop." What is wrong with this, and what should they have done?

Answer: "Document1" is a vague default name. The learner should have used a descriptive name that tells them what the document is, like "Letter to Landlord" or "Meeting Notes October 2026."

N. REFLECTION AND RECAP (5 minutes)

Plain-English Summary

Today you learned the fundamentals of Microsoft Word. You learned what Word is, how to open it, and how to identify the main parts of the interface. You learned how to type and edit text, select text, cut, copy and paste, and use undo and redo. You learned how to save a document with a descriptive file name and how to export it to PDF. These are the foundation skills for everything else you will do in Word.

Five Important Takeaways

1. Microsoft Word is a word processing program used for creating professional documents.
2. The Ribbon contains all the tools you need, organised into tabs.
3. Select text first, then format or edit it.
4. Save regularly with Ctrl+S. Use descriptive file names.
5. Export finished documents to PDF for sharing.

Questions That Help You Recognise What You Can Now Do

- Can you open Word and identify the main parts of the interface?
- Can you type, select, cut, copy and paste text?
- Can you save a document with a descriptive name?
- Can you export a document to PDF?
- Can you undo and redo your actions?

Connection to the Next Session

In the next session, you will learn how to format documents professionally. You will learn about fonts, sizes, bold, italic, underline, alignment, spacing and bullets. These are the tools that turn a plain document into a professional one. You now have the foundation. Next, you will learn to make your documents look great.

O. INSTRUCTOR EVALUATION NOTES

Observable Signs of Competence

- The learner opens Word independently.
- The learner identifies the Ribbon, tabs and document area.
- The learner types, selects, cuts, copies and pastes text.
- The learner saves with a descriptive file name.
- The learner exports to PDF.
- The learner uses Ctrl+Z and Ctrl+Y confidently.

Common Misconceptions

- **"Word is just for typing."** Some learners think Word is only for typing text. Correct this by showing them the formatting tools and explaining that Word is for creating professional documents.
- **"Saving and exporting are the same thing."** Some learners think saving a Word document is enough. Correct this by explaining that PDF is the format you share with others.
- **"Ctrl+Z is only for mistakes."** Some learners think undo is only for fixing errors. Correct this by explaining that undo is a tool for exploring — you can try something, undo it, and try something else.

Suggested Remedial Exercises

- For learners who cannot find the Ribbon: Print a labelled Word interface diagram and have the learner label each part.
- For learners who forget to save: Set a timer that goes off every 5 minutes. When it rings, everyone presses Ctrl+S.
- For learners who cannot export to PDF: Walk them through the process step by step. Let them practise multiple times.

Differentiation Guidance

Level 1 — Supported (for learners who struggle):

- Provide a labelled Word interface diagram for reference.
- Pair with a more confident learner for practical exercises.
- Allow extra time for the independent exercise.
- Focus on typing and saving before formatting.

Level 2 — Independent (standard):

- Complete all exercises as written.
- Answer all knowledge check questions.
- Complete the assignment to the standard shown in the rubric.

Level 3 — Challenge (for learners who finish early):

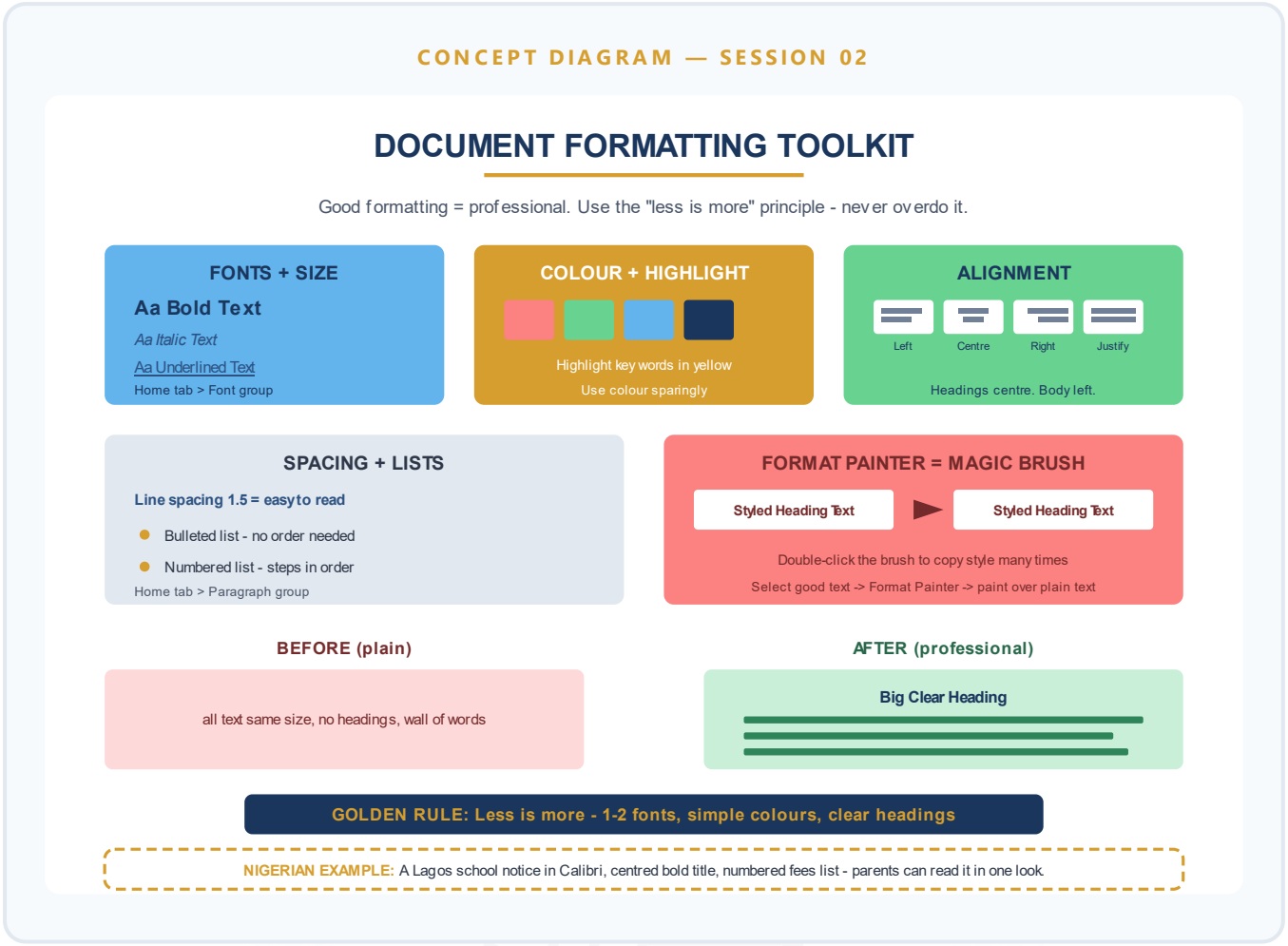
- Ask the learner to create a more complex document (e.g., a two-page report with headings and subheadings).
- Ask the learner to research and write down three more keyboard shortcuts.
- Ask the learner to explain the difference between Save and Save As.

End of Session 1

Next Session: Session 2 — Document Formatting **What to Bring:** Attention to detail, willingness to experiment with formatting **Preparation:** Prepare a plain-text document for formatting practice

SESSION 02 OF 9

Document Formatting



SESSION 2: DOCUMENT FORMATTING

Week: 1 **Module:** Microsoft Word **Duration:** 105 minutes **Difficulty:** Beginner **Prerequisite Knowledge:** Session 1 — Word fundamentals, interface navigation, typing, saving **Connection to Previous Session:** Builds on Session 1 by teaching the formatting tools that turn plain text into professional documents **Preparation Required Before Class:**

- Prepare a plain-text document (e.g., a sample CV or business letter) for formatting practice
- Ensure Microsoft Word is installed and functional on all machines
- Have a projector connected for instructor demonstrations

A. SESSION IDENTIFICATION

Field	Detail
Session Number	2
Title	Document Formatting
Week	1

Field	Detail
Module	Microsoft Word
Duration	105 minutes
Difficulty	Beginner
Prerequisite Knowledge	Session 1 — Word fundamentals
Connection to Previous Session	Builds on Session 1 by teaching formatting tools
Preparation	Plain-text document for formatting practice

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Change fonts and font sizes** to create a professional appearance.
2. **Apply bold, italic and underline** to emphasise text.
3. **Change text colour and highlighting** for visual impact.
4. **Set text alignment** (left, centre, right, justify) appropriately.
5. **Adjust line spacing and paragraph spacing** for readability.
6. **Create bulleted and numbered lists** for organised information.
7. **Use the Format Painter** to copy formatting from one section to another.
8. **Explain why** consistent formatting matters in professional documents.
9. **Apply the "less is more" principle** — using formatting sparingly and purposefully.

C. REQUIRED MATERIALS

Hardware:

- One desktop computer per learner
- One keyboard per learner
- One mouse per learner
- One monitor per learner
- Projector for instructor demonstration

Software:

- Windows 10 or Windows 11
- Microsoft Word 2019 or Microsoft 365

Printed Materials:

- "Formatting Tools Reference" handout (one per learner) — a table of formatting tools and their uses

Sample Files:

- A plain-text CV (provided by instructor) for formatting practice
- A plain-text business letter (provided by instructor) for formatting practice

Internet: Not required for this session.

D. OPENING ACTIVITY (10 minutes)

"Spot the Difference"

Instructor Setup: Display two versions of the same document on the projector: one plain (no formatting) and one professionally formatted.

Instructor Script: "Good morning, everyone. Look at these two documents. They contain the exact same text. But one looks like a school assignment, and the other looks like it was made by a professional. What is the difference? [Pause for answers — fonts, sizes, bold, spacing, alignment.]

The difference is formatting. Formatting is how you make text look — the font, the size, the colour, the spacing, the alignment. Good formatting makes a document easy to read and professional in appearance. Bad formatting makes a document hard to read and amateur.

Today, you will learn the formatting tools that turn a plain document into a professional one. By the end of this session, you will be able to take any plain text and make it look like it was designed by a professional. Let us begin."

Expected Learner Response: Learners will identify visible differences: fonts, bold text, spacing, alignment. This activates prior knowledge and sets the context for the session.

E. PREREQUISITE REVIEW (5 minutes)

Instructor Script: "Before we start, let us remember what we learned in the last session. Quick review."

Review Questions and Expected Answers:

1. "How do you open Microsoft Word?"
   - Click the Start button, type "Word," click on Microsoft Word. Or double-click a desktop shortcut.
2. "What is the Ribbon?"
   - The Ribbon is the area below the title bar that contains all the tools, organised into tabs.
3. "How do you select text?"
   - Click and drag over the text. Or double-click a word. Or triple-click a paragraph. Or press Ctrl+A to select all.
4. "What does Ctrl+S do?"
   - It saves the document.
5. "What does Ctrl+Z do?"
   - It undoes the last action.

How to Address Common Gaps:

- If a learner cannot remember how to select text, demonstrate it again quickly.
- If a learner forgot the keyboard shortcuts, write them on the whiteboard: Ctrl+C, Ctrl+V, Ctrl+X, Ctrl+Z, Ctrl+S.

F. FULL LECTURE NOTES (35 minutes)

Topic 1: Fonts and Font Sizes (10 minutes)

Instructor Script: "A font is the style of the letters. The font you choose affects how your document looks and feels. Some fonts are formal and professional. Others are casual and playful. Choosing the right font is an important part of document design.

Common Fonts:

- **Calibri** — the default font in modern Word. Clean, modern, professional. Good for most documents.
- **Arial** — similar to Calibri. Clean and professional. Good for business documents.
- **Times New Roman** — the traditional serif font. Formal and academic. Good for reports and essays.
- **Cambria** — a serif font. Formal and elegant. Good for professional documents.
- **Verdana** — a sans-serif font. Clear and readable. Good for documents that will be read on screen.

Font Size: Font size is measured in points (pt). The default size in Word is 11pt or 12pt. Here is a general guide:

- **Body text:** 11pt or 12pt

- **Headings:** 14pt, 16pt or 18pt
- **Title:** 18pt, 20pt or larger
- **Footnotes or captions:** 9pt or 10pt

How to Change Font and Size:

1. Select the text you want to format.
2. Go to the Home tab on the Ribbon.
3. In the Font group, click the font name dropdown and choose a font.
4. Click the font size dropdown and choose a size.
5. Or use the Increase Font Size (A with an up arrow) and Decrease Font Size (A with a down arrow) buttons.

The "Less Is More" Rule: Use a maximum of two fonts in a professional document. One font for headings and one for body text. Using more than two fonts makes a document look cluttered and unprofessional.

For example:

- Headings: Calibri, 16pt, bold
- Body text: Calibri, 11pt, regular

This is clean, consistent and professional."

Topic 2: Bold, Italic and Underline (5 minutes)

Instructor Script: "Bold, italic and underline are three ways to emphasise text.

Bold: Bold makes text thicker and darker. It is used for headings, subheadings and important words. To make text bold, select it and click the Bold button (B) on the Home tab. Or press Ctrl+B.

Italic: Italic slants the text to the right. It is used for titles of books, foreign words and emphasis. To make text italic, select it and click the Italic button (I) on the Home tab. Or press Ctrl+I.

Underline: Underline draws a line under the text. It is used for hyperlinks and sometimes for headings. To underline text, select it and click the Underline button (U) on the Home tab. Or press Ctrl+U.

When to Use Each:

- **Bold:** Headings, subheadings, important terms
- **Italic:** Book titles, foreign words, subtle emphasis
- **Underline:** Hyperlinks, some headings

Common Mistake — Over-Emphasis: Some learners make everything bold, italic and underlined. This is a mistake. When everything is emphasised, nothing is emphasised. Use bold, italic and underline sparingly. Emphasise only the most important words or phrases."

Topic 3: Text Colour and Highlighting (5 minutes)

Instructor Script: "Text colour and highlighting add visual impact to a document. But they should be used carefully.

Text Colour: To change the colour of text:

1. Select the text.
2. Go to the Home tab.
3. Click the Font Colour button (A with a coloured bar underneath).
4. Choose a colour from the palette.

Highlighting: Highlighting puts a coloured background behind text, like using a highlighter pen on paper. To highlight text:

1. Select the text.
2. Go to the Home tab.
3. Click the Text Highlight Colour button (a marker pen icon).
4. Choose a colour.

When to Use Colour:

- **Headings:** A dark colour (dark blue, dark green, dark red) can make headings stand out.

- **Important information:** Red or orange can draw attention to critical information.
- **Hyperlinks:** Blue is the standard colour for hyperlinks.

When NOT to Use Colour:

- Do not use bright colours (yellow, pink, lime green) for body text. They are hard to read.
- Do not use too many colours. Stick to one or two colours for the entire document.
- Do not use colour as the only way to convey meaning. Some people are colour-blind."

Topic 4: Alignment (5 minutes)

Instructor Script: "Alignment determines how text is positioned on the page. There are four types of alignment:

Left Align: Text is aligned to the left margin. The right edge is uneven. This is the default alignment and is used for most body text. It is easy to read and looks clean.

Centre Align: Text is centred between the left and right margins. Both edges are uneven. This is used for titles, headings and short lines of text. It is not used for long paragraphs because it is hard to read.

Right Align: Text is aligned to the right margin. The left edge is uneven. This is used for dates, page numbers and some design elements. It is not used for body text.

Justify: Text is aligned to both the left and right margins. Both edges are even. This is used for formal documents like reports and essays. It gives a clean, professional look but can create awkward spacing between words.

How to Change Alignment:

1. Select the text (or click in the paragraph you want to align).
2. Go to the Home tab.
3. In the Paragraph group, click one of the alignment buttons:
    - Align Left (Ctrl+L)
    - Centre (Ctrl+E)
    - Align Right (Ctrl+R)
    - Justify (Ctrl+J)

Best Practice: Use left align for body text. Use centre align for titles and headings. Use justify for formal reports if you want a clean, even look."

Topic 5: Line Spacing and Paragraph Spacing (5 minutes)

Instructor Script: "Spacing is the amount of space between lines of text and between paragraphs. Good spacing makes a document easy to read. Bad spacing makes it cramped or disjointed.

Line Spacing: Line spacing is the space between lines within a paragraph. The default is 1.15 or 1.5. For most documents, 1.5 is a good choice. It gives enough space to read comfortably.

To change line spacing:

1. Select the text (or click in the paragraph).
2. Go to the Home tab.
3. In the Paragraph group, click the Line and Paragraph Spacing button (lines with an up and down arrow).
4. Choose a spacing option (1.0, 1.15, 1.5, 2.0, etc.).

Paragraph Spacing: Paragraph spacing is the space between paragraphs. By default, Word adds a small space after each paragraph. You can increase or decrease this space.

To change paragraph spacing:

1. Select the text (or click in the paragraph).
2. Go to the Home tab.
3. In the Paragraph group, click the Line and Paragraph Spacing button.
4. Choose "Add Space Before Paragraph" or "Add Space After Paragraph."

Best Practice:

- Use 1.5 line spacing for most documents.
- Use a small space after each paragraph (6pt or 8pt).
- Do not press Enter multiple times to create space between paragraphs. Use the paragraph spacing options instead."

Topic 6: Bullets and Numbering (5 minutes)

Instructor Script: "Bullets and numbering are used to organise information into lists. They make information easier to scan and understand.

Bullets: Bullets are used for lists where the order does not matter. For example:

- Apples
- Oranges
- Bananas

Numbering: Numbering is used for lists where the order matters. For example:

1. Preheat the oven
2. Mix the ingredients
3. Bake for 30 minutes

How to Create a List:

1. Type each item on a separate line.
2. Select all the items.
3. Go to the Home tab.
4. In the Paragraph group, click the Bullets button (three dots) or the Numbering button (three numbers).
5. The list is created.

Best Practice:

- Use bullets for unordered lists.
- Use numbering for ordered lists.
- Keep list items short and parallel in structure.
- Do not mix bullets and numbering in the same list."

Topic 7: Format Painter (5 minutes)

Instructor Script: "The Format Painter is a tool that copies formatting from one section of text to another. It is like a paintbrush for formatting.

How to Use Format Painter:

1. Select the text that has the formatting you want to copy.
2. Go to the Home tab.
3. In the Clipboard group, click the Format Painter button (a paintbrush icon).
4. The cursor changes to a paintbrush.
5. Select the text you want to apply the formatting to.
6. The formatting is copied.

Double-Click Trick: If you want to apply the same formatting to multiple sections, double-click the Format Painter button. This keeps the Format Painter active until you press Escape or click the Format Painter button again.

When to Use Format Painter:

- When you want to apply the same formatting to multiple headings.
- When you want to copy the formatting of a paragraph to another paragraph.
- When you want to ensure consistency across a document."

G. INSTRUCTOR DEMONSTRATION (15 minutes)

Demonstration 1: Formatting a Document (10 minutes)

What the Instructor Prepares:

- A plain-text document open on the projector (e.g., a sample CV)

What the Instructor Says and Does: "I am going to take this plain document and make it professional. Watch each step."

1. "First, I select the name at the top. I change the font to Calibri, size 18, bold, and centre align. This is the heading."
2. "I select the contact information below the name. I change the font to Calibri, size 10, and centre align."
3. "I select the 'Education' heading. I change the font to Calibri, size 14, bold, dark blue colour, and left align."
4. "I select the body text under Education. I change the font to Calibri, size 11, and set line spacing to 1.5."
5. "I repeat the same formatting for the 'Experience' heading and its body text."
6. "I select the 'Skills' heading. I apply the same formatting as the other headings."
7. "I select the list of skills. I click the Bullets button to create a bulleted list."
8. "I click the Format Painter. I select the 'Education' heading to copy its formatting. I then select the 'Experience' and 'Skills' heads to apply the same formatting. Now all headings look consistent."

What Learners Should Observe: The instructor applying fonts, sizes, bold, colour, alignment, spacing, bullets and Format Painter.

Expected Result: Learners can describe the formatting tools used and explain why each choice was made.

Questions the Instructor Can Ask:

- "Why did I use the same formatting for all the headings?" (Answer: Consistency makes a document look professional.)
- "Why did I use 1.5 line spacing?" (Answer: It makes the document easier to read.)
- "Why did I use the Format Painter?" (Answer: It saves time and ensures consistency.)

Demonstration 2: Common Formatting Mistakes (5 minutes)

What the Instructor Prepares:

- A document with intentional formatting mistakes

What the Instructor Says and Does: "Now I am going to show you some common formatting mistakes. See if you can spot them."

1. "This document uses five different fonts. [Point to each font.] This looks cluttered and unprofessional. The fix: use a maximum of two fonts."
2. "This heading is in bright yellow. [Point to the heading.] Yellow is hard to read on a white background. The fix: use a dark colour like dark blue or dark green."
3. "This paragraph has no spacing between lines. [Point to the paragraph.] It looks cramped. The fix: use 1.5 line spacing."
4. "This list uses dashes instead of bullets. [Point to the list.] It looks unprofessional. The fix: use the Bullets button."
5. "This text is justified but has big gaps between words. [Point to the gaps.] Justify can create awkward spacing. The fix: use left align for most documents."

What Learners Should Observe: The instructor identifying and correcting common formatting mistakes.

Expected Result: Learners can identify common formatting mistakes and explain how to fix them.

H. GUIDED PRACTICAL EXERCISE (20 minutes)

Exercise: "Format My Document"

Practical Objective: The learner will format a plain-text document using fonts, sizes, bold, italic, underline, colour, alignment, spacing, bullets and Format Painter.

Starting Materials:

- A plain-text document open on each machine (provided by instructor — e.g., a sample CV)

Instructions:

Part 1: Format the Heading (5 minutes)

1. "Select the name at the top of the document."
2. "Change the font to Calibri, size 18, bold."
3. "Centre align the name."
4. "Change the font colour to dark blue."

Instructor Prompts:

- "Why did we centre align the name?" (Answer: It is a heading. Centre alignment makes it stand out.)
- "Why did we use dark blue instead of bright blue?" (Answer: Dark colours are more professional and easier to read.)

Part 2: Format the Contact Information (3 minutes)

5. "Select the contact information below the name."
6. "Change the font to Calibri, size 10."
7. "Centre align the contact information."

Part 3: Format the Section Headings (5 minutes)

8. "Select the first section heading (e.g., 'Education')."
9. "Change the font to Calibri, size 14, bold, dark blue."
10. "Left align the heading."
11. "Use the Format Painter to copy this formatting to all other section headings."

Instructor Prompts:

- "Why did we use the Format Painter?" (Answer: To save time and ensure all headings look the same.)

Part 4: Format the Body Text (5 minutes)

12. "Select all the body text under the first heading."
13. "Change the font to Calibri, size 11."
14. "Set line spacing to 1.5."
15. "Repeat for all body text sections."

Part 5: Create a Bulleted List (2 minutes)

16. "Find the list of skills or qualifications."
17. "Select the list items."
18. "Click the Bullets button on the Home tab."

Instructor Prompts:

- "Why do we use bullets instead of dashes?" (Answer: Bullets are the standard format for lists in professional documents.)

Expected Result: Every learner has formatted a plain-text document into a professional-looking document.

Troubleshooting Guidance:

- If the learner cannot find the Font group: "It is on the Home tab, in the middle of the Ribbon."
- If the learner cannot change the font size: "Click the font size dropdown and choose a number. Or type a number directly into the box."
- If the Format Painter does not work: "Make sure you select the text with the formatting first, then click Format Painter, then select the text you want to format."

Completion Checklist:

- ☐ Heading formatted (Calibri, 18pt, bold, dark blue, centred)
- ☐ Contact information formatted (Calibri, 10pt, centred)
- ☐ Section headings formatted (Calibri, 14pt, bold, dark blue, left-aligned)
- ☐ Body text formatted (Calibri, 11pt, 1.5 line spacing)
- ☐ Bulleted list created
- ☐ Format Painter used for consistency

I. INDEPENDENT CLASS EXERCISE (10 minutes)

Exercise: "Fix the Messy Document"

Scenario: "You receive a document from a colleague. It has many formatting mistakes. Your task is to fix the mistakes and make the document professional."

Task: Open the "Messy Document" file provided by the instructor. Fix the following issues:

1. The document uses four different fonts. Change all text to Calibri.
2. The headings are in bright red. Change them to dark blue.
3. The body text has no line spacing. Set line spacing to 1.5.
4. The list uses dashes. Change it to a bulleted list.
5. The title is left-aligned. Centre align it.
6. The contact information is in size 16. Change it to size 10.
7. Save the document as "Fixed Document" on the Desktop.

Requirements:

- Fix all 7 issues independently
- Do not change the content of the document — only the formatting
- Save the document with the correct name

Time Allowance: 10 minutes

Success Criteria:

- All 7 issues fixed correctly
- The document looks professional and consistent
- The document is saved with the correct name

Hints Available If the Learner Gets Stuck:

- Hint 1: "Select all text (Ctrl+A) and change the font to Calibri. This fixes issue 1."
- Hint 2: "Select the headings and use the Font Colour button to change them to dark blue."
- Hint 3: "Select the list items and click the Bullets button on the Home tab."

Complete Answer Key for the Instructor:

The final document should have:

- All text in Calibri font
- Headings in dark blue, bold, size 14
- Body text in size 11, 1.5 line spacing
- Title centred, size 18, bold
- Contact information centred, size 10
- List formatted with bullets
- Consistent formatting throughout

J. REAL-WORLD CASE STUDY (5 minutes)

Case: "Chinedu's CV Disaster"

Situation: Chinedu is applying for a job. He creates a CV in Word, but he makes many formatting mistakes. He uses five different fonts, bright colours, no spacing, and dashes instead of bullets. He sends the CV to the employer. He does not get an interview.

The Problem: Chinedu's CV looks unprofessional. The employer receives hundreds of applications. A poorly formatted CV is an easy one to reject.

Why It Matters: Your CV is your first impression. A well-formatted CV gets you an interview. A poorly formatted CV gets you rejected. It is that simple.

Relevant Concept: Professional formatting. If Chinedu knew how to use fonts, sizes, alignment, spacing and bullets, his CV would stand out — for the right reasons.

Worked Solution: If Chinedu had taken this course, here is what he would do:

1. Use one font (Calibri) for the entire document.
2. Use size 18 for the name, size 14 for headings, size 11 for body text.
3. Use bold for headings, regular for body text.
4. Use dark blue for headings, black for body text.
5. Use 1.5 line spacing for readability.
6. Use bullets for lists.
7. Centre align the name and contact information.
8. Left align everything else.

Reasoning: Consistent formatting shows that Chinedu is detail-oriented and professional. It makes the CV easy to read and visually appealing. The employer can quickly find the information they need.

Possible Mistake: Chinedu might think that more formatting makes a better CV. He might use bright colours, multiple fonts and lots of bold text. This makes the CV look cluttered and amateur.

Correction: Professional does not mean fancy. Professional means clean, consistent and easy to read. Use formatting sparingly and purposefully.

Transfer Exercise: "Your friend Adaeze has a CV with no formatting at all — just plain text. What are three things you would tell her to do to make it look professional?"

Professional Connection: Every job application requires a CV. Every CV should be professional. The formatting skills you learn in this course will serve you every time you create a document for work, school or personal use.

K. COMMON MISTAKES AND TROUBLESHOOTING (5 minutes)

Mistake 1: Using Too Many Fonts

Symptoms: The learner uses four or five different fonts in one document. **Likely Cause:** The learner thinks more fonts make the document look interesting. **How to Investigate:** Count the number of fonts in the document. **Safe Corrective Steps:** Explain the "less is more" rule. Use a maximum of two fonts — one for headings and one for body text. **How to Confirm the Problem Is Resolved:** The learner uses two or fewer fonts. **How to Prevent Recurrence:** Show examples of professional documents and count the fonts together.

Mistake 2: Using Bright Colours for Body Text

Symptoms: The learner uses bright yellow, pink or lime green for body text. **Likely Cause:** The learner thinks bright colours make the document stand out. **How to Investigate:** Look at the document. Is the body text in a bright colour? **Safe Corrective Steps:** Explain that bright colours are hard to read. Use black or dark grey for body text. Use dark colours (dark blue, dark green, dark red) for headings. **How to Confirm the Problem Is Resolved:** The learner uses appropriate colours. **How to Prevent Recurrence:** Show examples of professional documents and discuss the colour choices.

Mistake 3: Pressing Enter Multiple Times for Spacing

Symptoms: The learner presses Enter three or four times to create space between paragraphs. **Likely Cause:** The learner does not know about paragraph spacing options. **How to Investigate:** Click the Show/Hide button (¶) on the Home tab. Count the paragraph marks between paragraphs. **Safe Corrective Steps:** Teach the learner to use the Line and Paragraph Spacing options instead of pressing Enter multiple times. **How to Confirm the Problem Is Resolved:** The learner uses paragraph spacing options. **How to Prevent Recurrence:** Show the learner the Show/Hide button and explain what the paragraph marks mean.

Mistake 4: Not Using the Format Painter

Symptoms: The learner manually formats each heading, resulting in inconsistent formatting. **Likely Cause:** The learner does not know about the Format Painter. **How to Investigate:** Compare the formatting of different headings. Are they consistent? **Safe Corrective Steps:** Teach the learner to use the Format Painter to copy formatting from one heading to another. **How to Confirm the Problem Is Resolved:** The learner uses the Format Painter. **How to Prevent Recurrence:** Make the Format Painter a standard step in every formatting exercise.

Mistake 5: Over-Emphasising Text

Symptoms: The learner makes everything bold, italic and underlined. **Likely Cause:** The learner thinks more emphasis makes the text more important. **How to Investigate:** Look at the document. Is everything bold, italic and underlined? **Safe Corrective Steps:** Explain that when everything is emphasised, nothing is emphasised. Use bold, italic and underline sparingly. **How to Confirm the Problem Is Resolved:** The learner uses emphasis sparingly. **How to Prevent Recurrence:** Show examples of professional documents and discuss how emphasis is used.

L. SESSION ASSIGNMENT (5 minutes)

Assignment Title: "Format My CV"

Purpose: To reinforce formatting skills by having the learner transform a plain-text CV into a professional document.

Realistic Scenario: "You have a CV, but it is just plain text with no formatting. You need to make it look professional before you send it to potential employers. Use the formatting tools you learned in this session to transform your CV."

Detailed Instructions:

1. Open the "Plain CV" file provided by the instructor.
2. Apply the following formatting:
    - Name: Calibri, 18pt, bold, dark blue, centred
    - Contact information: Calibri, 10pt, centred
    - Section headings (Education, Experience, Skills): Calibri, 14pt, bold, dark blue, left-aligned
    - Body text: Calibri, 11pt, 1.5 line spacing
    - Skills list: Bulleted list
3. Use the Format Painter to ensure all section headings have the same formatting.
4. Save the document as "Formatted CV" on the Desktop.
5. Export the document to PDF.
6. Show both files to the instructor before leaving.

Required Deliverables:

- A Word document named "Formatted CV" saved on the Desktop
- A PDF version of the same document on the Desktop

Submission Format:

- Both files must be visible on the Desktop of the academy machine
- The learner must show the instructor both files before leaving

Deadline Recommendation:

- Before leaving the session (in-class assignment)

Marking Rubric:

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Name formatting	Calibri, 18pt, bold, dark blue, centred	Most elements correct	Few elements correct
Contact info formatting	Calibri, 10pt, centred	Mostly correct	Incorrect

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Section headings	All consistent (Calibri, 14pt, bold, dark blue)	Mostly consistent	Inconsistent
Body text	Calibri, 11pt, 1.5 spacing	Mostly correct	Incorrect
Bulleted list	Properly formatted	Mostly correct	Not formatted
Format Painter used	Used for consistency	Used but not consistently	Not used
File saved correctly	Saved on Desktop with correct name	Saved but wrong location or name	Not saved
PDF exported	PDF exported correctly	PDF exported but wrong location	Not exported

Total: 40 marks

Common Mistakes to Avoid:

- Using more than two fonts
- Using bright colours for body text
- Not using the Format Painter
- Pressing Enter multiple times for spacing
- Not exporting to PDF

Instructor Reference Solution:

The final CV should have:

- Name: Calibri, 18pt, bold, dark blue, centred
- Contact information: Calibri, 10pt, centred
- Section headings: Calibri, 14pt, bold, dark blue, left-aligned
- Body text: Calibri, 11pt, 1.5 line spacing
- Skills list: Bulleted list
- Consistent formatting throughout
- Saved as "Formatted CV.docx" and "Formatted CV.pdf" on the Desktop

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M. KNOWLEDGE CHECK (5 minutes)

Questions

1. Multiple Choice: What is the maximum number of fonts you should use in a professional document? a) One b) Two c) Three d) Five

Answer: b) Two

2. Short Answer: What are the four types of text alignment?

Answer: Left align, centre align, right align, justify.

3. Multiple Choice: What does the Format Painter do? a) Copies text from one place to another b) Copies formatting from one section to another c) Changes the font of the entire document d) Creates a bulleted list

Answer: b) Copies formatting from one section to another

4. Explain in Your Own Words: Why is it important to use consistent formatting in a document?

Answer: Consistent formatting makes a document look professional and organised. It shows that the creator is detail-oriented and takes pride in their work. Inconsistent formatting makes a document look cluttered and amateur.

5. Practical Identification: Open a Word document. Identify the Font group, the Paragraph group, and the Format Painter button on the Home tab.

Answer: The learner correctly identifies all three elements.

6. Scenario-Based: Your friend says, "I used five different fonts in my CV because I wanted it to look interesting." What advice would you give?

Answer: Using too many fonts makes a document look cluttered and unprofessional. Use a maximum of two fonts — one for headings and one for body text. Keep it clean and consistent.

7. Error-Detecting: A learner writes in their assignment: "I pressed Enter four times to create space between paragraphs." What is wrong with this, and what should they have done instead?

Answer: Pressing Enter multiple times creates inconsistent spacing. The learner should have used the Line and Paragraph Spacing options on the Home tab to set a specific amount of space after each paragraph.

N. REFLECTION AND RECAP (5 minutes)

Plain-English Summary

Today you learned how to format documents professionally in Microsoft Word. You learned about fonts, font sizes, bold, italic, underline, text colour, highlighting, alignment, line spacing, paragraph spacing, bullets, numbering and the Format Painter. You also learned the "less is more" rule — use formatting sparingly and purposefully. These are the tools that turn a plain document into a professional one.

Five Important Takeaways

1. Use a maximum of two fonts in a professional document.
2. Use bold, italic and underline sparingly. Emphasise only the most important text.
3. Use left align for body text and centre align for titles and headings.
4. Use 1.5 line spacing for readability.
5. Use the Format Painter to ensure consistency across your document.

Questions That Help You Recognise What You Can Now Do

- Can you change fonts and font sizes?
- Can you apply bold, italic and underline?
- Can you change text alignment?
- Can you adjust line spacing and paragraph spacing?
- Can you create bulleted and numbered lists?
- Can you use the Format Painter?

Connection to the Next Session

In the next session, you will learn how to create professional documents with page setup, headers, footers, page numbers and tables. You will also learn how to print and export to PDF. These are the skills that complete your Word toolkit. You now know how to format text. Next, you will learn to format entire pages.

O. INSTRUCTOR EVALUATION NOTES

Observable Signs of Competence

- The learner changes fonts and font sizes confidently.
- The learner applies bold, italic and underline appropriately.
- The learner sets alignment correctly.
- The learner adjusts line spacing and paragraph spacing.
- The learner creates bulleted and numbered lists.
- The learner uses the Format Painter for consistency.

- The learner follows the "less is more" principle.

Common Misconceptions

- **"More formatting is better."** Some learners think that using lots of fonts, colours and effects makes a document look professional. Correct this by explaining the "less is more" rule.
- **"Format Painter is only for experts."** Some learners think the Format Painter is an advanced tool. Correct this by showing them how simple it is to use.
- **"Justify is always the best alignment."** Some learners think justify is the most professional alignment. Correct this by explaining that justify can create awkward spacing and that left align is usually the best choice for body text.

Suggested Remedial Exercises

- For learners who use too many fonts: Give them a document with five fonts and ask them to reduce it to two.
- For learners who over-emphasise text: Give them a document where everything is bold, italic and underlined. Ask them to remove the unnecessary emphasis.
- For learners who cannot use the Format Painter: Walk them through the process step by step. Let them practise multiple times.

Differentiation Guidance

Level 1 — Supported (for learners who struggle):

- Provide a "Formatting Tools Reference" handout with screenshots.
- Pair with a more confident learner for practical exercises.
- Allow extra time for the independent exercise.
- Focus on fonts, sizes and bold before moving to more advanced tools.

Level 2 — Independent (standard):

- Complete all exercises as written.
- Answer all knowledge check questions.
- Complete the assignment to the standard shown in the rubric.

Level 3 — Challenge (for learners who finish early):

- Ask the learner to create a formatted flyer for a fictional event.
- Ask the learner to research and write down three more formatting tips.
- Ask the learner to explain the difference between serif and sans-serif fonts.

End of Session 2

Next Session: Session 3 — Professional Documents **What to Bring:** Attention to detail, willingness to experiment with page layout **Preparation:** Prepare a business letter template and table examples

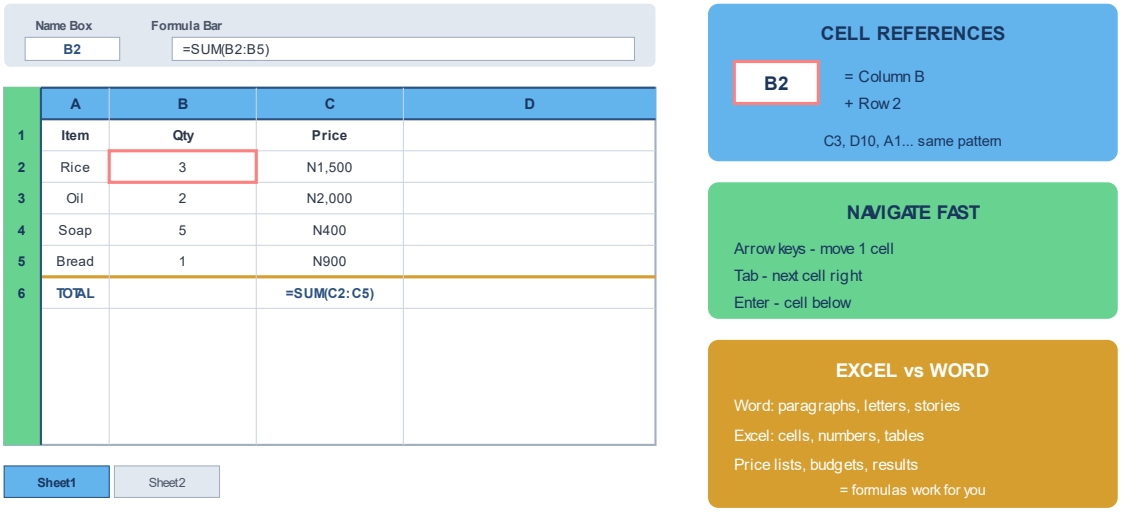
SESSION 03 OF 9

Professional Documents

CONCEPT DIAGRAM — SESSION 03

EXCEL: CELLS, ROWS, COLUMNS

Excel is for numbers and data. Every box is a CELL named by column letter + row number.



BASIC FORMATTING: Bold headers | Borders around data | Fill colour | Number format = Naira (N)

NIGERIAN EXAMPLE

A shopkeeper tracks daily sales in Excel: Item (A), Qty (B), Price (C). Cell B2 shows how many bags of rice were sold on Monday.

SESSION 3: PROFESSIONAL DOCUMENTS

Week: 1 **Module:** Microsoft Word **Duration:** 105 minutes **Difficulty:** Beginner **Prerequisite Knowledge:** Sessions 1–2 — Word fundamentals, formatting **Connection to Previous Session:** Builds on Sessions 1–2 by teaching page layout, headers, footers, tables and printing **Preparation Required Before Class:**

- Prepare a business letter template for practice
- Prepare table examples for demonstration
- Ensure Microsoft Word is installed and functional on all machines

A. SESSION IDENTIFICATION

Field	Detail
Session Number	3
Title	Professional Documents
Week	1

Field	Detail
Module	Microsoft Word
Duration	105 minutes
Difficulty	Beginner
Prerequisite Knowledge	Sessions 1–2 — Word fundamentals, formatting
Connection to Previous Session	Builds on Sessions 1–2 by teaching page layout and advanced features
Preparation	Business letter template, table examples

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Set page size and orientation** (portrait vs landscape) appropriately.
2. **Adjust margins** to control the amount of white space around the text.
3. **Insert and edit headers and footers** with page numbers, dates and document titles.
4. **Insert and format tables** for organised information.
5. **Insert page breaks** to control where new pages begin.
6. **Print a document** with the correct settings.
7. **Export a document to PDF** for professional sharing.
8. **Explain why** page layout matters in professional documents.

C. REQUIRED MATERIALS

Hardware:

- One desktop computer per learner
- One keyboard per learner
- One mouse per learner
- One monitor per learner
- Projector for instructor demonstration

Software:

- Windows 10 or Windows 11
- Microsoft Word 2019 or Microsoft 365

Printed Materials:

- "Page Layout Reference" handout (one per learner)

Sample Files:

- A multi-page document for header/footer practice
- A document with tabular data for table practice

Internet: Not required for this session.

D. OPENING ACTIVITY (10 minutes)

"The Unprofessional Report"

Instructor Setup: Display two versions of the same report on the projector: one with no page layout (default margins, no headers, no page numbers) and one with professional page layout.

Instructor Script: "Good morning, everyone. Look at these two reports. They contain the exact same text and formatting. But one looks like a draft, and the other looks like a finished, professional document. What is the difference? [Pause for answers — margins, headers, footers, page numbers.]

The difference is page layout. Page layout is how you set up the page itself — the margins, the orientation, the headers, the footers and the page numbers. Good page layout makes a document look polished and professional. Bad page layout makes it look unfinished.

Today, you will learn how to set up pages like a professional. You will learn about margins, orientation, headers, footers, page numbers, tables and printing. These are the final skills you need to create complete, professional documents in Word. Let us begin."

Expected Learner Response: Learners will identify visible differences: margins, headers, footers, page numbers. This sets the context for the session.

E. PREREQUISITE REVIEW (5 minutes)

Instructor Script: "Before we start, let us remember what we learned in the last session. Quick review."

Review Questions and Expected Answers:

1. "How do you change the font of selected text?"
    - Select the text, go to the Home tab, and choose a font from the Font group.
2. "What does the Format Painter do?"
    - It copies formatting from one section of text to another.
3. "What are the four types of text alignment?"
    - Left align, centre align, right align, justify.
4. "What is the "less is more" rule?"
    - Use formatting sparingly. Use a maximum of two fonts. Emphasise only the most important text.
5. "How do you create a bulleted list?"
    - Select the list items and click the Bullets button on the Home tab.

How to Address Common Gaps:

- If a learner cannot remember how to change fonts, demonstrate it quickly.
- If a learner forgot the Format Painter, remind them: "It is the paintbrush icon on the Home tab."

F. FULL LECTURE NOTES (35 minutes)

Topic 1: Page Size and Orientation (5 minutes)

Instructor Script: "Page size and orientation determine the shape and size of your document.

Page Size: The default page size in Word is A4 (210 x 297 mm), which is the standard size in Nigeria and most of the world. In the United States, the standard is Letter (8.5 x 11 inches). For most documents, A4 is the correct choice.

To change the page size:

1. Go to the Layout tab on the Ribbon.
2. Click the Size button.
3. Choose A4, Letter, or another size.

Orientation: Orientation determines whether the page is taller than it is wide (portrait) or wider than it is tall (landscape).

- **Portrait:** The default. Used for most documents — letters, reports, CVs, essays.

- **Landscape:** Used for wide documents — tables, charts, certificates, flyers.

To change the orientation:

1. Go to the Layout tab.
2. Click the Orientation button.
3. Choose Portrait or Landscape.

When to Use Each:

- Use portrait for text-heavy documents.
- Use landscape for documents with wide tables or images.
- You can even mix orientations in the same document by using section breaks (advanced topic)."

Topic 2: Margins (5 minutes)

Instructor Script: "Margins are the white space between the text and the edge of the page. They make a document look clean and professional. They also make it easier to read — text that goes all the way to the edge of the page is hard to read.

Default Margins: The default margins in Word are 1 inch (2.54 cm) on all sides. This is a good standard for most documents.

Changing Margins:

1. Go to the Layout tab.
2. Click the Margins button.
3. Choose a preset (Normal, Narrow, Moderate, Wide) or click "Custom Margins" to set your own.

Common Margin Settings:

- **Normal:** 1 inch on all sides. Good for most documents.
- **Narrow:** 0.5 inch on all sides. Good for documents with lots of content.
- **Moderate:** 1 inch top and bottom, 0.75 inch left and right. Good for reports.
- **Wide:** 1 inch top and bottom, 2 inch left and right. Good for documents with comments.

Best Practice: Use the default Normal margins unless you have a specific reason to change them. Consistent margins make a document look professional."

Topic 3: Headers and Footers (10 minutes)

Instructor Script: "Headers and footers are areas at the top and bottom of every page. They are used for information that should appear on every page of a document — the document title, the author's name, the date, the page number, the company logo.

Headers: The header is the area at the top of the page. It is used for:

- Document title
- Company name or logo
- Chapter or section name
- Author's name

Footers: The footer is the area at the bottom of the page. It is used for:

- Page numbers
- Date
- Company contact information
- Confidentiality notices

Inserting a Header or Footer:

1. Go to the Insert tab on the Ribbon.
2. Click the Header button (for headers) or the Footer button (for footers).
3. Choose a style from the gallery (e.g., Blank, Austin, Banded).
4. The header or footer area opens. Type your text.
5. Click "Close Header and Footer" or double-click in the document area to close.

Editing a Header or Footer:

1. Double-click in the header or footer area.
2. Make your changes.
3. Click "Close Header and Footer" or double-click in the document area.

Page Numbers: Page numbers are usually placed in the footer. To insert page numbers:

1. Go to the Insert tab.
2. Click the Page Number button.
3. Choose a position (Top of Page, Bottom of Page, Page Margins, Current Position).
4. Choose a style.
5. The page number appears on every page.

Different First Page: If you do not want a header or footer on the first page (e.g., a title page), you can set the first page to be different:

1. Double-click in the header or footer area.
2. Go to the Header & Footer Tools Design tab.
3. Check the "Different First Page" box.
4. The header and footer will not appear on the first page."

Topic 4: Tables (10 minutes)

Instructor Script: "Tables are used to organise information into rows and columns. They are useful for:

- CVs (education, experience)
- Invoices (item, quantity, price)
- Schedules (date, time, event)
- Comparisons (feature, option A, option B)

Inserting a Table:

1. Go to the Insert tab.
2. Click the Table button.
3. A grid appears. Drag your mouse to select the number of rows and columns you need.
4. Click to insert the table.

Entering Data in a Table:

1. Click in a cell.
2. Type your text.
3. Press Tab to move to the next cell.
4. Press Enter to create a new line within a cell.

Formatting a Table:

1. Click anywhere in the table.
2. The Table Tools tabs appear (Design and Layout).
3. Use the Design tab to apply a table style (e.g., Grid Table, Light Shading).
4. Use the Layout tab to add or delete rows and columns, merge cells, and adjust cell size.

Table Tips:

- Use a table style to make your table look professional.
- Keep your table simple — do not use too many colours or borders.
- Make sure your table fits within the margins of the page."

Topic 5: Page Breaks and Printing (5 minutes)

Instructor Script: "A page break is where one page ends and the next one begins. Word creates page breaks automatically when a page is full. But sometimes you want to control where a page break occurs.

Inserting a Manual Page Break:

1. Click where you want the new page to begin.
2. Go to the Layout tab.
3. Click the Breaks button.
4. Choose "Page Break."
5. A new page begins at the cursor position.

When to Use Manual Page Breaks:

- To start a new chapter on a new page.
- To prevent a heading from being separated from the text that follows it.
- To create a title page that is separate from the rest of the document.

Printing a Document:

1. Click "File."
2. Click "Print."
3. Choose your printer.
4. Choose the number of copies.
5. Choose which pages to print (All, Current Page, or a range).
6. Click "Print."

Print Preview: Before printing, always check the Print Preview. This shows you exactly how the document will look on paper. To see the Print Preview, click "File" and then "Print." The preview appears on the right side of the screen."

G. INSTRUCTOR DEMONSTRATION (15 minutes)

Demonstration 1: Setting Up a Professional Page Layout (5 minutes)

What the Instructor Prepares:

- A multi-page document open on the projector

What the Instructor Says and Does: "I am going to set up a professional page layout for this document. Watch each step."

1. "I go to the Layout tab. I click Margins and choose Normal (1 inch on all sides)."
2. "I click Orientation and choose Portrait."
3. "I click Size and choose A4."
4. "Now I will add a header. I go to the Insert tab. I click Header. I choose the 'Blank' style. I type 'CEA Report — October 2026.' I click 'Close Header and Footer.'"
5. "Now I will add a footer with page numbers. I go to the Insert tab. I click Footer. I choose the 'Blank' style. I go to the Insert tab again. I click Page Number. I choose 'Bottom of Page' and then 'Plain Number 2' (centred). I click 'Close Header and Footer.'"
6. "Now I will insert a page break before the second chapter. I click at the beginning of the chapter heading. I go to the Layout tab. I click Breaks. I choose Page Break. The chapter now starts on a new page."

What Learners Should Observe: The instructor setting margins, orientation, size, header, footer, page numbers and page breaks.

Expected Result: Learners can describe the steps to set up a professional page layout.

Demonstration 2: Creating and Formatting a Table (5 minutes)

What the Instructor Prepares:

- A document with tabular data open on the projector

What the Instructor Says and Does: "Now I will create a table to organise this information."

1. "I go to the Insert tab. I click Table. I drag my mouse to select 3 columns and 4 rows. I click to insert the table."

2. "I click in the first cell and type 'Item.' I press Tab and type 'Quantity.' I press Tab and type 'Price.'"
3. "I fill in the rest of the table with data."
4. "I click anywhere in the table. The Table Tools tabs appear. I go to the Design tab. I choose a table style — 'Grid Table 4' — to make it look professional."
5. "I go to the Layout tab. I click 'Distribute Rows' to make all rows the same height."

What Learners Should Observe: The instructor inserting a table, entering data, applying a table style and formatting the table.

Expected Result: Learners can describe the steps to create and format a table.

Demonstration 3: Printing and Exporting to PDF (5 minutes)

What the Instructor Prepares:

- The completed document from Demonstrations 1 and 2

What the Instructor Says and Does: "Now I will show you how to print and export to PDF."

1. "I click 'File.' I click 'Print.' The Print Preview appears. I can see exactly how the document will look on paper. I check that the margins, header, footer and page numbers are correct."
2. "I click 'Print.' The document prints."
3. "Now I will export to PDF. I click 'File.' I click 'Export.' I click 'Create PDF/XPS Document.' I choose where to save. I type a file name. I click 'Publish.'"
4. "I now have a PDF version of the document. I can send this to anyone and they will see it exactly as I designed it."

What Learners Should Observe: The instructor using Print Preview, printing and exporting to PDF.

Expected Result: Learners can describe the steps to print and export to PDF.

H. GUIDED PRACTICAL EXERCISE (20 minutes)

Exercise: "Create a Professional Business Letter"

Practical Objective: The learner will create a professional business letter with proper page layout, header, footer, page numbers and a table.

Starting Materials:

- A blank Word document on each machine

Instructions:

Part 1: Set Up the Page Layout (5 minutes)

1. "Go to the Layout tab. Set margins to Normal (1 inch on all sides)."
2. "Set orientation to Portrait."
3. "Set size to A4."

Instructor Prompts:

- "Why do we use Normal margins?" (Answer: They are the standard for professional documents.)
- "Why do we use Portrait orientation?" (Answer: Business letters are usually taller than they are wide.)

Part 2: Add a Header and Footer (5 minutes)

4. "Go to the Insert tab. Click Header. Choose 'Blank.' Type your name and the name of your school or organisation. Click 'Close Header and Footer.'"
5. "Go to the Insert tab. Click Footer. Choose 'Blank.' Go to the Insert tab. Click Page Number. Choose 'Bottom of Page' and 'Plain Number 2' (centred). Click 'Close Header and Footer.'"

Instructor Prompts:

- "What is the purpose of a header?" (Answer: To show the document title or author's name on every page.)
- "What is the purpose of a footer?" (Answer: To show page numbers or other information on every page.)

Part 3: Write the Letter (5 minutes)

6. "Type a short business letter. Include:
    - Today's date
    - Recipient's name and address
    - Subject line
    - Greeting (e.g., 'Dear Mr. Okafor,')
    - Body (2–3 paragraphs)
    - Sign-off (e.g., 'Yours sincerely,')
    - Your name"

Part 4: Insert a Table (5 minutes)

7. "At the end of the letter, insert a table with 3 columns and 4 rows."
8. "Fill in the table with sample data (e.g., Item, Quantity, Price)."
9. "Apply a table style from the Design tab."

Instructor Prompts:

- "Why do we use a table instead of typing the data in paragraphs?" (Answer: Tables make information easier to read and compare.)

Expected Result: Every learner has created a professional business letter with page layout, header, footer, page numbers and a table.

Troubleshooting Guidance:

- If the learner cannot find the Layout tab: "It is on the Ribbon, between the Design and References tabs."
- If the learner cannot insert a header: "Go to the Insert tab. Click Header. Choose a style."
- If the learner cannot insert a table: "Go to the Insert tab. Click Table. Drag your mouse to select rows and columns."

Completion Checklist:

- ☐ Page layout set (Normal margins, Portrait, A4)
- ☐ Header inserted with name and organisation
- ☐ Footer inserted with page numbers
- ☐ Business letter written
- ☐ Table inserted and formatted
- ☐ Table style applied

I. INDEPENDENT CLASS EXERCISE (10 minutes)

Exercise: "Create a Professional Report Cover Page"

Scenario: "You are creating a report for your school or workplace. You need a professional cover page with the title, your name, the date and your organisation. You also need to insert a table of contents placeholder and page numbers that start from the second page."

Task: Create a cover page in Word with the following elements:

1. A title (e.g., "Annual Report 2026")
2. A subtitle (e.g., "Prepared for Cyber Elias Academy")
3. Your name
4. The date
5. A table with your organisation's contact information
6. Page numbers that start from the second page (no page number on the cover page)

Requirements:

- Use proper formatting (fonts, sizes, alignment)
- Use a table for the contact information

- Set the page numbers to start from the second page
- Save the document as "Report Cover Page" on the Desktop

Time Allowance: 10 minutes

Success Criteria:

- All 6 elements present and properly formatted
- Page numbers start from the second page
- Document saved with the correct name

Hints Available If the Learner Gets Stuck:

- Hint 1: "To set page numbers to start from the second page, double-click in the footer area, go to the Header & Footer Tools Design tab, and check 'Different First Page.'"
- Hint 2: "Use the Insert tab to add a table for the contact information."
- Hint 3: "Centre align the title and subtitle for a professional look."

Complete Answer Key for the Instructor:

The cover page should have:

- Title: Calibri, 24pt, bold, centred
- Subtitle: Calibri, 16pt, centred
- Name: Calibri, 12pt, centred
- Date: Calibri, 12pt, centred
- Contact information table: 2 columns, 3 rows, with a table style applied
- Page numbers in the footer, starting from the second page (no page number on the cover page)

J. REAL-WORLD CASE STUDY (5 minutes)

Case: "The Unfinished Report"

Situation: Chinedu is a student at a university in Port Harcourt. He writes a 10-page report for his final year project. He spends weeks researching and writing the content. But when he prints the report, it looks terrible. The margins are too narrow, there are no page numbers, the header is missing, and the table of contents is not formatted. His supervisor is not impressed.

The Problem: Chinedu spent weeks on the content but no time on the presentation. The report looks unfinished because it lacks professional page layout.

Why It Matters: In academic and professional settings, presentation matters. A well-written report with poor presentation suggests the author is careless. A well-written report with good presentation suggests the author is thorough and professional.

Relevant Concept: Page layout. If Chinedu knew how to set margins, insert headers and footers, add page numbers and format tables, his report would look polished and professional.

Worked Solution: If Chinedu had taken this course, here is what he would do:

1. Set margins to Normal (1 inch on all sides).
2. Insert a header with the report title and his name.
3. Insert a footer with page numbers.
4. Format the table of contents with proper styles.
5. Insert page breaks before each major section.
6. Print the report and check the Print Preview.

Reasoning: Good page layout is like good packaging. The content is the product, but the packaging is what people see first. A well-packaged product looks more valuable.

Possible Mistake: Chinedu might think that page layout is not important — that the content is all that matters. This is a mistake. In the real world, people judge by appearances. A poorly presented report may not be read at all, no matter how good the content is.

Correction: Always spend time on page layout. It does not take long — maybe 15 minutes for a 10-page report. But it makes a huge difference in how the report is received.

Transfer Exercise: "Your friend Adaeze has a 5-page essay with no formatting. What are three things you would tell her to do to make it look more professional?"

Professional Connection: Every report, proposal, CV and business letter needs good page layout. This is not optional — it is expected. The skills you learn in this session will serve you every time you create a multi-page document.

K. COMMON MISTAKES AND TROUBLESHOOTING (5 minutes)

Mistake 1: Forgetting to Check Print Preview

Symptoms: The learner prints a document without checking the Print Preview. The printed document looks wrong. **Likely Cause:** The learner does not know about Print Preview or does not think to use it. **How to Investigate:** Ask the learner, "Did you check the Print Preview before printing?" **Safe Corrective Steps:** Teach the learner to always check the Print Preview before printing. Show them how to access it (File > Print). **How to Confirm the Problem Is Resolved:** The learner checks Print Preview before printing. **How to Prevent Recurrence:** Make Print Preview a standard step in every printing exercise.

Mistake 2: Not Using Headers and Footers

Symptoms: The learner creates a multi-page document with no header, footer or page numbers. **Likely Cause:** The learner does not know how to insert headers and footers. **How to Investigate:** Ask the learner, "Does your document have a header or footer?" **Safe Corrective Steps:** Teach the learner to insert headers and footers from the Insert tab. **How to Confirm the Problem Is Resolved:** The learner inserts headers and footers. **How to Prevent Recurrence:** Make headers and footers a standard requirement for multi-page documents.

Mistake 3: Using Too Many Page Breaks

Symptoms: The learner inserts a page break before every paragraph. **Likely Cause:** The learner does not understand that Word creates page breaks automatically. **How to Investigate:** Click the Show/Hide button (¶) on the Home tab. Count the manual page breaks. **Safe Corrective Steps:** Explain that Word creates page breaks automatically when a page is full. Manual page breaks should only be used when you want to start a new section on a new page. **How to Confirm the Problem Is Resolved:** The learner uses manual page breaks sparingly. **How to Prevent Recurrence:** Show the learner the Show/Hide button and explain what the page break symbol means.

Mistake 4: Not Applying a Table Style

Symptoms: The learner creates a table but does not apply a table style. The table looks plain and unprofessional. **Likely Cause:** The learner does not know about table styles. **How to Investigate:** Click on the table. Do the Table Tools tabs appear? **Safe Corrective Steps:** Teach the learner to apply a table style from the Design tab. **How to Confirm the Problem Is Resolved:** The learner applies table styles. **How to Prevent Recurrence:** Make table styles a standard step in every table exercise.

Mistake 5: Not Exporting to PDF

Symptoms: The learner saves the Word document but does not export it to PDF. **Likely Cause:** The learner does not know how or does not understand why PDF is important. **How to Investigate:** Ask the learner, "Did you export to PDF?" **Safe Corrective Steps:** Show the learner how to export to PDF. Explain that PDF is the standard format for sharing finished documents. **How to Confirm the Problem Is Resolved:** The learner exports to PDF. **How to Prevent Recurrence:** Make PDF export a standard step in every assignment.

L. SESSION ASSIGNMENT (5 minutes)

Assignment Title: "My Professional Business Letter"

Purpose: To reinforce page layout skills by having the learner create a complete professional business letter.

Realistic Scenario: "You need to write a formal business letter to a company in Port Harcourt. The letter should be professional in every way — content, formatting and page layout. Create this letter in Microsoft Word."

Detailed Instructions:

1. Open Microsoft Word. Create a new blank document.
2. Set up the page layout:
    - Margins: Normal (1 inch on all sides)
    - Orientation: Portrait
    - Size: A4
3. Insert a header with your name and the name of your school or organisation.
4. Insert a footer with page numbers (centred).
5. Write a formal business letter that includes:
    - Today's date
    - Recipient's name and address
    - Subject line
    - Greeting
    - Body (2–3 paragraphs)
    - Sign-off
    - Your name
6. Insert a table at the end of the letter with your contact information (2 columns, 3 rows).
7. Apply a table style.
8. Save the document as "Professional Business Letter" on the Desktop.
9. Export the document to PDF.
10. Show both files to the instructor before leaving.

Required Deliverables:

- A Word document named "Professional Business Letter" saved on the Desktop
- A PDF version of the same document on the Desktop

Submission Format:

- Both files must be visible on the Desktop of the academy machine
- The learner must show the instructor both files before leaving

Deadline Recommendation:

- Before leaving the session (in-class assignment)

Marking Rubric:

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Page layout	Normal margins, Portrait, A4	Mostly correct	Incorrect
Header	Present with name and organisation	Present but incomplete	Missing
Footer	Page numbers centred	Page numbers present but not centred	Missing
Letter content	All parts present and well-written	Most parts present	Several parts missing
Table	Present, formatted, with style	Present but not formatted	Missing
File saved correctly	Saved on Desktop with correct name	Saved but wrong location or name	Not saved
PDF exported	PDF exported correctly	PDF exported but wrong location	Not exported

Total: 35 marks

Common Mistakes to Avoid:

- Forgetting to set margins

- Not inserting a header or footer
- Not applying a table style
- Not exporting to PDF

Instructor Reference Solution:

The final document should have:

- Normal margins, Portrait orientation, A4 size
- Header with name and organisation
- Footer with centred page numbers
- Complete business letter with all parts
- Contact information table with a table style applied
- Saved as "Professional Business Letter.docx" and "Professional Business Letter.pdf" on the Desktop

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M. KNOWLEDGE CHECK (5 minutes)

Questions

1. Multiple Choice: What is the default page size in Microsoft Word? a) Letter b) A4 c) Legal d) A3

Answer: b) A4

2. Short Answer: What are headers and footers used for?

Answer: Headers are used for information at the top of every page (e.g., document title, author's name). Footers are used for information at the bottom of every page (e.g., page numbers, date).

3. Multiple Choice: How do you insert a table in Word? a) Go to the Home tab and click Table b) Go to the Insert tab and click Table c) Go to the Layout tab and click Table d) Go to the Design tab and click Table

Answer: b) Go to the Insert tab and click Table

4. Explain in Your Own Words: Why is it important to check the Print Preview before printing?

Answer: Print Preview shows you exactly how the document will look on paper. It helps you catch formatting errors, check margins, and ensure that headers, footers and page numbers are correct before you waste paper and ink.

5. Practical Identification: Open a Word document. Identify the Layout tab, the Insert tab, and the Header & Footer Tools Design tab.

Answer: The learner correctly identifies all three tabs.

6. Scenario-Based: Your friend says, "I printed my report but the margins are too narrow and there are no page numbers. What did I do wrong?" What advice would you give?

Answer: You should have set the margins to Normal (1 inch on all sides) before printing. You should also have inserted page numbers in the footer. Always check the Print Preview before printing to catch these issues.

7. Error-Detecting: A learner writes in their assignment: "I inserted a page break before every paragraph." What is wrong with this, and what should they have done instead?

Answer: Word creates page breaks automatically when a page is full. Manual page breaks should only be used when you want to start a new section on a new page. Inserting a page break before every paragraph creates unnecessary blank space and makes the document hard to edit.

N. REFLECTION AND RECAP (5 minutes)

Plain-English Summary

Today you learned how to create professional documents in Microsoft Word. You learned about page size, orientation, margins, headers, footers, page numbers, tables, page breaks, printing and PDF export. These are the skills that complete your Word toolkit. You now know how to create documents that look like they were made by a professional.

Five Important Takeaways

1. Use Normal margins (1 inch on all sides) for most documents.
2. Insert headers and footers for a professional appearance.
3. Use tables to organise information into rows and columns.
4. Always check the Print Preview before printing.
5. Export finished documents to PDF for sharing.

Questions That Help You Recognise What You Can Now Do

- Can you set page size, orientation and margins?
- Can you insert and edit headers and footers?
- Can you insert page numbers?
- Can you create and format tables?
- Can you insert page breaks?
- Can you print and export to PDF?

Connection to the Next Session

This is the last session on Microsoft Word. You now have all the skills you need to create professional documents. In the next session, you will start Microsoft Excel — the world's most powerful spreadsheet program. You will learn how to enter data, use formulas and create charts. This is the skill that employers specifically ask for. You are ready.

O. INSTRUCTOR EVALUATION NOTES

Observable Signs of Competence

- The learner sets page size, orientation and margins correctly.
- The learner inserts and edits headers and footers.
- The learner inserts page numbers.
- The learner creates and formats tables.
- The learner inserts page breaks appropriately.
- The learner checks Print Preview before printing.
- The learner exports to PDF.

Common Misconceptions

- **"Page layout is not important."** Some learners think that only the content matters. Correct this by showing them the difference between a well-laid-out document and a poorly laid-out one.
- **"Headers and footers are only for books."** Some learners think headers and footers are only for published books. Correct this by showing them how headers and footers are used in business letters, reports and CVs.
- **"Tables are only for data."** Some learners think tables are only for numerical data. Correct this by showing them how tables can be used for any organised information.

Suggested Remedial Exercises

- For learners who cannot insert headers and footers: Walk them through the process step by step. Let them practise multiple times.
- For learners who cannot create tables: Show them the Insert tab and the Table button. Let them practise creating tables with different numbers of rows and columns.
- For learners who do not check Print Preview: Make Print Preview a mandatory step before every printing exercise.

Differentiation Guidance

Level 1 — Supported (for learners who struggle):

- Provide a "Page Layout Reference" handout with screenshots.

- Pair with a more confident learner for practical exercises.
- Allow extra time for the independent exercise.
- Focus on margins and headers before moving to tables.

Level 2 — Independent (standard):

- Complete all exercises as written.
- Answer all knowledge check questions.
- Complete the assignment to the standard shown in the rubric.

Level 3 — Challenge (for learners who finish early):

- Ask the learner to create a multi-page report with a table of contents.
- Ask the learner to research and write down three more page layout tips.
- Ask the learner to explain the difference between a section break and a page break.

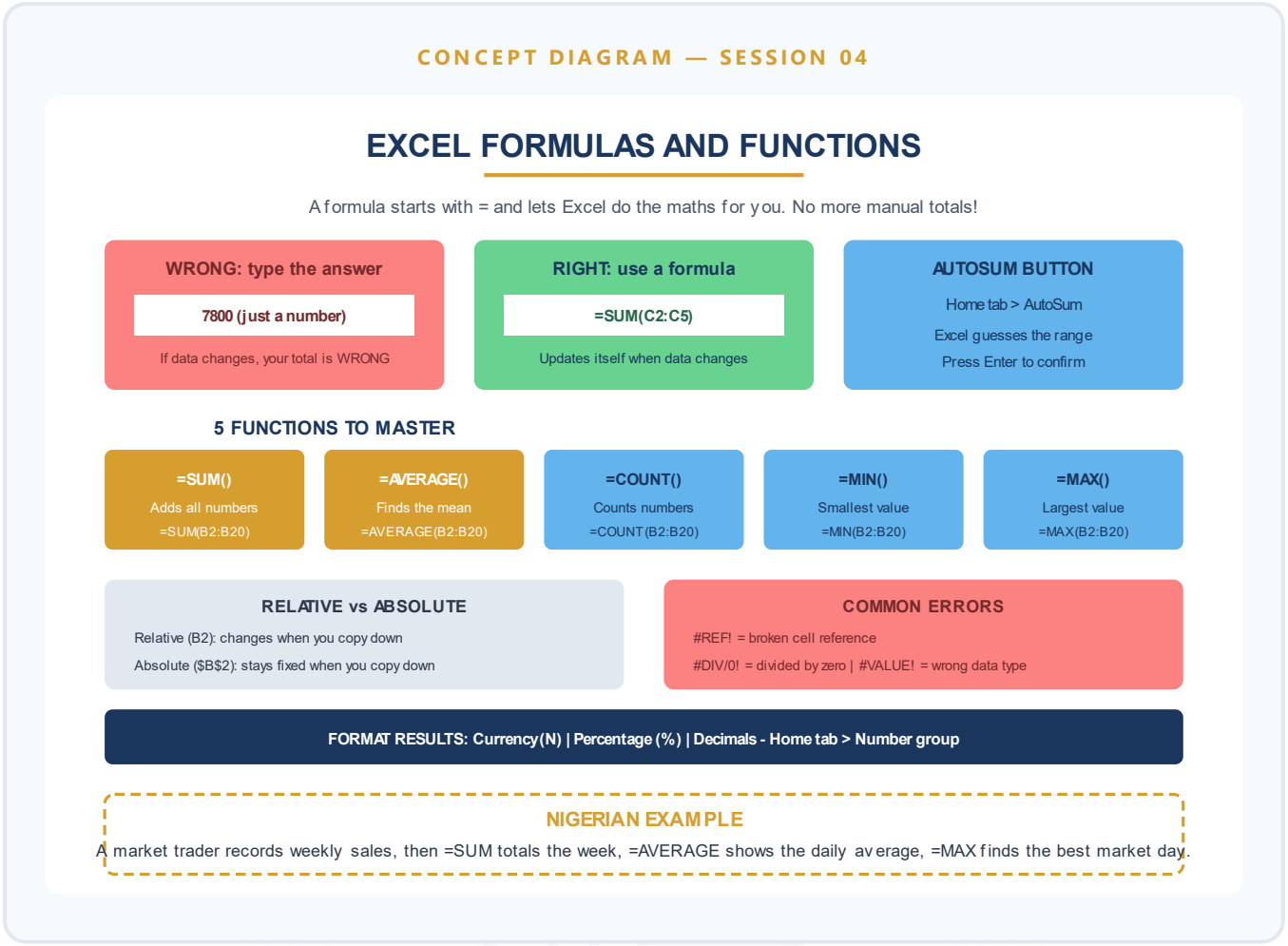
End of Session 3 — End of Week 1 (Microsoft Word)

Next Session: Session 4 — Excel Fundamentals (Week 2) **What to Bring:** Attention to detail, willingness to work with numbers **Preparation:** Prepare a sample dataset for data entry practice



SESSION 04 OF 9

Excel Fundamentals



SESSION 4: EXCEL FUNDAMENTALS

Week: 2 **Module:** Microsoft Excel **Duration:** 105 minutes **Difficulty:** Beginner **Prerequisite Knowledge:** Sessions 1–3 — Word fundamentals, formatting, page layout **Connection to Previous Session:** Transitions from Word to Excel, introducing a new application **Preparation Required Before Class:**

- Ensure Microsoft Excel is installed and functional on all machines
- Prepare a sample dataset for data entry practice (provided in this document)

A. SESSION IDENTIFICATION

Field	Detail
Session Number	4
Title	Excel Fundamentals
Week	2

Field	Detail
Module	Microsoft Excel
Duration	105 minutes
Difficulty	Beginner
Prerequisite Knowledge	Sessions 1—3 — Word skills
Connection to Previous Session	Transitions from Word to Excel
Preparation	Sample dataset prepared

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Explain what Microsoft Excel is** and what it is used for.
2. **Open Microsoft Excel** from the Start menu.
3. **Identify the main parts of the Excel interface:** the Ribbon, worksheet tabs, formula bar, column headers, row headers, and cells.
4. **Navigate between cells** using the mouse and keyboard (arrow keys, Tab, Enter).
5. **Enter and edit data** in cells (text, numbers, dates).
6. **Understand cell references** (e.g., A1, B2, C3).
7. **Save and open workbooks.**
8. **Apply basic cell formatting:** bold, borders, fill colour, number format.
9. **Explain why** Excel is different from Word and when to use each.

C. REQUIRED MATERIALS

Hardware:

- One desktop computer per learner
- One keyboard per learner
- One mouse per learner
- One monitor per learner
- Projector for instructor demonstration

Software:

- Windows 10 or Windows 11
- Microsoft Excel 2019 or Microsoft 365

Printed Materials:

- "Excel Interface Diagram" handout (one per learner)

Sample Files:

- A blank Excel workbook for practice
- A sample dataset (provided below) for data entry practice

Sample Dataset for Practice:

```
Name, Department, Position, Salary (₦)
Adaeze Okafor, Sales, Sales Officer, 85,000
Chinedu Obi, IT, IT Support, 120,000
Funmi Adewale, HR, HR Officer, 95,000
Tunde Bakare, Finance, Accountant, 110,000
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```
Ngozi Eze, Marketing, Marketing Officer, 90,000
Emmanuel Okon, Sales, Sales Manager, 150,000
Amina Bello, IT, Software Developer, 180,000
Kunle Adebayo, Finance, Financial Analyst, 130,000
```

Internet: Not required for this session.

D. OPENING ACTIVITY (10 minutes)

"Word vs Excel — Which Tool for Which Job?"

Instructor Setup: Open Word side by side with Excel on the projector.

Instructor Script: "Good morning, everyone. Welcome to Week 2. Today we start Microsoft Excel. Before we open anything, I want you to think about this: when would you use Word, and when would you use Excel?

Word is for documents — letters, reports, CVs, essays. It is designed for text. Excel is for numbers — budgets, calculations, data analysis, records. It is designed for data.

If I asked you to write a letter to your landlord, which would you use? [Pause.] Word. If I asked you to create a monthly budget for a small business, which would you use? [Pause.] Excel.

Excel is the most powerful tool in the Office suite. It is used by accountants, managers, analysts, business owners and anyone who works with numbers. When a job advert says "proficient in Excel," they mean you should be able to enter data, use formulas and create charts. That is exactly what you will learn this week.

Let us begin."

Expected Learner Response: Learners will correctly identify Word for text and Excel for numbers. This sets the context for the session.

E. PREREQUISITE REVIEW (5 minutes)

Instructor Script: "Before we start, let us remember what we learned in the Word sessions. Quick review."

Review Questions and Expected Answers:

1. "How do you open a program?"
    - Click the Start button, type the program name, and click it.
2. "What is the Ribbon?"
    - The Ribbon is the area below the title bar that contains all the tools, organised into tabs.
3. "How do you save a file?"
    - Press Ctrl+S or click File > Save.
4. "What does the Format Painter do?"
    - It copies formatting from one section to another.
5. "What is the "less is more" rule?"
    - Use formatting sparingly. Keep it clean and professional.

How to Address Common Gaps:

- If a learner is still struggling with basic computer skills, pair them with a more confident learner.
- If a learner cannot open Excel, guide them through the Start menu.

F. FULL LECTURE NOTES (35 minutes)

Topic 1: What Is Microsoft Excel? (5 minutes)

Instructor Script: "Microsoft Excel is a spreadsheet program. A spreadsheet is a grid of rows and columns that organises data into cells. Each cell can contain text, numbers, dates or formulas.

Excel is used for:

- Budgets and financial calculations
- Sales records and inventory tracking
- Student grades and attendance
- Data analysis and reporting
- Charts and graphs
- And much more

Excel vs Word:

- Word is for text. Excel is for numbers.
- Word is linear (top to bottom). Excel is a grid (rows and columns).
- Word is for reading. Excel is for calculating.

Why Excel Matters: Excel is one of the most requested skills in job adverts. Employers want people who can work with numbers, create budgets and analyse data. If you can use Excel, you are more valuable than someone who cannot."

Topic 2: Opening Excel and the Interface (10 minutes)

Instructor Script: "Let us open Excel and explore the interface.

Opening Excel:

1. Click the Start button.
2. Type "Excel" in the search box.
3. Click on "Microsoft Excel."
4. Excel opens. You will see a start screen with templates. Click "Blank workbook" to create a new spreadsheet.

The Excel Interface — Main Parts:

1. The Title Bar (top): Shows the name of the workbook and the program. By default, a new workbook is called "Book1."

2. The Ribbon (below the title bar): Like Word, the Ribbon contains all the tools, organised into tabs:

- **File:** Open, save, print, export
- **Home:** Font, alignment, number format, cells, editing
- **Insert:** Tables, charts, images, shapes
- **Page Layout:** Margins, orientation, size
- **Formulas:** Functions, named ranges, formula auditing
- **Data:** Sort, filter, data tools
- **Review:** Spelling, comments, protect
- **View:** Zoom, layout views, freeze panes

3. The Formula Bar (below the Ribbon): This shows the contents of the selected cell. If the cell contains a formula, the formula appears here. If the cell contains a value, the value appears here.

4. The Column Headers (letters): Columns are labelled with letters: A, B, C, D, and so on. There are 16,384 columns in a modern Excel worksheet.

5. The Row Headers (numbers): Rows are labelled with numbers: 1, 2, 3, 4, and so on. There are 1,048,576 rows in a modern Excel worksheet.

6. Cells: A cell is the intersection of a column and a row. Each cell has a unique address called a cell reference. For example, the cell in column A, row 1 is called A1. The cell in column B, row 3 is called B3.

7. The Worksheet Tabs (bottom): A workbook can contain multiple worksheets. Each worksheet is like a separate page. The tabs at the bottom show the worksheet names (Sheet1, Sheet2, Sheet3). You can add, delete and rename worksheets.

8. The Status Bar (bottom): Shows information about the selected cells, such as the sum, average and count."

Topic 3: Entering and Editing Data (10 minutes)

Instructor Script: "Now let us learn how to enter and edit data in Excel.

Entering Data:

1. Click on a cell. The cell is selected (highlighted with a green border).
2. Type your data (text, number or date).
3. Press Enter to confirm and move to the cell below.
4. Press Tab to confirm and move to the cell to the right.
5. Press Escape to cancel.

Editing Data:

1. Double-click on a cell to edit its contents.
2. Or click on a cell and press F2.
3. Make your changes.
4. Press Enter to confirm.

Deleting Data:

1. Select a cell or range of cells.
2. Press Delete to remove the contents.
3. The cell becomes empty.

Selecting Multiple Cells:

- **Adjacent cells:** Click and drag over the cells.
- **Non-adjacent cells:** Hold Ctrl and click on each cell.
- **Entire column:** Click on the column header (letter).
- **Entire row:** Click on the row header (number).
- **Entire worksheet:** Click the Select All button (the triangle at the top-left corner of the grid).

AutoFill: Excel has a feature called AutoFill that helps you quickly enter data. For example, if you type "January" in a cell and drag the fill handle (the small square at the bottom-right corner of the cell), Excel will automatically fill in February, March, April, and so on."

Topic 4: Cell References (5 minutes)

Instructor Script: "Every cell in Excel has a unique address called a cell reference. The reference is made up of the column letter and the row number.

For example:

- A1 is the cell in column A, row 1.
- B3 is the cell in column B, row 3.
- C10 is the cell in column C, row 10.

Cell references are important because they allow you to refer to specific cells in formulas. For example, if you want to add the values in A1 and B1, you would write a formula like =A1+B1. We will learn about formulas in the next session.

Types of Cell References:

- **Relative reference:** A1 — changes when you copy the formula to another cell.
- **Absolute reference:** A1 — does not change when you copy the formula.
- **Mixed reference:** A$1 or A1 — part changes, part does not.

We will learn more about cell references in the next session."

Topic 5: Saving and Opening Workbooks (5 minutes)

Instructor Script: "Saving and opening Excel workbooks works the same way as Word.

Saving a Workbook:

1. Press Ctrl+S or click File > Save As.
2. Choose where to save (Desktop, Documents, etc.).
3. Type a file name.
4. Click Save.

Opening a Workbook:

1. Click File > Open.
2. Navigate to where you saved the workbook.
3. Click on the workbook name.
4. Click Open.

File Extensions:

- Excel workbooks have the extension .xlsx (modern format).
- Older Excel files have the extension .xls.
- When you export to PDF, the extension is .pdf.

Good File Names:

- Good: "Monthly Budget 2026," "Sales Report October," "Employee Records"
- Bad: "Book1," "Spreadsheet1," "Stuff""

G. INSTRUCTOR DEMONSTRATION (15 minutes)

Demonstration 1: Entering Data in Excel (5 minutes)

What the Instructor Prepares:

- Excel open on the projector with a blank workbook

What the Instructor Says and Does: "I am going to enter some data into Excel. Watch carefully."

1. "I click on cell A1. I type 'Name' and press Enter."
2. "I type 'Department' in B1 and press Enter."
3. "I type 'Position' in C1 and press Enter."
4. "I type 'Salary' in D1 and press Enter."
5. "Now I type the data for each employee, using Tab to move across and Enter to move down."

What Learners Should Observe: The instructor entering data using Tab to move across columns and Enter to move down rows.

Expected Result: Learners can describe how to enter data in Excel.

Demonstration 2: Basic Formatting (5 minutes)

What the Instructor Prepares:

- The data from Demonstration 1

What the Instructor Says and Does: "Now I will format this data to make it look professional."

1. "I click on cell A1. I hold Shift and click on cell D1. This selects the header row."
2. "I press Ctrl+B to make the headers bold."
3. "I go to the Home tab. I click the Fill Color button and choose a dark blue colour for the header row."
4. "I click the Font Color button and choose white for the header text."
5. "I select the Salary column (D2:D9). I go to the Home tab. I click the Number Format dropdown and choose 'Currency' and then '₦ English (Nigeria).'"
6. "I select all the data (A1:D9). I go to the Home tab. I click the Borders button and choose 'All Borders.'"

What Learners Should Observe: The instructor applying bold, fill colour, font colour, currency format and borders.

Expected Result: Learners can describe how to apply basic formatting in Excel.

Demonstration 3: Saving and Opening (5 minutes)

What the Instructor Prepares:

- The formatted data from Demonstration 2

What the Instructor Says and Does: "Now I will save and open this workbook."

1. "I press Ctrl+S. A dialog box appears. I navigate to the Desktop. I type 'Employee Records' in the File name box. I click Save."
2. "I close Excel. I reopen it. I click File > Open. I navigate to the Desktop. I click on 'Employee Records.xlsx.' I click Open. The workbook reopens with all the data and formatting intact."

What Learners Should Observe: The instructor saving and reopening a workbook.

Expected Result: Learners can describe how to save and open an Excel workbook.

H. GUIDED PRACTICAL EXERCISE (20 minutes)

Exercise: "My First Spreadsheet"

Practical Objective: The learner will enter data into Excel, apply basic formatting, and save the workbook.

Starting Materials:

- Excel open on each machine with a blank workbook
- The sample dataset provided in this document

Instructions:

Part 1: Enter the Data (8 minutes)

1. "Click on cell A1. Type 'Name' and press Tab."
2. "Type 'Department' and press Tab."
3. "Type 'Position' and press Tab."
4. "Type 'Salary' and press Enter."
5. "Now enter the data for all 8 employees from the sample dataset. Use Tab to move across and Enter to move down."

Instructor Prompts:

- "What happens when you press Tab?" (Answer: The cursor moves to the cell on the right.)
- "What happens when you press Enter?" (Answer: The cursor moves to the cell below.)

Part 2: Format the Header Row (4 minutes)

6. "Click on cell A1. Hold Shift and click on cell D1. This selects the header row."
7. "Press Ctrl+B to make the headers bold."
8. "Go to the Home tab. Click the Fill Color button and choose a dark blue colour."
9. "Click the Font Color button and choose white."

Instructor Prompts:

- "Why did we select the entire header row?" (Answer: To format all headers at once.)
- "Why did we use white text on a dark blue background?" (Answer: It looks professional and is easy to read.)

Part 3: Format the Salary Column (4 minutes)

10. "Click on cell D2. Hold Shift and click on cell D9. This selects the salary data."
11. "Go to the Home tab. Click the Number Format dropdown. Choose 'Currency' and then '₦ English (Nigeria).'"

Instructor Prompts:

- "Why do we format the salary as currency?" (Answer: To show that the numbers are in naira, not just random digits.)

Part 4: Add Borders and Save (4 minutes)

12. "Select all the data (A1:D9)."
13. "Go to the Home tab. Click the Borders button. Choose 'All Borders.'"
14. "Press Ctrl+S. Navigate to the Desktop. Type 'My First Spreadsheet' in the File name box. Click Save."

Instructor Prompts:

- "Why do we add borders?" (Answer: To make the data easier to read and separate cells visually.)

Expected Result: Every learner has created a formatted spreadsheet with employee data and saved it to the Desktop.

Troubleshooting Guidance:

- If the learner cannot select multiple cells: "Click the first cell, hold Shift, and click the last cell."
- If the learner cannot find the Number Format dropdown: "It is on the Home tab, in the Number group. It usually shows 'General' by default."
- If the learner cannot save: "Press Ctrl+S. A dialog box appears. Navigate to the Desktop. Type a file name. Click Save."

Completion Checklist:

- ☐ Data entered for all 8 employees
- ☐ Header row formatted (bold, dark blue fill, white text)
- ☐ Salary column formatted as currency (₦)
- ☐ Borders added to all data
- ☐ Workbook saved as "My First Spreadsheet" on the Desktop

I. INDEPENDENT CLASS EXERCISE (10 minutes)

Exercise: "Excel Scavenger Hunt"

Scenario: "Your colleague sends you an Excel file and asks you to find and use several tools. Your task is to find each tool in the Excel interface and use it."

Task: Without looking at your notes or your neighbour's screen, complete the following tasks in Excel:

1. Open Excel and create a blank workbook.
2. Type your name in cell A1.
3. Type your school or workplace in cell A2.
4. Type today's date in cell A3.
5. Type "Sales Report" in cell B1.
6. Enter the following numbers in cells B2 through B6: 5000, 7500, 3200, 8900, 6100.
7. Make the text in cell A1 bold.
8. Make the text in cell B1 bold.
9. Add borders to cells A1 through B6.
10. Save the workbook as "Scavenger Hunt" on the Desktop.

Requirements:

- Complete all 10 tasks independently
- Do not ask the instructor for help unless you are truly stuck

Time Allowance: 10 minutes

Success Criteria:

- All 10 tasks completed correctly
- The learner can explain what each tool does

Hints Available If the Learner Gets Stuck:

- Hint 1: "To select multiple cells, click the first cell, hold Shift, and click the last cell."
- Hint 2: "The Borders button is on the Home tab, in the Font group."
- Hint 3: "To save, press Ctrl+S."

Complete Answer Key for the Instructor:

1. The learner opens Excel and creates a blank workbook.
2. The learner clicks on cell A1 and types their name.
3. The learner clicks on cell A2 and types their school or workplace.
4. The learner clicks on cell A3 and types today's date.
5. The learner clicks on cell B1 and types "Sales Report."
6. The learner enters the numbers in cells B2 through B6.
7. The learner selects cell A1 and presses Ctrl+B.
8. The learner selects cell B1 and presses Ctrl+B.
9. The learner selects cells A1 through B6 and clicks the Borders button.
10. The learner saves the workbook as "Scavenger Hunt" on the Desktop.

J. REAL-WORLD CASE STUDY (5 minutes)

Case: "Adaeze's Budget"

Situation: Adaeze runs a small fashion design business in Port Harcourt. She keeps her financial records in a notebook. She writes down her income and expenses by hand. At the end of each month, she tries to add up the numbers, but she often makes mistakes. She loses track of how much money she is making or losing.

One day, a bank manager tells her she needs a proper financial statement to apply for a loan. Adaeze does not know how to create one. She hires someone to do it for her. It costs her ₦5,000.

The Problem: Adaeze does not know how to use Excel. She wastes time and money keeping records by hand.

Why It Matters: Excel is the standard tool for financial records. If Adaeze knew how to use Excel, she could create her own budgets, track her income and expenses, and generate professional financial statements. She would save time and money.

Relevant Concept: Excel basics. If Adaeze knew how to enter data, format cells and save workbooks, she could manage her business finances herself.

Worked Solution: If Adaeze had taken this course, here is what she would do:

1. Open Excel and create a new workbook.
2. Type "Income" in cell A1 and "Expenses" in cell B1.
3. Enter her monthly income and expenses in the cells below.
4. Format the headers as bold with a fill colour.
5. Format the numbers as currency (₦).
6. Save the workbook as "Adaeze's Fashion Budget."
7. Update it every month.
8. Create a professional financial statement for the bank.

Reasoning: Excel makes financial management easy and accurate. You do not have to do mental arithmetic. The formulas do the math for you (we will learn about formulas in the next session).

Possible Mistake: Adaeze might think Excel is too complicated for a small business. She might think a notebook is good enough. But a notebook cannot calculate totals, create charts or generate professional reports.

Correction: Excel is not just for accountants and big companies. It is for anyone who works with numbers — and that includes small business owners.

Transfer Exercise: "Your friend Chinedu sells phone accessories in a shop. He keeps his sales records in a notebook. What advice would you give him?"

Professional Connection: Every business, no matter how small, needs to track income and expenses. Excel is the most common tool for this. Knowing how to use Excel makes you more employable and more capable as a business owner.

K. COMMON MISTAKES AND TROUBLESHOOTING (5 minutes)

Mistake 1: Typing Data in the Wrong Cell

Symptoms: The learner types data but it appears in a different cell than expected. **Likely Cause:** The learner did not click on the correct cell before typing. **How to Investigate:** Ask the learner, "Which cell did you click on before typing?" **Safe Corrective Steps:** Teach the learner to always click on the cell first, then type. Show them how the selected cell is highlighted with a green border. **How to Confirm the Problem Is Resolved:** The learner clicks on the correct cell before typing. **How to Prevent Recurrence:** Remind the learner: "Click first, then type."

Mistake 2: Not Using Tab and Enter to Navigate

Symptoms: The learner uses the mouse to click on every cell before typing. **Likely Cause:** The learner does not know about Tab and Enter shortcuts. **How to Investigate:** Watch the learner's hands. Are they using the mouse to navigate? **Safe Corrective Steps:** Teach the learner to use Tab to move right and Enter to move down. This is much faster than clicking. **How to Confirm the Problem Is Resolved:** The learner uses Tab and Enter to navigate. **How to Prevent Recurrence:** Make Tab and Enter a standard part of every data entry exercise.

Mistake 3: Not Formatting Numbers as Currency

Symptoms: The learner enters salary data as plain numbers (e.g., 85000) instead of currency (e.g., ₦85,000). **Likely Cause:** The learner does not know how to format numbers as currency. **How to Investigate:** Look at the salary column. Are the numbers formatted as currency? **Safe Corrective Steps:** Teach the learner to select the numbers and apply currency format from the Home tab. **How to Confirm the Problem Is Resolved:** The learner formats numbers as currency. **How to Prevent Recurrence:** Make currency formatting a standard step in every financial data exercise.

Mistake 4: Not Saving the Workbook

Symptoms: The learner enters data but does not save. The computer crashes and the data is lost. **Likely Cause:** The learner forgets to save. **How to Investigate:** Ask the learner, "Did you save your workbook?" **Safe Corrective Steps:** Teach the learner to press Ctrl+S regularly. Make it a habit. **How to Confirm the Problem Is Resolved:** The learner saves regularly. **How to Prevent Recurrence:** Set a timer that goes off every 5 minutes. When it rings, everyone presses Ctrl+S.

Mistake 5: Confusing Excel with Word

Symptoms: The learner tries to format Excel like a Word document (e.g., centre-aligning all data, using paragraph spacing). **Likely Cause:** The learner is applying Word habits to Excel. **How to Investigate:** Ask the learner, "Are you trying to make Excel look like Word?" **Safe Corrective Steps:** Explain the differences between Word and Excel. Word is for text. Excel is for numbers and data. **How to Confirm the Problem Is Resolved:** The learner understands the differences. **How to Prevent Recurrence:** Remind the learner: "Excel is a grid, not a page."

L. SESSION ASSIGNMENT (5 minutes)

Assignment Title: "My Sales Record"

Purpose: To reinforce Excel fundamentals by having the learner create a simple sales record.

Realistic Scenario: "You sell phone accessories in a shop in Port Harcourt. You want to keep a record of your daily sales for one week. Create this record in Microsoft Excel."

Detailed Instructions:

1. Open Microsoft Excel. Create a new blank workbook.
2. Create a table with the following columns: Date, Product, Quantity, Price (₦), Total (₦)
3. Enter sales data for 5 days (you can make up the data).
4. Format the header row as bold with a fill colour and white text.
5. Format the Price and Total columns as currency (₦).
6. Add borders to all the data.

7. Save the workbook as "My Sales Record" on the Desktop.
8. Show the saved file to the instructor before leaving.

Required Deliverables:

- An Excel workbook named "My Sales Record" saved on the Desktop
- The workbook contains: 5 columns, 5 rows of data, formatted headers, currency format, borders

Submission Format:

- The file must be saved on the Desktop of the academy machine
- The learner must show the instructor the saved file before leaving

Deadline Recommendation:

- Before leaving the session (in-class assignment)

Marking Rubric:

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Table structure	5 correct columns	4 correct columns	3 or fewer correct columns
Data entry	5 rows of data, all accurate	4 rows, mostly accurate	3 or fewer rows, inaccurate
Header formatting	Bold, fill colour, white text	Mostly formatted	Not formatted
Currency format	Price and Total in ₦	One column in ₦	Not formatted
Borders	All data has borders	Most data has borders	No borders
File saved correctly	Saved on Desktop with correct name	Saved but wrong location or name	Not saved

Total: 30 marks

Common Mistakes to Avoid:

- Forgetting to format the header row
- Not formatting numbers as currency
- Not adding borders
- Not saving the file

Instructor Reference Solution:

The final workbook should have:

- Columns: Date, Product, Quantity, Price (₦), Total (₦)
- 5 rows of data (one for each day)
- Header row: bold, dark blue fill, white text
- Price and Total columns: formatted as ₦ currency
- All data: bordered
- Saved as "My Sales Record.xlsx" on the Desktop

M. KNOWLEDGE CHECK (5 minutes)

Questions

1. Multiple Choice: What is Microsoft Excel? a) A word processing program b) A spreadsheet program c) A presentation program d) A web browser

Answer: b) A spreadsheet program

2. Short Answer: What is a cell reference?

Answer: A cell reference is the unique address of a cell, made up of the column letter and the row number (e.g., A1, B2, C3).

3. Multiple Choice: What is the difference between a workbook and a worksheet? a) A workbook is a single page, and a worksheet is a collection of pages b) A workbook is a collection of worksheets, and a worksheet is a single page c) A workbook and a worksheet are the same thing d) A worksheet is a type of workbook

Answer: b) A workbook is a collection of worksheets, and a worksheet is a single page

4. Explain in Your Own Words: Why is it important to format numbers as currency in a financial spreadsheet?

Answer: Formatting numbers as currency shows that the numbers represent money, not just random digits. It makes the data easier to read and understand. It also prevents confusion — ₦85,000 is clearly eighty-five thousand naira, while 85000 could be anything.

5. Practical Identification: Open Excel. Identify the Ribbon, the formula bar, the column headers, the row headers, and the worksheet tabs.

Answer: The learner correctly identifies all five elements.

6. Scenario-Based: Your friend says, "I entered data in Excel but I cannot find it. Where did it go?" What are two things you would tell your friend to check?

Answer: (1) Check if the data was saved. If not, it may have been lost. (2) Check if the data is in a different worksheet. Click on the worksheet tabs at the bottom to see if the data is in Sheet2 or Sheet3.

7. Error-Detecting: A learner writes in their assignment: "I used Word to create my budget because Excel is too complicated." What is wrong with this, and what should they have done instead?

Answer: Word is not designed for calculations. Excel is the correct tool for budgets because it can perform calculations, format numbers as currency and create charts. The learner should have used Excel.

N. REFLECTION AND RECAP (5 minutes)

Plain-English Summary

Today you learned the fundamentals of Microsoft Excel. You learned what Excel is, how to open it, and how to identify the main parts of the interface. You learned how to enter and edit data, navigate between cells, and understand cell references. You learned how to apply basic formatting (bold, fill colour, currency format, borders) and how to save and open workbooks. These are the foundation skills for everything else you will do in Excel.

Five Important Takeaways

1. Microsoft Excel is a spreadsheet program for working with numbers and data.
2. A cell is the intersection of a column and a row. Each cell has a unique reference (e.g., A1, B2).
3. Use Tab to move right and Enter to move down when entering data.
4. Format numbers as currency when working with financial data.
5. Save regularly with Ctrl+S and use descriptive file names.

Questions That Help You Recognise What You Can Now Do

- Can you open Excel and identify the main parts of the interface?
- Can you enter and edit data in cells?
- Can you navigate between cells using Tab and Enter?
- Can you apply basic formatting (bold, fill colour, currency, borders)?
- Can you save and open a workbook?

Connection to the Next Session

In the next session, you will learn about formulas and functions — the most powerful feature of Excel. You will learn how to use SUM, AVERAGE, COUNT and other functions to perform calculations automatically. This is the skill that makes Excel truly powerful. You now have the foundation. Next, you will learn to make Excel do the math for you.

O. INSTRUCTOR EVALUATION NOTES

Observable Signs of Competence

- The learner opens Excel independently.
- The learner identifies the Ribbon, formula bar, column headers, row headers and cells.
- The learner enters and edits data confidently.
- The learner navigates between cells using Tab and Enter.
- The learner applies basic formatting (bold, fill colour, currency, borders).
- The learner saves with a descriptive file name.

Common Misceptions

- **"Excel is just a bigger Word."** Some learners think Excel is a more complex version of Word. Correct this by explaining that Excel is a grid for numbers, while Word is a page for text.
- **"I can just click on every cell with the mouse."** Some learners use the mouse to navigate instead of Tab and Enter. Correct this by teaching the keyboard shortcuts.
- **"Formatting is not important in Excel."** Some learners think that only the data matters. Correct this by showing them how formatting makes data easier to read and more professional.

Suggested Remedial Exercises

- For learners who cannot navigate between cells: Have them practise entering data using only Tab and Enter (no mouse).
- For learners who cannot format cells: Walk them through the formatting process step by step. Let them practise multiple times.
- For learners who confuse Excel with Word: Show them a side-by-side comparison of Word and Excel. Explain the differences.

Differentiation Guidance

Level 1 — Supported (for learners who struggle):

- Provide a labelled Excel interface diagram for reference.
- Pair with a more confident learner for practical exercises.
- Allow extra time for the independent exercise.
- Focus on data entry and navigation before formatting.

Level 2 — Independent (standard):

- Complete all exercises as written.
- Answer all knowledge check questions.
- Complete the assignment to the standard shown in the rubric.

Level 3 — Challenge (for learners who finish early):

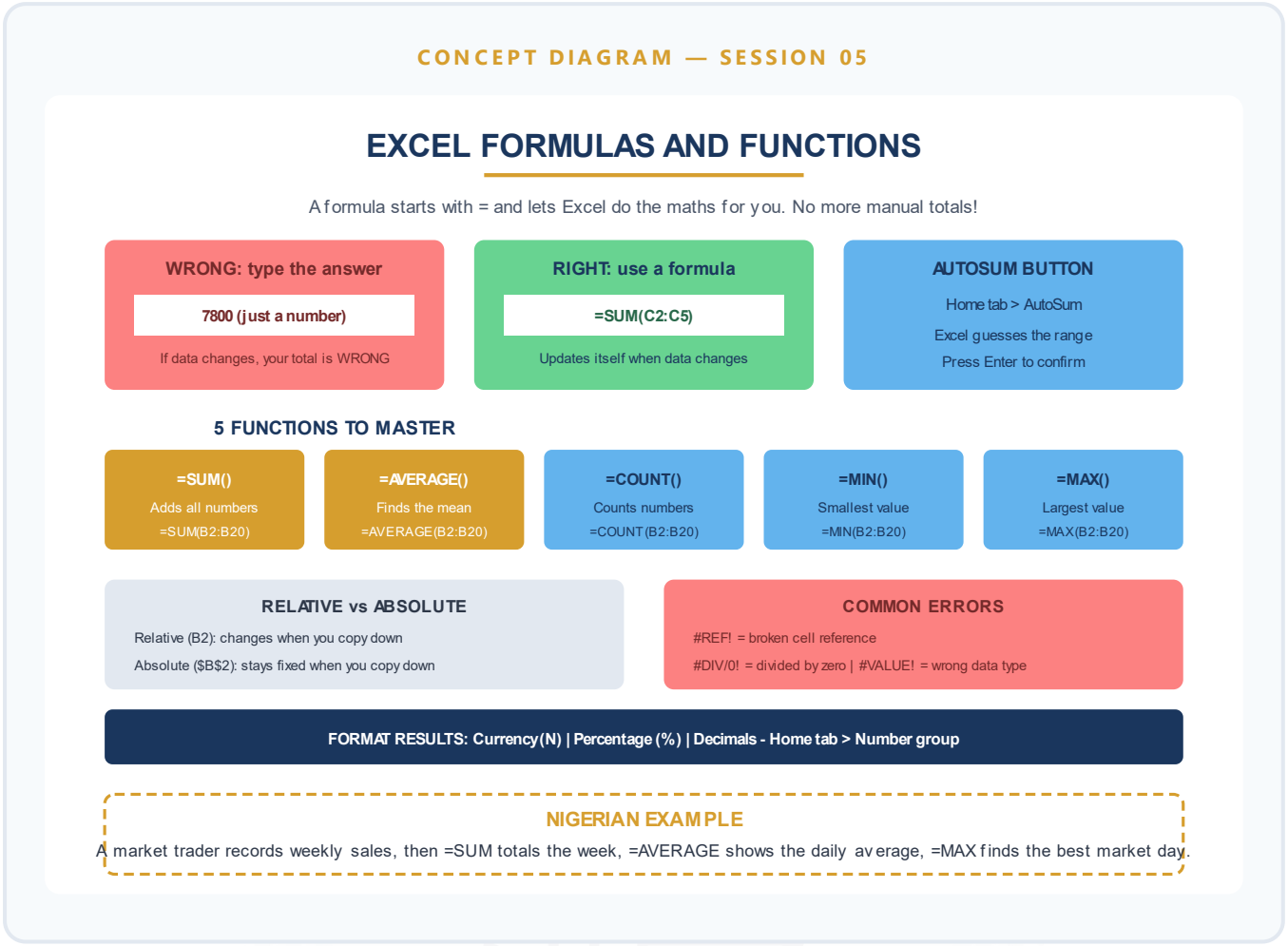
- Ask the learner to create a more complex spreadsheet (e.g., a grade book with student names, scores and averages).
- Ask the learner to research and write down three more Excel features.
- Ask the learner to explain the difference between a workbook and a worksheet to a partner.

End of Session 4

Next Session: Session 5 — Formulas and Functions **What to Bring:** Attention to detail, willingness to work with numbers
Preparation: Prepare formula exercises and sample datasets

SESSION 05 OF 9

Formulas Functions



SESSION 5: FORMULAS AND FUNCTIONS

Week: 2 **Module:** Microsoft Excel **Duration:** 105 minutes **Difficulty:** Beginner **Prerequisite Knowledge:** Session 4 — Excel fundamentals, data entry, cell references **Connection to Previous Session:** Builds on Session 4 by teaching the most powerful feature of Excel — formulas **Preparation Required Before Class:**

- Prepare formula exercises and sample datasets
- Ensure Microsoft Excel is installed and functional on all machines

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A. SESSION IDENTIFICATION

Field	Detail
Session Number	5
Title	Formulas and Functions
Week	2

Field	Detail
Module	Microsoft Excel
Duration	105 minutes
Difficulty	Beginner
Prerequisite Knowledge	Session 4 — Excel fundamentals
Connection to Previous Session	Builds on Session 4 by teaching formulas
Preparation	Formula exercises and sample datasets prepared

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Explain what a formula is** and how it differs from manually typed values.
2. **Create basic formulas** using the equals sign (=) and cell references.
3. **Use the SUM function** to add a range of cells.
4. **Use the AVERAGE function** to calculate the mean of a range of cells.
5. **Use the COUNT, MIN and MAX functions** for basic data analysis.
6. **Use AutoSum** to quickly calculate totals.
7. **Understand relative vs absolute cell references** and when to use each.
8. **Identify and fix common formula errors** (#REF!, #DIV/0!, #VALUE!).
9. **Apply number formatting** (currency, percentage, decimals) to formula results.
10. **Explain why** formulas are more efficient than manual calculations.

C. REQUIRED MATERIALS

Hardware:

- One desktop computer per learner
- One keyboard per learner
- One mouse per learner
- One monitor per learner
- Projector for instructor demonstration

Software:

- Windows 10 or Windows 11
- Microsoft Excel 2019 or Microsoft 365

Printed Materials:

- "Formula Reference" handout (one per learner)

Sample Files:

- A sample dataset with sales data for formula practice

Sample Dataset:

```
Product, Qty Sold, Unit Price (₦), Total (₦)
Rice (50kg), 25, 35000, =B2*C2
Beans (50kg), 40, 18000, =B3*C3
Garri (50kg), 30, 12000, =B4*C4
Cooking Oil (5L), 50, 8500, =B5*C5
Sugar (50kg), 20, 28000, =B6*C6
```

Internet: Not required for this session.

D. OPENING ACTIVITY (10 minutes)

"The Magic of Formulas"

Instructor Setup: Open Excel on the projector with a simple dataset. Type a formula and show how it automatically calculates the result.

Instructor Script: "Good morning, everyone. Today we are going to learn the most powerful feature of Excel — formulas. A formula is a set of instructions that tells Excel to perform a calculation. Instead of typing a number manually, you type a formula and Excel calculates it for you.

Watch this. [Type 10 in cell A1 and 20 in cell B1.] Now, if I want to add these two numbers, I could use a calculator and type the answer. But watch what happens when I use a formula. [Type =A1+B1 in cell C1 and press Enter.] Excel automatically calculates the answer — 30.

Now watch what happens when I change the numbers. [Change A1 to 15.] The answer in C1 automatically updates to 35. I did not have to do anything. Excel recalculated the formula.

This is the magic of formulas. Once you set them up, they update automatically whenever the data changes. This saves you time and prevents errors.

Today, you will learn how to use formulas to perform calculations automatically. By the end of this session, you will be able to use Excel like a professional. Let us begin."

Expected Learner Response: Amazement at the automatic calculation. Curiosity about how formulas work.

E. PREREQUISITE REVIEW (5 minutes)

Instructor Script: "Before we start, let us remember what we learned in the last session. Quick review."

Review Questions and Expected Answers:

1. "How do you open Microsoft Excel?"
   - Click the Start button, type "Excel," and click on Microsoft Excel.
2. "What is a cell reference?"
   - The unique address of a cell, made up of the column letter and the row number (e.g., A1, B2).
3. "How do you navigate between cells?"
   - Use Tab to move right and Enter to move down. Or use the arrow keys.
4. "How do you format numbers as currency?"
   - Select the numbers, go to the Home tab, click the Number Format dropdown, and choose Currency.
5. "What does Ctrl+S do?"
   - It saves the workbook.

How to Address Common Gaps:

- If a learner cannot remember cell references, remind them: "Column letter + row number = cell reference."
- If a learner forgot how to format currency, demonstrate it quickly.

F. FULL LECTURE NOTES (35 minutes)

Topic 1: What Is a Formula? (5 minutes)

Instructor Script: "A formula is a set of instructions that tells Excel to perform a calculation. Every formula starts with an equals sign (=). This tells Excel that what follows is a formula, not just text.

For example:

- =A1+B1 means "add the value in A1 to the value in B1."
- =A1*B1 means "multiply the value in A1 by the value in B1."
- =A1-B1 means "subtract the value in B1 from the value in A1."
- =A1/B1 means "divide the value in A1 by the value in B1."

Why Formulas Matter: If you type a number manually (e.g., 30), it is fixed. If the data changes, you have to recalculate and retype the number. But if you use a formula (e.g., =A1+B1), Excel recalculates automatically whenever the data changes.

This is why formulas are the most important feature of Excel. They save time, prevent errors and make your spreadsheets dynamic."

Topic 2: Basic Calculations (10 minutes)

Instructor Script: "Let us learn the four basic calculations: addition, subtraction, multiplication and division.

Addition (+): =A1+B1 adds the values in A1 and B1.

Subtraction (-): =A1-B1 subtracts the value in B1 from the value in A1.

Multiplication (*): =A1*B1 multiplies the values in A1 and B1. Note: The multiplication sign in Excel is the asterisk (*), not the letter x.

Division (/): =A1/B1 divides the value in A1 by the value in B1. Note: The division sign in Excel is the forward slash (/), not the division symbol.

How to Create a Formula:

1. Click on the cell where you want the result to appear.
2. Type = (equals sign).
3. Click on the first cell you want to include in the formula.
4. Type the operator (+, -, *, /).
5. Click on the second cell.
6. Press Enter.

For example, to add the values in A1 and B1:

1. Click on cell C1.
2. Type =
3. Click on cell A1.
4. Type +
5. Click on cell B1.
6. Press Enter.

The result appears in C1. If you click on C1, you can see the formula =A1+B1 in the formula bar."

Topic 3: SUM, AVERAGE, COUNT, MIN, MAX (10 minutes)

Instructor Script: "Excel has built-in functions for common calculations. A function is a predefined formula that performs a specific task. Let us learn the five most important functions.

SUM: The SUM function adds a range of cells. Syntax: =SUM(range) Example: =SUM(A1:A10) adds all the values from A1 to A10.

AVERAGE: The AVERAGE function calculates the mean of a range of cells. Syntax: =AVERAGE(range) Example: =AVERAGE(A1:A10) calculates the average of all the values from A1 to A10.

COUNT: The COUNT function counts the number of cells that contain numbers. Syntax: =COUNT(range) Example: =COUNT(A1:A10) counts how many cells from A1 to A10 contain numbers.

MIN: The MIN function finds the smallest value in a range. Syntax: =MIN(range) Example: =MIN(A1:A10) finds the smallest value from A1 to A10.

MAX: The MAX function finds the largest value in a range. Syntax: =MAX(range) Example: =MAX(A1:A10) finds the largest value from A1 to A10.

How to Use a Function:

1. Click on the cell where you want the result to appear.
2. Type = (equals sign).
3. Type the function name (e.g., SUM).
4. Type an opening parenthesis.
5. Select the range of cells you want to include.
6. Type a closing parenthesis.
7. Press Enter.

For example, to calculate the sum of A1 to A10:

1. Click on cell B1.
2. Type =SUM(
3. Select cells A1 to A10.
4. Type)
5. Press Enter.

The result appears in B1."

Topic 4: AutoSum (5 minutes)

Instructor Script: "AutoSum is a shortcut that automatically creates a SUM formula for you. It is the fastest way to calculate totals.

How to Use AutoSum:

1. Select the cell below the column of numbers you want to add (or to the right of the row of numbers).
2. Go to the Home tab.
3. Click the AutoSum button (Σ) in the Editing group.
4. Excel automatically creates a SUM formula for the adjacent cells.
5. Press Enter to confirm.

AutoSum Tips:

- AutoSum works best when you have a column or row of numbers with no gaps.
- If AutoSum selects the wrong range, you can adjust the selection before pressing Enter.
- You can also access AutoSum from the Formulas tab."

Topic 5: Common Formula Errors (5 minutes)

Instructor Script: "Sometimes formulas produce errors instead of results. Here are the most common errors and how to fix them.

#DIV/0!: This error occurs when you try to divide by zero. For example, =A1/B1 where B1 is empty or contains zero. Fix: Check that the cell you are dividing by contains a number, not zero or empty.

#REF!: This error occurs when a formula refers to a cell that does not exist. This usually happens when you delete a row or column that a formula refers to. Fix: Check the formula and make sure all cell references are valid.

#VALUE!: This error occurs when a formula contains the wrong type of data. For example, =A1+B1 where A1 contains text instead of a number. Fix: Check that all cells in the formula contain the correct type of data.

#NAME?: This error occurs when Excel does not recognise a function name. This usually happens when you misspell a function. Fix: Check the spelling of the function name.

How to Debug a Formula:

1. Click on the cell with the error.
2. Look at the formula bar. Check the formula for mistakes.

3. Check the cell references. Are they correct?
4. Check the data types. Are they correct?
5. Fix the mistake and press Enter."

G. INSTRUCTOR DEMONSTRATION (15 minutes)

Demonstration 1: Creating Formulas (5 minutes)

What the Instructor Prepares:

- Excel open on the projector with a simple dataset

What the Instructor Says and Does: "I am going to create some formulas. Watch carefully."

1. "I type 10 in cell A1 and 20 in cell B1."
2. "I click on cell C1. I type =A1+B1 and press Enter. The result is 30."
3. "I click on cell C2. I type =A1*B1 and press Enter. The result is 200."
4. "I click on cell C3. I type =A1/B1 and press Enter. The result is 0.5."
5. "I click on cell C4. I type =A1-B1 and press Enter. The result is -10."

What Learners Should Observe: The instructor creating formulas using cell references and operators.

Expected Result: Learners can describe how to create basic formulas.

Demonstration 2: Using SUM and AVERAGE (5 minutes)

What the Instructor Prepares:

- A column of numbers in Excel

What the Instructor Says and Does: "Now I will use the SUM and AVERAGE functions."

1. "I have a column of numbers in A1 to A5: 10, 20, 30, 40, 50."
2. "I click on cell A6. I type =SUM(A1:A5) and press Enter. The result is 150."
3. "I click on cell A7. I type =AVERAGE(A1:A5) and press Enter. The result is 30."
4. "I click on cell A8. I type =COUNT(A1:A5) and press Enter. The result is 5."
5. "I click on cell A9. I type =MIN(A1:A5) and press Enter. The result is 10."
6. "I click on cell A10. I type =MAX(A1:A5) and press Enter. The result is 50."

What Learners Should Observe: The instructor using SUM, AVERAGE, COUNT, MIN and MAX functions.

Expected Result: Learners can describe how to use these functions.

Demonstration 3: AutoSum (5 minutes)

What the Instructor Prepares:

- A column of numbers in Excel

What the Instructor Says and Does: "Now I will use AutoSum to quickly calculate the total."

1. "I have a column of numbers in A1 to A5."
2. "I click on cell A6 (the cell below the numbers)."
3. "I go to the Home tab. I click the AutoSum button (Σ)."
4. "Excel automatically creates the formula =SUM(A1:A5)."
5. "I press Enter. The result is 150."

What Learners Should Observe: The instructor using AutoSum to quickly create a SUM formula.

Expected Result: Learners can describe how to use AutoSum.

H. GUIDED PRACTICAL EXERCISE (20 minutes)

Exercise: "My Sales Calculations"

Practical Objective: The learner will create formulas to calculate totals, averages and other statistics for a sales dataset.

Starting Materials:

- Excel open on each machine with the sample dataset provided in this document

Instructions:

Part 1: Enter the Data (5 minutes)

1. "Open Excel. Create a new blank workbook."
2. "Enter the following data:
    - A1: Product, B1: Qty Sold, C1: Unit Price (₦), D1: Total (₦)
    - A2: Rice (50kg), B2: 25, C2: 35000
    - A3: Beans (50kg), B3: 40, C3: 18000
    - A4: Garri (50kg), B4: 30, C4: 12000
    - A5: Cooking Oil (5L), B5: 50, C5: 8500
    - A6: Sugar (50kg), B6: 20, C6: 28000"

Part 2: Create Formulas for Totals (5 minutes)

3. "Click on cell D2. Type =B2*C2 and press Enter. This calculates the total for Rice."
4. "Click on cell D3. Type =B3*C3 and press Enter. This calculates the total for Beans."
5. "Repeat for D4, D5 and D6."

Instructor Prompts:

- "What does the formula =B2*C2 calculate?" (Answer: The total sales for Rice — quantity multiplied by unit price.)

Part 3: Use SUM and AVERAGE (5 minutes)

6. "Click on cell B7. Type =SUM(B2:B6) and press Enter. This calculates the total quantity sold."
7. "Click on cell D7. Type =SUM(D2:D6) and press Enter. This calculates the total sales."
8. "Click on cell D8. Type =AVERAGE(D2:D6) and press Enter. This calculates the average sales per product."

Instructor Prompts:

- "What does =SUM(B2:B6) calculate?" (Answer: The total quantity of all products sold.)
- "What does =AVERAGE(D2:D6) calculate?" (Answer: The average sales per product.)

Part 4: Use COUNT, MIN, MAX and Format (5 minutes)

9. "Click on cell B9. Type =COUNT(B2:B6) and press Enter. This counts the number of products."
10. "Click on cell D10. Type =MIN(D2:D6) and press Enter. This finds the lowest sales."
11. "Click on cell D11. Type =MAX(D2:D6) and press Enter. This finds the highest sales."
12. "Select cells C2:D11. Format as currency (₦)."

Instructor Prompts:

- "What does =COUNT(B2:B6) calculate?" (Answer: The number of products — 5.)
- "What does =MIN(D2:D6) calculate?" (Answer: The lowest sales figure.)
- "What does =MAX(D2:D6) calculate?" (Answer: The highest sales figure.)

Expected Result: Every learner has created a sales spreadsheet with formulas for totals, averages, counts, minimum and maximum.

Troubleshooting Guidance:

- If the formula does not work: "Check that you typed the equals sign (=) first. Check that the cell references are correct."
- If the result is an error: "Check the error message. #DIV/0! means you are dividing by zero. #VALUE! means you are using the wrong data type."
- If the currency format does not apply: "Select the cells first, then apply the format."

Completion Checklist:

- [ ] Data entered for all 5 products
- [ ] Formulas created for totals (D2:D6)
- [ ] SUM used for total quantity and total sales
- [ ] AVERAGE used for average sales
- [ ] COUNT, MIN, MAX used
- [ ] Currency format applied

I. INDEPENDENT CLASS EXERCISE (10 minutes)

Exercise: "My Grade Book"

Scenario: "You are a teacher. You have the scores of 5 students in 3 subjects. You need to calculate the total score, average score and grade for each student."

Task: Create a grade book in Excel with the following:

1. Columns: Student Name, English, Maths, Science, Total, Average, Grade
2. Enter data for 5 students (you can make up the scores)
3. Use formulas to calculate the Total and Average for each student
4. Use the IF function to assign a grade (A for 70+, B for 60+, C for 50+, D for 40+, F for below 40)
5. Format the header row as bold with a fill colour
6. Save the workbook as "My Grade Book" on the Desktop

Requirements:

- Use formulas for Total and Average (not manual calculations)
- Use the IF function for grades
- Format the header row
- Save with the correct name

Time Allowance: 10 minutes

Success Criteria:

- All formulas work correctly
- Grades are assigned correctly
- Header row is formatted
- File is saved with the correct name

Hints Available If the Learner Gets Stuck:

- Hint 1: "The IF function syntax is =IF(condition, value_if_true, value_if_false)."
- Hint 2: "For the grade, you can use a nested IF: =IF(D2>=70,"A",IF(D2>=60,"B",IF(D2>=50,"C",IF(D2>=40,"D","F"))))"
- Hint 3: "To calculate the Total, use =SUM(B2:D2). To calculate the Average, use =AVERAGE(B2:D2)."

Complete Answer Key for the Instructor:

The grade book should have:

- Columns: Student Name, English, Maths, Science, Total, Average, Grade
- 5 rows of student data
- Total: =SUM(B2:D2) for each student
- Average: =AVERAGE(B2:D2) for each student
- Grade: =IF(E2>=70,"A",IF(E2>=60,"B",IF(E2>=50,"C",IF(E2>=40,"D","F"))))
- Header row: bold, fill colour
- Saved as "My Grade Book.xlsx" on the Desktop

J. REAL-WORLD CASE STUDY (5 minutes)

Case: "Chinedu's Business"

Situation: Chinedu runs a small provision shop in Port Harcourt. He sells rice, beans, garri, cooking oil and sugar. He keeps his sales records in a notebook. At the end of each month, he tries to calculate his total sales, but he often makes mistakes. He loses track of which products are making money and which are not.

One day, a friend shows him how to use Excel. Chinedu creates a spreadsheet with his sales data. He uses formulas to calculate totals, averages and profits. For the first time, he can see exactly how his business is performing.

The Problem: Chinedu was doing manual calculations, which were slow and error-prone. He could not see patterns in his sales data.

Why It Matters: Formulas are the most powerful feature of Excel. They automate calculations, prevent errors and give you insights into your data. If you know how to use formulas, you can make better business decisions.

Relevant Concept: Formulas and functions. If Chinedu had known about formulas from the beginning, he would have saved hours of work and made fewer mistakes.

Worked Solution: If Chinedu had taken this course, here is what he would do:

1. Open Excel and create a new workbook.
2. Enter his product names, quantities and prices.
3. Use formulas to calculate totals (=Qty*Price).
4. Use SUM to calculate total sales.
5. Use AVERAGE to calculate average sales per product.
6. Use MIN and MAX to find the best and worst performing products.
7. Update the spreadsheet every day.
8. Use the data to make better business decisions.

Reasoning: Formulas turn Excel from a simple data entry tool into a powerful analysis tool. They give you insights that you cannot get from manual calculations.

Possible Mistake: Chinedu might type the totals manually instead of using formulas. This means he has to recalculate every time the data changes.

Correction: Always use formulas for calculations. If the data changes, the formula updates automatically.

Transfer Exercise: "Your friend Adaeze sells clothes in a boutique. She keeps her sales records in a notebook. What advice would you give her?"

Professional Connection: Every business that works with numbers needs Excel formulas. Accountants, managers, sales officers, business owners — everyone. Knowing how to use formulas makes you more employable and more capable.

K. COMMON MISTAKES AND TROUBLESHOOTING (5 minutes)

Mistake 1: Forgetting the Equals Sign

Symptoms: The learner types a formula but Excel treats it as text. **Likely Cause:** The learner forgot to type the equals sign (=) at the beginning. **How to Investigate:** Click on the cell. Does the formula bar show the equals sign? **Safe Corrective Steps:** Remind the learner that every formula must start with =. **How to Confirm the Problem Is Resolved:** The learner includes the equals sign. **How to Prevent Recurrence:** Remind the learner: "Every formula starts with =."

Mistake 2: Using the Wrong Operator

Symptoms: The learner uses the letter x for multiplication instead of the asterisk (*). **Likely Cause:** The learner is used to writing x for multiplication on paper. **How to Investigate:** Check the formula. Does it use x or *? **Safe Corrective Steps:** Explain that Excel uses * for multiplication and / for division. **How to Confirm the Problem Is Resolved:** The learner uses the correct operators. **How to Prevent Recurrence:** Write the operators on the whiteboard: +, -, *, /.

Mistake 3: Not Using Cell References

Symptoms: The learner types =10+20 instead of =A1+B1. **Likely Cause:** The learner does not understand the importance of cell references. **How to Investigate:** Check the formula. Does it use cell references or typed values? **Safe Corrective Steps:** Explain that using cell references allows the formula to update automatically when the data changes. **How to Confirm the Problem Is Resolved:** The learner uses cell references. **How to Prevent Recurrence:** Show the learner what happens when they change the data. The formula updates automatically.

Mistake 4: Dividing by Zero

Symptoms: The formula produces a #DIV/0! error. **Likely Cause:** The learner is dividing by a cell that is empty or contains zero. **How to Investigate:** Check the formula. What cell is the divisor? **Safe Corrective Steps:** Check that the divisor cell contains a number, not zero or empty. **How to Confirm the Problem Is Resolved:** The formula works correctly. **How to Prevent Recurrence:** Remind the learner to check for zero and empty cells before dividing.

Mistake 5: Misspelling Function Names

Symptoms: The formula produces a #NAME? error. **Likely Cause:** The learner misspelled the function name (e.g., =SUMM instead of =SUM). **How to Investigate:** Check the formula. Is the function name spelled correctly? **Safe Corrective Steps:** Check the spelling of the function name. **How to Confirm the Problem Is Resolved:** The formula works correctly. **How to Prevent Recurrence:** Remind the learner to double-check function names.

L. SESSION ASSIGNMENT (5 minutes)

Assignment Title: "My Monthly Budget"

Purpose: To reinforce formula skills by having the learner create a monthly budget with calculations.

Realistic Scenario: "You want to create a monthly budget for your personal expenses. You need to track your income, expenses and savings. Create this budget in Microsoft Excel."

Detailed Instructions:

1. Open Microsoft Excel. Create a new blank workbook.
2. Create a table with the following columns: Item, Amount (₦)
3. Enter your income sources (e.g., Salary, Side Business, Gifts)
4. Enter your expenses (e.g., Rent, Food, Transport, Utilities, Entertainment)
5. Use the SUM function to calculate total income and total expenses.
6. Use a formula to calculate savings (Total Income - Total Expenses).
7. Use the AVERAGE function to calculate average monthly expenses.
8. Format the Amount column as currency (₦).
9. Format the header row as bold with a fill colour.
10. Save the workbook as "My Monthly Budget" on the Desktop.
11. Show the saved file to the instructor before leaving.

Required Deliverables:

- An Excel workbook named "My Monthly Budget" saved on the Desktop
- The workbook contains: income sources, expenses, total income, total expenses, savings, average expenses

Submission Format:

- The file must be saved on the Desktop of the academy machine
- The learner must show the instructor the saved file before leaving

Deadline Recommendation:

- Before leaving the session (in-class assignment)

Marking Rubric:

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Income sources	3+ sources listed	2 sources listed	1 or fewer sources
Expenses	5+ expenses listed	3-4 expenses listed	2 or fewer expenses
SUM for total income	Correct formula	Mostly correct	Incorrect
SUM for total expenses	Correct formula	Mostly correct	Incorrect
Savings formula	Correct (Income - Expenses)	Mostly correct	Incorrect
AVERAGE for expenses	Correct formula	Mostly correct	Incorrect
Currency format	All amounts in ₦	Most amounts in ₦	Not formatted
Header formatting	Bold, fill colour	Mostly formatted	Not formatted
File saved correctly	Saved on Desktop with correct name	Saved but wrong location or name	Not saved

Total: 45 marks

Common Mistakes to Avoid:

- Typing totals manually instead of using formulas
- Forgetting the equals sign
- Not formatting numbers as currency
- Not saving the file

Instructor Reference Solution:

The final workbook should have:

- Income sources with amounts
- Expenses with amounts
- Total Income: =SUM(income range)
- Total Expenses: =SUM(expense range)
- Savings: =Total Income - Total Expenses
- Average Expenses: =AVERAGE(expense range)
- All amounts formatted as ₦ currency
- Header row: bold, fill colour
- Saved as "My Monthly Budget.xlsx" on the Desktop

M. KNOWLEDGE CHECK (5 minutes)

Questions

1. Multiple Choice: What must every Excel formula start with? a) A letter b) A number c) An equals sign (=) d) A function name

Answer: c) An equals sign (=)

2. Short Answer: What is the difference between a formula and a function?

Answer: A formula is any calculation that starts with = (e.g., =A1+B1). A function is a predefined formula that performs a specific task (e.g., =SUM(A1:A10)).

3. Multiple Choice: What does the SUM function do? a) Calculates the average of a range b) Adds a range of cells c) Counts the number of cells d) Finds the maximum value

Answer: b) Adds a range of cells

4. Explain in Your Own Words: Why is it better to use formulas instead of typing calculated values manually?

Answer: Formulas update automatically when the data changes. If you type a value manually, you have to recalculate and retype it every time the data changes. Formulas save time and prevent errors.

5. Practical Identification: Open Excel. Create a formula that adds the values in cells A1, A2 and A3.

Answer: The learner creates the formula =SUM(A1:A3) or =A1+A2+A3.

6. Scenario-Based: Your friend says, "I typed =A1+B1 in Excel but it shows the text '=A1+B1' instead of the result. What did I do wrong?" What advice would you give?

Answer: You probably forgot to press Enter after typing the formula. Or, the cell is formatted as text. Check the cell format and make sure it is set to General or Number, not Text.

7. Error-Detecting: A learner writes in their assignment: "I used the formula =A1+B1+C1 to calculate the total, but it shows #VALUE!." What is wrong, and how can they fix it?

Answer: The #VALUE! error means that one of the cells (A1, B1 or C1) contains text instead of a number. The learner should check each cell and make sure they all contain numbers.

N. REFLECTION AND RECAP (5 minutes)

Plain-English Summary

Today you learned about formulas and functions in Microsoft Excel. You learned how to create basic formulas using the equals sign and cell references. You learned how to use SUM, AVERAGE, COUNT, MIN and MAX functions. You learned how to use AutoSum for quick calculations. You learned how to identify and fix common formula errors. These are the skills that make Excel truly powerful.

Five Important Takeaways

1. Every formula starts with an equals sign (=).
2. Use cell references (e.g., A1, B1) instead of typed values in formulas.
3. SUM adds a range, AVERAGE calculates the mean, COUNT counts cells, MIN finds the minimum, MAX finds the maximum.
4. AutoSum (Σ) is the fastest way to calculate totals.
5. Common errors: #DIV/0! (divide by zero), #REF! (invalid reference), #VALUE! (wrong data type), #NAME? (misspelled function).

Questions That Help You Recognise What You Can Now Do

- Can you create a basic formula using cell references?
- Can you use SUM, AVERAGE, COUNT, MIN and MAX?
- Can you use AutoSum?
- Can you identify and fix common formula errors?
- Can you explain why formulas are better than manual calculations?

Connection to the Next Session

In the next session, you will learn how to manage data in Excel — sorting, filtering, conditional formatting and charts. These are the skills that turn raw data into useful information. You now know how to calculate. Next, you will learn to analyse and present.

O. INSTRUCTOR EVALUATION NOTES

Observable Signs of Competence

- The learner creates formulas with the equals sign.
- The learner uses cell references instead of typed values.
- The learner uses SUM, AVERAGE, COUNT, MIN and MAX correctly.
- The learner uses AutoSum.
- The learner identifies and fixes common formula errors.
- The learner applies number formatting to formula results.

Common Misconceptions

- **"Formulas are only for mathematicians."** Some learners think formulas are too advanced. Correct this by showing them how simple formulas are — just =A1+B1.
- **"I can just use a calculator."** Some learners think manual calculation is faster. Correct this by showing them how formulas update automatically when the data changes.
- **"AutoSum is the only way to add numbers."** Some learners think AutoSum is the only option. Correct this by showing them how to create SUM formulas manually.

Suggested Remedial Exercises

- For learners who forget the equals sign: Have them write "Every formula starts with =" on a sticky note and place it on their monitor.
- For learners who use typed values instead of cell references: Show them what happens when they change the data. The formula updates automatically; the typed value does not.
- For learners who cannot use functions: Walk them through the process step by step. Let them practise multiple times.

Differentiation Guidance

Level 1 — Supported (for learners who struggle):

- Provide a "Formula Reference" handout with examples.
- Pair with a more confident learner for practical exercises.
- Allow extra time for the independent exercise.
- Focus on basic formulas (+, -, *, /) before functions.

Level 2 — Independent (standard):

- Complete all exercises as written.
- Answer all knowledge check questions.
- Complete the assignment to the standard shown in the rubric.

Level 3 — Challenge (for learners who finish early):

- Ask the learner to create a more complex budget with multiple income sources and expense categories.
- Ask the learner to research and write down three more Excel functions.
- Ask the learner to explain the difference between relative and absolute cell references.

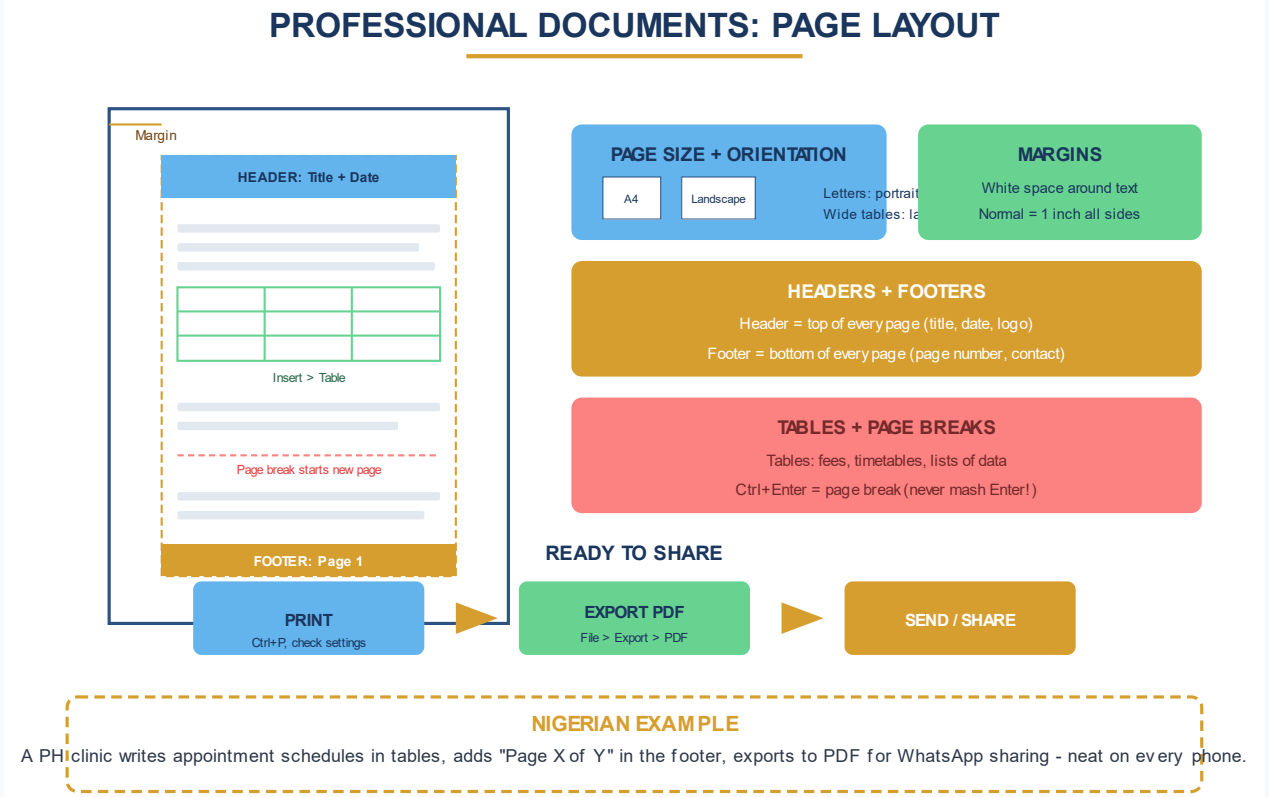
End of Session 5

Next Session: Session 6 — Data Management **What to Bring:** Attention to detail, willingness to explore data tools **Preparation:** Prepare a sales dataset for sorting, filtering and charting

SESSION 06 OF 9

Data Management

CONCEPT DIAGRAM — SESSION 06



SESSION 6: DATA MANAGEMENT

Week: 2 **Module:** Microsoft Excel **Duration:** 105 minutes **Difficulty:** Beginner **Prerequisite Knowledge:** Sessions 4–5 — Excel fundamentals, formulas and functions **Connection to Previous Session:** Builds on Sessions 4–5 by teaching data analysis and visualisation tools **Preparation Required Before Class:**

- Prepare a sales dataset for sorting, filtering and charting
- Ensure Microsoft Excel is installed and functional on all machines

A. SESSION IDENTIFICATION

Field	Detail
Session Number	6
Title	Data Management
Week	2

Field	Detail
Module	Microsoft Excel
Duration	105 minutes
Difficulty	Beginner
Prerequisite Knowledge	Sessions 4–5 — Excel fundamentals, formulas
Connection to Previous Session	Builds on Sessions 4–5 by teaching data analysis tools
Preparation	Sales dataset prepared

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Sort data** alphabetically, numerically and by custom criteria.
2. **Filter data** to show only specific records.
3. **Apply conditional formatting** to highlight important data.
4. **Create simple charts** (bar, pie, line) from data.
5. **Format charts** with titles, labels and colours.
6. **Explain why** data management tools are essential for analysis.
7. **Choose the right chart type** for different data.

C. REQUIRED MATERIALS

Hardware:

- One desktop computer per learner
- One keyboard per learner
- One mouse per learner
- One monitor per learner
- Projector for instructor demonstration

Software:

- Windows 10 or Windows 11
- Microsoft Excel 2019 or Microsoft 365

Printed Materials:

- "Chart Types Reference" handout (one per learner)

Sample Files:

- A sales dataset with 20+ records for sorting, filtering and charting

Sample Dataset:

```
Date, Product, Category, Qty, Unit Price (₦), Total (₦), Salesperson
01/10/2026, Rice (50kg), Food, 25, 35000, 875000, Adaeze
01/10/2026, Beans (50kg), Food, 40, 18000, 720000, Chinedu
02/10/2026, Cooking Oil (5L), Food, 50, 8500, 425000, Funmi
02/10/2026, Sugar (50kg), Food, 20, 28000, 560000, Tunde
03/10/2026, Rice (50kg), Food, 30, 35000, 1050000, Adaeze
03/10/2026, Beans (50kg), Food, 35, 18000, 630000, Chinedu
04/10/2026, Cooking Oil (5L), Food, 45, 8500, 382500, Funmi
04/10/2026, Sugar (50kg), Food, 25, 28000, 700000, Tunde
```

```
05/10/2026, Rice (50kg), Food, 20, 35000, 700000, Adaeze
05/10/2026, Beans (50kg), Food, 30, 18000, 540000, Chinedu
```

Internet: Not required for this session.

D. OPENING ACTIVITY (10 minutes)

"Find the Pattern"

Instructor Setup: Display a large dataset on the projector. Ask learners to find specific information.

Instructor Script: "Good morning, everyone. Look at this dataset. It contains sales records for a small business. I want you to find the answer to this question: Which product generated the most revenue? [Pause.] How long would it take you to find the answer by reading through all the records? [Pause.] A while, right?

Now watch what I do. [Sort the data by Total in descending order.] The answer is right at the top — Rice (50kg) with ₦1,050,000. It took me two seconds.

Excel has powerful tools for managing and analysing data. Today, you will learn how to sort, filter, format and chart data. These are the skills that turn raw data into useful information. Let us begin."

Expected Learner Response: Recognition that manual data analysis is slow and error-prone. Curiosity about Excel's data tools.

E. PREREQUISITE REVIEW (5 minutes)

Instructor Script: "Before we start, let us remember what we learned in the last session. Quick review."

Review Questions and Expected Answers:

1. "What must every formula start with?"
   - An equals sign (=).
2. "What does the SUM function do?"
   - It adds a range of cells.
3. "What is the difference between a formula and a function?"
   - A formula is any calculation that starts with =. A function is a predefined formula (e.g., SUM, AVERAGE).
4. "What does AutoSum do?"
   - It automatically creates a SUM formula for adjacent cells.
5. "What does the #VALUE! error mean?"
   - The formula contains the wrong type of data (e.g., text instead of a number).

How to Address Common Gaps:

- If a learner cannot remember the functions, write them on the whiteboard: SUM, AVERAGE, COUNT, MIN, MAX.
- If a learner forgot the equals sign, remind them: "Every formula starts with =."

F. FULL LECTURE NOTES (35 minutes)

Topic 1: Sorting Data (10 minutes)

Instructor Script: "Sorting means arranging data in a specific order. You can sort alphabetically (A to Z or Z to A), numerically (smallest to largest or largest to smallest), or by date (oldest to newest or newest to oldest).

How to Sort Data:

1. Click anywhere in the column you want to sort by.

2. Go to the Data tab on the Ribbon.
3. Click the Sort A to Z button (for ascending order) or the Sort Z to A button (for descending order).
4. Excel sorts the entire dataset based on the selected column.

Custom Sort: If you want to sort by multiple criteria (e.g., first by Category, then by Total), use the Custom Sort:

1. Go to the Data tab.
2. Click the Sort button.
3. In the Sort dialog box, choose the first column to sort by.
4. Click "Add Level" to add a second criterion.
5. Choose the second column to sort by.
6. Click OK.

Sort Tips:

- Always include the header row in your selection. Excel will ask if your data has headers. Say "Yes."
- Sorting does not change the data — it only changes the order.
- You can undo a sort with Ctrl+Z."

Topic 2: Filtering Data (10 minutes)

Instructor Script: "Filtering means showing only the records that meet specific criteria. For example, you might want to see only the sales for a specific product, or only the sales above a certain amount.

How to Filter Data:

1. Click anywhere in your dataset.
2. Go to the Data tab.
3. Click the Filter button.
4. Dropdown arrows appear in the header row.
5. Click on a dropdown arrow to see the filter options.
6. Check or uncheck the items you want to show or hide.
7. Click OK.

Filter Tips:

- You can filter by multiple columns at the same time.
- To clear a filter, click the Filter button again or click "Clear" in the dropdown.
- Filtering does not delete data — it only hides the records that do not meet the criteria.
- You can use the Search box in the filter dropdown to quickly find specific items."

Topic 3: Conditional Formatting (5 minutes)

Instructor Script: "Conditional formatting automatically applies formatting (e.g., colour, bold, icon) to cells that meet specific criteria. For example, you could highlight all sales above ₦500,000 in green, or all sales below ₦300,000 in red.

How to Apply Conditional Formatting:

1. Select the cells you want to format.
2. Go to the Home tab.
3. Click the Conditional Formatting button.
4. Choose a rule type (e.g., Highlight Cells Rules, Top/Bottom Rules, Data Bars, Color Scales).
5. Set the criteria (e.g., "Greater than 500000").
6. Choose the formatting (e.g., green fill).
7. Click OK.

Common Uses:

- Highlight cells greater than or less than a value.
- Highlight the top 10 or bottom 10 items.

- Use data bars to show relative values visually.
- Use colour scales to show a gradient from low to high."

Topic 4: Creating Charts (10 minutes)

Instructor Script: "Charts are visual representations of data. They make it easy to see patterns, trends and comparisons. Excel can create many types of charts, but we will focus on the three most common: bar charts, pie charts and line charts.

Bar Charts: Bar charts are used to compare values across categories. For example, you could use a bar chart to compare the total sales of different products.

Pie Charts: Pie charts are used to show proportions of a whole. For example, you could use a pie chart to show what percentage of total sales comes from each product.

Line Charts: Line charts are used to show trends over time. For example, you could use a line chart to show how sales have changed over the past 12 months.

How to Create a Chart:

1. Select the data you want to chart (including headers).
2. Go to the Insert tab.
3. Click the type of chart you want (e.g., Clustered Column for a bar chart, Pie, Line).
4. Excel creates the chart.
5. Use the Chart Tools tabs (Design and Format) to customise the chart.

Chart Tips:

- Always include headers in your selection. Excel uses them for chart labels.
- Choose the right chart type for your data.
- Add a chart title and axis labels.
- Keep it simple — do not use too many colours or effects."

G. INSTRUCTOR DEMONSTRATION (15 minutes)

Demonstration 1: Sorting and Filtering (5 minutes)

What the Instructor Prepares:

- The sample dataset open on the projector

What the Instructor Says and Does: "I am going to sort and filter this data."

1. "I click anywhere in the dataset. I go to the Data tab. I click the Sort Z to A button. The data is now sorted by Total in descending order — highest to lowest."
2. "I click the Filter button. Dropdown arrows appear in the header row."
3. "I click the dropdown arrow in the Product column. I uncheck 'Select All.' I check only 'Rice (50kg).' I click OK. Now I only see the Rice sales."
4. "I clear the filter by clicking the Filter button again."

What Learners Should Observe: The instructor sorting data and filtering by product.

Expected Result: Learners can describe how to sort and filter data.

Demonstration 2: Conditional Formatting (5 minutes)

What the Instructor Prepares:

- The sample dataset open on the projector

What the Instructor Says and Does: "Now I will apply conditional formatting."

1. "I select the Total column (F2:F11)."

2. "I go to the Home tab. I click Conditional Formatting. I choose Highlight Cells Rules. I choose 'Greater Than.'"
3. "I type 700000 and choose 'Green Fill with Dark Green Text.' I click OK."
4. "All sales above ₦700,000 are now highlighted in green."

What Learners Should Observe: The instructor applying conditional formatting to highlight high-value sales.

Expected Result: Learners can describe how to apply conditional formatting.

Demonstration 3: Creating a Chart (5 minutes)

What the Instructor Prepares:

- A summary table with product names and total sales

What the Instructor Says and Does: "Now I will create a chart."

1. "I have a summary table with product names in column A and total sales in column B."
2. "I select the data (including headers). I go to the Insert tab. I click the Clustered Column chart button. A bar chart appears."
3. "I click on the chart. The Chart Tools tabs appear. I go to the Design tab. I click 'Add Chart Element.' I choose 'Chart Title.' I type 'Total Sales by Product.'"
4. "I click 'Add Chart Element' again. I choose 'Axis Titles.' I type 'Product' for the horizontal axis and 'Sales (₦)' for the vertical axis."

What Learners Should Observe: The instructor creating a bar chart and adding titles and labels.

Expected Result: Learners can describe how to create and format a chart.

H. GUIDED PRACTICAL EXERCISE (20 minutes)

Exercise: "Analyse My Sales Data"

Practical Objective: The learner will sort, filter, apply conditional formatting and create a chart from a sales dataset.

Starting Materials:

- Excel open on each machine with the sample dataset

Instructions:

Part 1: Sort the Data (5 minutes)

1. "Click anywhere in the dataset. Go to the Data tab. Click the Sort Z to A button. The data is now sorted by Total in descending order."

Instructor Prompts:

- "What is the highest sales figure?" (Answer: ₦1,050,000 for Rice.)
- "What is the lowest sales figure?" (Answer: ₦382,500 for Cooking Oil.)

Part 2: Filter the Data (5 minutes)

2. "Click the Filter button. Dropdown arrows appear in the header row."
3. "Click the dropdown arrow in the Product column. Uncheck 'Select All.' Check only 'Rice (50kg).' Click OK."
4. "How many records are shown?" (Answer: 3 — all Rice sales.)
5. "Clear the filter by clicking the Filter button again."

Instructor Prompts:

- "What happened to the other records?" (Answer: They are hidden, not deleted.)
- "How do you show all records again?" (Answer: Click the Filter button again.)

Part 3: Apply Conditional Formatting (5 minutes)

6. "Select the Total column (F2:F11)."
7. "Go to the Home tab. Click Conditional Formatting. Choose Highlight Cells Rules. Choose 'Greater Than.'"
8. "Type 700000 and choose 'Green Fill with Dark Green Text.' Click OK."

9. "How many cells are highlighted?" (Answer: 5 — all sales above ₦700,000.)

Instructor Prompts:

- "Why did we use conditional formatting?" (Answer: To quickly identify high-value sales.)

Part 4: Create a Chart (5 minutes)

10. "Create a summary table with product names and total sales:
    - A1: Product, B1: Total Sales (₦)
    - A2: Rice, B2: =SUMIF(B2:B11,"Rice (50kg)",F2:F11)
    - A3: Beans, B3: =SUMIF(B2:B11,"Beans (50kg)",F2:F11)
    - A4: Cooking Oil, B4: =SUMIF(B2:B11,"Cooking Oil (5L)",F2:F11)
    - A5: Sugar, B5: =SUMIF(B2:B11,"Sugar (50kg)",F2:F11)"
11. "Select the summary table (A1:B5). Go to the Insert tab. Click the Clustered Column chart button."
12. "Add a chart title: 'Total Sales by Product.'"

Instructor Prompts:

- "What type of chart did we create?" (Answer: A bar chart / clustered column chart.)
- "Why did we use a bar chart?" (Answer: To compare values across categories.)

Expected Result: Every learner has sorted, filtered, applied conditional formatting and created a chart from the sales dataset.

Troubleshooting Guidance:

- If the sort does not work: "Make sure you clicked somewhere in the dataset first. Check that your data has headers."
- If the filter does not work: "Make sure you clicked the Filter button. Check that dropdown arrows appear in the header row."
- If the chart does not work: "Make sure you selected the data including headers. Check that you chose the correct chart type."

Completion Checklist:

- ☐ Data sorted by Total in descending order
- ☐ Data filtered to show only Rice sales
- ☐ Conditional formatting applied to highlight sales above ₦700,000
- ☐ Summary table created with SUMIF formulas
- ☐ Bar chart created with title

I. INDEPENDENT CLASS EXERCISE (10 minutes)

Exercise: "My Sales Report"

Scenario: "You are the sales manager of a small business. You have a dataset of 20 sales transactions. Your task is to analyse the data and create a summary report."

Task: Using the sample dataset, complete the following:

1. Sort the data by Date in ascending order (oldest to newest).
2. Filter the data to show only sales by a specific salesperson (e.g., Adaeze).
3. Apply conditional formatting to highlight all sales above ₦600,000 in yellow.
4. Create a summary table showing the total sales for each salesperson.
5. Create a bar chart from the summary table.
6. Add a chart title and axis labels.
7. Save the workbook as "My Sales Report" on the Desktop.

Requirements:

- Complete all 7 tasks independently
- Use the sample dataset provided by the instructor

Time Allowance: 10 minutes

Success Criteria:

- All 7 tasks completed correctly
- The chart is clear and properly labelled
- The file is saved with the correct name

Hints Available If the Learner Gets Stuck:

- Hint 1: "To sort by date, click the Date column and use the Sort Oldest to Newest button."
- Hint 2: "To filter by salesperson, click the dropdown arrow in the Salesperson column and select the name."
- Hint 3: "To create a summary table, use the SUMIF function: =SUMIF(range, criteria, sum_range)."

Complete Answer Key for the Instructor:

The final workbook should have:

- Data sorted by Date in ascending order
- Data filtered to show only one salesperson
- Conditional formatting highlighting sales above ₦600,000
- Summary table with SUMIF formulas for each salesperson
- Bar chart with title and axis labels
- Saved as "My Sales Report.xlsx" on the Desktop

J. REAL-WORLD CASE STUDY (5 minutes)

Case: "Adaeze's Sales Analysis"

Situation: Adaeze is the sales manager of a provision shop in Port Harcourt. She has a dataset of 100 sales transactions. She needs to analyse the data to find out which products are selling well, which salespeople are performing best, and what the total revenue is.

The Problem: Adaeze has 100 records to analyse. Doing it manually would take hours and be error-prone.

Why It Matters: Data management tools save time and provide insights that manual analysis cannot. If Adaeze can sort, filter and chart her data, she can make better business decisions.

Relevant Concept: Data management. If Adaeze knew how to use sorting, filtering, conditional formatting and charts, she could analyse her data in minutes.

Worked Solution: If Adaeze had taken this course, here is what she would do:

1. Sort the data by Total in descending order to find the highest sales.
2. Filter the data by salesperson to see individual performance.
3. Apply conditional formatting to highlight high-value sales.
4. Create a summary table with total sales by product.
5. Create a bar chart to visualise the data.
6. Use the insights to make better business decisions.

Reasoning: Data management tools turn raw data into actionable insights. They help you see patterns, identify problems and make informed decisions.

Possible Mistake: Adaeze might create a chart without a title or labels. This makes the chart hard to understand.

Correction: Always add a chart title and axis labels. A chart without labels is just a pretty picture — it does not communicate anything.

Transfer Exercise: "Your friend Chinedu has a dataset of student grades. He wants to find out which students are performing well and which need help. What advice would you give him?"

Professional Connection: Every job that involves data requires data management skills. Sales managers, accountants, teachers, administrators — everyone. Knowing how to sort, filter and chart data makes you more employable and more capable.

K. COMMON MISTAKES AND TROUBLESHOOTING (5 minutes)

Mistake 1: Sorting Without Headers

Symptoms: The learner sorts the data but the header row is included in the sort. **Likely Cause:** The learner did not tell Excel that the data has headers. **How to Investigate:** Ask the learner, "Did Excel ask if your data has headers? What did you say?" **Safe Corrective Steps:** When sorting, Excel asks "Does your data have headers?" Make sure to say "Yes." This keeps the header row at the top. **How to Confirm the Problem Is Resolved:** The header row stays at the top after sorting. **How to Prevent Recurrence:** Remind the learner to always say "Yes" when Excel asks about headers.

Mistake 2: Filtering the Wrong Column

Symptoms: The learner filters the data but the wrong records are shown. **Likely Cause:** The learner filtered by the wrong column. **How to Investigate:** Ask the learner, "Which column did you filter by?" **Safe Corrective Steps:** Clear the filter and apply it to the correct column. **How to Confirm the Problem Is Resolved:** The correct records are shown. **How to Prevent Recurrence:** Remind the learner to check which column they are filtering by.

Mistake 3: Creating a Chart Without Headers

Symptoms: The chart has no labels or the labels are wrong. **Likely Cause:** The learner did not include headers in the chart selection. **How to Investigate:** Check the chart. Are there labels? **Safe Corrective Steps:** Select the data including headers, then create the chart. **How to Confirm the Problem Is Resolved:** The chart has correct labels. **How to Prevent Recurrence:** Remind the learner to always include headers in the chart selection.

Mistake 4: Using the Wrong Chart Type

Symptoms: The chart does not effectively communicate the data. **Likely Cause:** The learner chose the wrong chart type. **How to Investigate:** Ask the learner, "What type of chart did you create? Is it the best choice for this data?" **Safe Corrective Steps:** Explain the different chart types and when to use each. Help the learner choose the right one. **How to Confirm the Problem Is Resolved:** The learner chooses the correct chart type. **How to Prevent Recurrence:** Provide a "Chart Types Reference" handout.

Mistake 5: Not Adding Chart Titles and Labels

Symptoms: The chart has no title or axis labels. **Likely Cause:** The learner does not know how to add them. **How to Investigate:** Look at the chart. Is there a title? Are there axis labels? **Safe Corrective Steps:** Teach the learner to add chart titles and axis labels from the Chart Tools Design tab. **How to Confirm the Problem Is Resolved:** The chart has a title and axis labels. **How to Prevent Recurrence:** Make chart titles and labels a standard requirement for every chart exercise.

L. SESSION ASSIGNMENT (5 minutes)

Assignment Title: "My Sales Analysis Report"

Purpose: To reinforce data management skills by having the learner analyse a sales dataset and create a summary report.

Realistic Scenario: "You are the sales manager of a small business. You have a dataset of sales transactions. Your task is to analyse the data and create a summary report for your boss."

Detailed Instructions:

1. Open the "Sales Dataset" file provided by the instructor.
2. Sort the data by Total in descending order.
3. Filter the data to show only sales above ₦500,000.
4. Apply conditional formatting to highlight the top 3 sales in green.
5. Create a summary table showing the total sales for each product.
6. Create a bar chart from the summary table.
7. Add a chart title and axis labels.
8. Save the workbook as "My Sales Analysis Report" on the Desktop.

9. Show the saved file to the instructor before leaving.

Required Deliverables:

- An Excel workbook named "My Sales Analysis Report" saved on the Desktop
- The workbook contains: sorted data, filtered data, conditional formatting, summary table, bar chart

Submission Format:

- The file must be saved on the Desktop of the academy machine
- The learner must show the instructor the saved file before leaving

Deadline Recommendation:

- Before leaving the session (in-class assignment)

Marking Rubric:

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Sorting	Data sorted by Total descending	Mostly correct	Incorrect
Filtering	Only sales above ₦500,000 shown	Mostly correct	Incorrect
Conditional formatting	Top 3 sales highlighted in green	Mostly correct	Incorrect
Summary table	Correct totals for each product	Mostly correct	Incorrect
Bar chart	Clear, with title and labels	Mostly correct	Incorrect
File saved correctly	Saved on Desktop with correct name	Saved but wrong location or name	Not saved

Total: 30 marks

Common Mistakes to Avoid:

- Sorting without headers
- Filtering the wrong column
- Creating a chart without headers
- Not adding chart titles and labels
- Not saving the file

Instructor Reference Solution:

The final workbook should have:

- Data sorted by Total in descending order
- Data filtered to show only sales above ₦500,000
- Conditional formatting highlighting the top 3 sales in green
- Summary table with correct totals for each product
- Bar chart with title and axis labels
- Saved as "My Sales Analysis Report.xlsx" on the Desktop

M. KNOWLEDGE CHECK (5 minutes)

Questions

1. Multiple Choice: What does sorting do? a) Deletes data that is not needed b) Arranges data in a specific order c) Highlights important data d) Creates a visual representation of data

Answer: b) Arranges data in a specific order

2. Short Answer: What is the difference between sorting and filtering?

Answer: Sorting arranges all the data in a specific order. Filtering shows only the records that meet specific criteria and hides the rest.

3. Multiple Choice: What type of chart is best for comparing values across categories? a) Pie chart b) Line chart c) Bar chart d) Scatter plot

Answer: c) Bar chart

4. Explain in Your Own Words: Why is it important to add titles and labels to a chart?

Answer: Titles and labels tell the viewer what the chart is about and what the axes represent. Without them, the chart is just a pretty picture that does not communicate anything.

5. Practical Identification: Open Excel. Identify the Data tab, the Sort button, the Filter button, and the Conditional Formatting button.

Answer: The learner correctly identifies all four elements.

6. Scenario-Based: Your friend says, "I created a chart but it does not have any labels. What did I do wrong?" What advice would you give?

Answer: You probably did not include headers in your chart selection. To fix this, select the data including headers and create a new chart. Then add a chart title and axis labels from the Chart Tools Design tab.

7. Error-Detecting: A learner writes in their assignment: "I sorted my data but the header row is now mixed in with the data." What went wrong, and how can they fix it?

Answer: The learner did not tell Excel that the data has headers. To fix this, undo the sort (Ctrl+Z), then sort again and make sure to say "Yes" when Excel asks if your data has headers.

N. REFLECTION AND RECAP (5 minutes)

Plain-English Summary

Today you learned how to manage and analyse data in Microsoft Excel. You learned how to sort data, filter data, apply conditional formatting and create charts. These are the skills that turn raw data into useful information. You now know how to make Excel do the analysis for you.

Five Important Takeaways

1. Sorting arranges data in a specific order (alphabetical, numerical, date).
2. Filtering shows only the records that meet specific criteria.
3. Conditional formatting automatically highlights important data.
4. Charts make data visual and easy to understand.
5. Always add titles and labels to charts.

Questions That Help You Recognise What You Can Now Do

- Can you sort data alphabetically and numerically?
- Can you filter data to show specific records?
- Can you apply conditional formatting?
- Can you create a bar chart, pie chart and line chart?
- Can you add titles and labels to a chart?

Connection to the Next Session

This is the last session on Microsoft Excel. You now have all the skills you need to create professional spreadsheets. In the next session, you will start Microsoft PowerPoint — the world's most common presentation program. You will learn how to create slides, apply themes and deliver professional presentations. You are ready.

O. INSTRUCTOR EVALUATION NOTES

Observable Signs of Competence

- The learner sorts data correctly.
- The learner filters data to show specific records.
- The learner applies conditional formatting.
- The learner creates charts with titles and labels.
- The learner chooses the right chart type for the data.

Common Misconceptions

- **"Sorting and filtering are the same thing."** Some learners confuse sorting and filtering. Correct this by explaining that sorting arranges all data, while filtering hides some data.
- **"Charts are just for decoration."** Some learners think charts are only for making documents look pretty. Correct this by explaining that charts communicate data visually.
- **"Conditional formatting is too advanced."** Some learners think conditional formatting is an advanced feature. Correct this by showing them how simple it is to apply.

Suggested Remedial Exercises

- For learners who cannot sort: Have them practise sorting a small dataset (5 rows) multiple times.
- For learners who cannot filter: Have them practise filtering by different criteria (e.g., by product, by salesperson, by date).
- For learners who cannot create charts: Walk them through the process step by step. Let them practise creating different chart types.

Differentiation Guidance

Level 1 — Supported (for learners who struggle):

- Provide a "Chart Types Reference" handout.
- Pair with a more confident learner for practical exercises.
- Allow extra time for the independent exercise.
- Focus on sorting and filtering before charts.

Level 2 — Independent (standard):

- Complete all exercises as written.
- Answer all knowledge check questions.
- Complete the assignment to the standard shown in the rubric.

Level 3 — Challenge (for learners who finish early):

- Ask the learner to create a pie chart and a line chart from the same dataset.
- Ask the learner to research and write down three more Excel data management tools.
- Ask the learner to explain when to use each chart type.

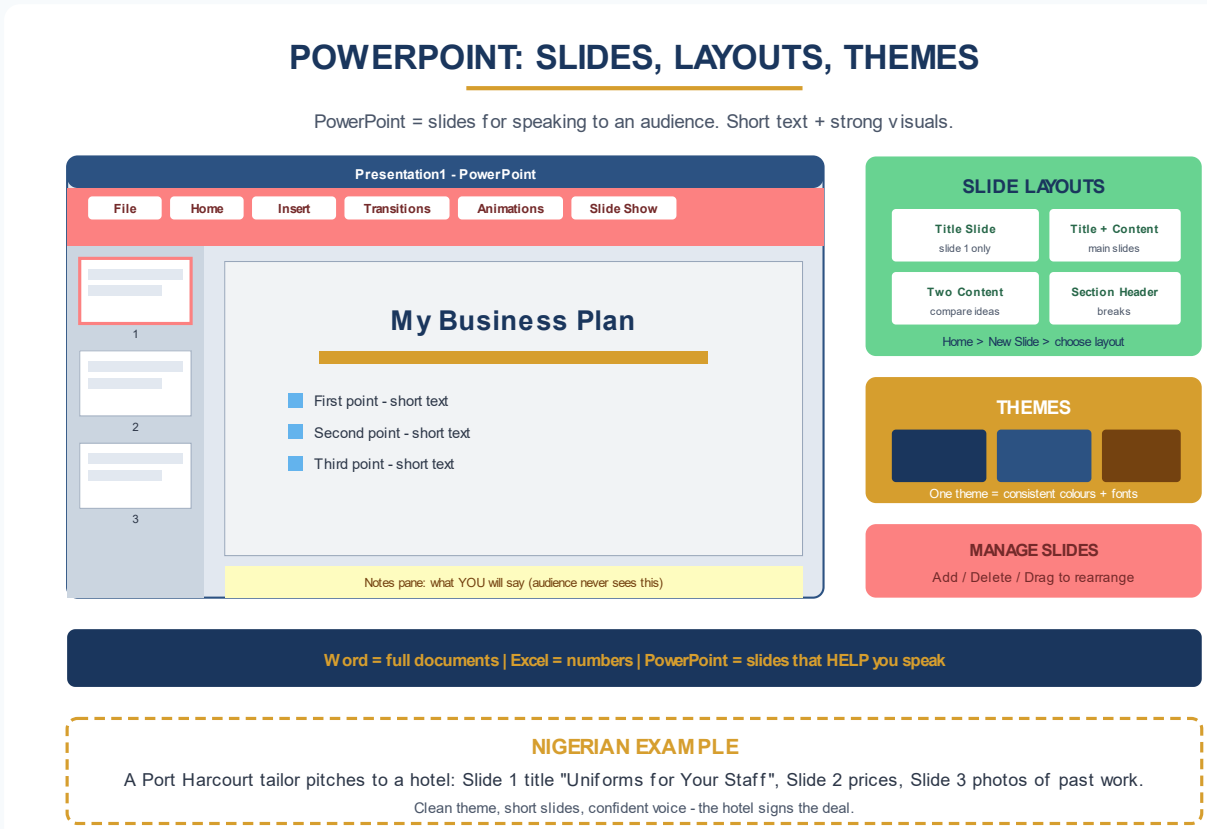
End of Session 6 — End of Week 2 (Microsoft Excel)

Next Session: Session 7 — PowerPoint Fundamentals (Week 3) **What to Bring:** Creativity, willingness to present **Preparation:** Prepare sample images and presentation templates

SESSION 07 OF 9

PowerPoint Fundamentals

CONCEPT DIAGRAM — SESSION 07



SESSION 7: POWERPOINT FUNDAMENTALS

Week: 3 **Module:** Microsoft PowerPoint **Duration:** 105 minutes **Difficulty:** Beginner **Prerequisite Knowledge:** Sessions 1–6 — Word and Excel skills **Connection to Previous Session:** Transitions from Excel to PowerPoint, introducing the third core Office application **Preparation Required Before Class:**

- Ensure Microsoft PowerPoint is installed and functional on all machines
- Prepare sample images for insertion practice
- Have a projector connected for instructor demonstrations

A. SESSION IDENTIFICATION

Field	Detail
Session Number	7
Title	PowerPoint Fundamentals
Week	3

Field	Detail
Module	Microsoft PowerPoint
Duration	105 minutes
Difficulty	Beginner
Prerequisite Knowledge	Sessions 1–6 — Word and Excel skills
Connection to Previous Session	Transitions from Excel to PowerPoint
Preparation	Sample images prepared

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Explain what Microsoft PowerPoint is** and what it is used for.
2. **Open Microsoft PowerPoint** from the Start menu.
3. **Identify the main parts of the PowerPoint interface:** the Ribbon, slides pane, slide area, notes pane, and status bar.
4. **Create a new presentation** from a blank slide or a template.
5. **Add, delete and rearrange slides.**
6. **Apply slide layouts** (Title Slide, Title and Content, Two Content, etc.).
7. **Apply themes and colour schemes** for a professional appearance.
8. **Save and open presentations.**
9. **Explain why** PowerPoint is different from Word and Excel and when to use each.

C. REQUIRED MATERIALS

Hardware:

- One desktop computer per learner
- One keyboard per learner
- One mouse per learner
- One monitor per learner
- Projector for instructor demonstration

Software:

- Windows 10 or Windows 11
- Microsoft PowerPoint 2019 or Microsoft 365

Printed Materials:

- "PowerPoint Interface Diagram" handout (one per learner)

Sample Files:

- Sample images for insertion practice (provided by instructor)
- A sample presentation for demonstration

Internet: Not required for this session.

D. OPENING ACTIVITY (10 minutes)

"The Power of a Good Presentation"

Instructor Setup: Display two presentations on the projector: one plain (no design, too much text) and one professional (clean design, minimal text, good images).

Instructor Script: "Good morning, everyone. Welcome to Week 3. Today we start Microsoft PowerPoint. Before we open anything, look at these two presentations. They contain the same information. But one is boring and hard to follow. The other is engaging and professional. What is the difference? [Pause for answers — design, images, too much text, colours.]

The difference is how the presentation was designed. PowerPoint is the world's most common presentation tool. It is used in schools, businesses, churches and conferences. A good presentation can inspire, inform and persuade. A bad presentation can bore, confuse and lose the audience.

Today, you will learn the fundamentals of PowerPoint. You will learn how to create slides, apply themes and design professional presentations. By the end of this course, you will be able to create presentations that hold a room. Let us begin."

Expected Learner Response: Recognition that they have seen both good and bad presentations. Curiosity about how to create good ones.

E. PREREQUISITE REVIEW (5 minutes)

Instructor Script: "Before we start, let us remember what we learned in the Excel sessions. Quick review."

Review Questions and Expected Answers:

1. "How do you open a program?"
   - Click the Start button, type the program name, and click it.
2. "What is the Ribbon?"
   - The Ribbon is the area below the title bar that contains all the tools, organised into tabs.
3. "How do you save a file?"
   - Press Ctrl+S or click File > Save.
4. "What does the SUM function do?"
   - It adds a range of cells.
5. "How do you create a chart in Excel?"
   - Select the data, go to the Insert tab, and choose a chart type.

How to Address Common Gaps:

- If a learner is still struggling with basic computer skills, pair them with a more confident learner.
- If a learner cannot open PowerPoint, guide them through the Start menu.

F. FULL LECTURE NOTES (35 minutes)

Topic 1: What Is Microsoft PowerPoint? (5 minutes)

Instructor Script: "Microsoft PowerPoint is a presentation program. It is used to create slideshows — a series of slides that contain text, images, charts and other content. Presentations are used for:

- Business pitches and sales presentations
- School projects and lectures
- Church sermons and community meetings
- Training sessions and workshops
- Conferences and seminars

PowerPoint vs Word vs Excel:

- Word is for documents (letters, reports, CVs).
- Excel is for spreadsheets (budgets, data, calculations).

- PowerPoint is for presentations (slides, speeches, visual aids).

Why PowerPoint Matters: In the professional world, you will often need to present information to others. A good presentation can help you win a job, secure a loan, sell a product or teach a class. Knowing how to use PowerPoint is a valuable professional skill."

Topic 2: Opening PowerPoint and the Interface (10 minutes)

Instructor Script: "Let us open PowerPoint and explore the interface.

Opening PowerPoint:

1. Click the Start button.
2. Type "PowerPoint" in the search box.
3. Click on "Microsoft PowerPoint."
4. PowerPoint opens. You will see a start screen with templates. Click "Blank Presentation" to create a new presentation.

The PowerPoint Interface — Main Parts:

1. The Title Bar (top): Shows the name of the presentation and the program. By default, a new presentation is called "Presentation1."

2. The Ribbon (below the title bar): Like Word and Excel, the Ribbon contains all the tools, organised into tabs:

- **File:** Open, save, print, export
- **Home:** Font, paragraph, drawing, editing
- **Insert:** Tables, images, shapes, charts, text boxes
- **Design:** Themes, variants, slide size
- **Transitions:** Slide transitions
- **Animations:** Object animations
- **Slide Show:** Presenter view, setup
- **Review:** Spelling, comments
- **View:** Layout views, zoom

3. The Slides Pane (left): This shows thumbnails of all the slides in your presentation. You can click on a thumbnail to select and edit a slide. You can also drag thumbnails to rearrange slides.

4. The Slide Area (centre): This is the large area where you create and edit the current slide. It shows the slide exactly as it will appear during the presentation.

5. The Notes Pane (bottom): This is where you can add speaker notes for each slide. These notes are visible to the presenter but not to the audience.

6. The Status Bar (bottom): Shows the slide number, total slides, language and zoom level."

Topic 3: Creating and Managing Slides (10 minutes)

Instructor Script: "Now let us learn how to create and manage slides.

Adding a New Slide:

1. Go to the Home tab.
2. Click the New Slide button.
3. A dropdown menu appears with different slide layouts.
4. Choose a layout (e.g., Title Slide, Title and Content, Two Content).
5. A new slide is added to the presentation.

Slide Layouts: A slide layout is a pre-designed arrangement of text boxes, images and other elements. Common layouts include:

- **Title Slide:** A title and a subtitle. Used for the first slide.
- **Title and Content:** A title and a content area (text, images, charts, etc.). Used for most slides.
- **Two Content:** A title and two content areas. Used for comparisons.
- **Section Header:** A title and a subtitle. Used to separate sections.

- **Blank:** No pre-designed elements. Used for custom layouts.

Deleting a Slide:

1. Click on the slide thumbnail in the Slides Pane.
2. Press Delete or right-click and choose "Delete Slide."

Rearranging Slides:

1. Click and drag a slide thumbnail to a new position in the Slides Pane.
2. The slide moves to the new position.

Duplicating a Slide:

1. Right-click on a slide thumbnail.
2. Choose "Duplicate Slide."
3. An exact copy of the slide is created."

Topic 4: Themes and Design (10 minutes)

Instructor Script: "Themes are pre-designed colour schemes, fonts and effects that give your presentation a professional, consistent appearance. Instead of choosing colours and fonts manually, you can apply a theme and PowerPoint does the work for you.

Applying a Theme:

1. Go to the Design tab.
2. The Themes gallery appears. Hover over a theme to see a preview.
3. Click on a theme to apply it to the entire presentation.
4. The theme changes the colours, fonts and background of all slides.

Theme Variants: Each theme has variants — different colour schemes based on the same design. To see the variants:

1. Go to the Design tab.
2. Click the Variants dropdown.
3. Choose a colour variant.

Customising a Theme: You can customise a theme by changing the colours, fonts and effects:

1. Go to the Design tab.
2. Click the Variants dropdown.
3. Click "Colours" to choose a colour scheme.
4. Click "Fonts" to choose a font scheme.
5. Click "Effects" to choose an effect scheme.

Best Practice: Choose a theme that matches the tone of your presentation. For business presentations, use clean, professional themes. For creative presentations, use more colourful themes. Avoid themes that are too busy or distracting."

G. INSTRUCTOR DEMONSTRATION (15 minutes)

Demonstration 1: Creating a Presentation (5 minutes)

What the Instructor Prepares:

- PowerPoint open on the projector with a blank presentation

What the Instructor Says and Does: "I am going to create a simple presentation. Watch carefully."

1. "I click the Start button. I type 'PowerPoint.' I click on Microsoft PowerPoint."
2. "The start screen appears. I click 'Blank Presentation.'"
3. "A new presentation opens with a Title Slide layout. I click on the title box and type 'My Business Pitch.'"
4. "I click on the subtitle box and type 'Prepared by [Instructor Name].'"
5. "I go to the Home tab. I click the New Slide button. I choose 'Title and Content.' A new slide is added."

6. "I click on the title box and type 'About Us.' I click on the content box and type a few bullet points."

What Learners Should Observe: The instructor creating a new presentation, adding slides and entering text.

Expected Result: Learners can describe the steps to create a presentation and add slides.

Demonstration 2: Applying a Theme (5 minutes)

What the Instructor Prepares:

- The presentation from Demonstration 1

What the Instructor Says and Does: "Now I will apply a theme to make the presentation look professional."

1. "I go to the Design tab. The Themes gallery appears."
2. "I hover over a theme called 'Ion.' The slides change colour. I like this theme. I click on it."
3. "The entire presentation now has a consistent colour scheme and font."
4. "I click the Variants dropdown. I choose a different colour variant. The colours change but the design stays the same."

What Learners Should Observe: The instructor applying a theme and choosing a variant.

Expected Result: Learners can describe how to apply a theme.

Demonstration 3: Saving and Opening (5 minutes)

What the Instructor Prepares:

- The themed presentation from Demonstration 2

What the Instructor Says and Does: "Now I will save and open this presentation."

1. "I press Ctrl+S. A dialog box appears. I navigate to the Desktop. I type 'My Business Pitch' in the File name box. I click Save."
2. "I close PowerPoint. I reopen it. I click File > Open. I navigate to the Desktop. I click on 'My Business Pitch.pptx.' I click Open. The presentation reopens with all the slides and the theme intact."

What Learners Should Observe: The instructor saving and reopening a presentation.

Expected Result: Learners can describe how to save and open a PowerPoint presentation.

H. GUIDED PRACTICAL EXERCISE (20 minutes)

Exercise: "My First Presentation"

Practical Objective: The learner will create a simple presentation with multiple slides, apply a theme and save the presentation.

Starting Materials:

- PowerPoint open on each machine with a blank presentation

Instructions:

Part 1: Create the Title Slide (5 minutes)

1. "Click on the title box. Type your name."
2. "Click on the subtitle box. Type 'Introduction to Microsoft PowerPoint.'"

Instructor Prompts:

- "What is the purpose of the title slide?" (Answer: To introduce the presentation and the presenter.)

Part 2: Add Content Slides (10 minutes)

3. "Go to the Home tab. Click the New Slide button. Choose 'Title and Content.'"
4. "Click on the title box. Type 'What I Have Learned.' Click on the content box. Type three bullet points about what you have learned in this course so far."
5. "Add another slide. Choose 'Title and Content.' Type 'My Goals' in the title box. Type two or three goals in the content box."
6. "Add another slide. Choose 'Title and Content.' Type 'Thank You' in the title box. Type 'Questions?' in the content box."

Instructor Prompts:

- "How many slides do you have now?" (Answer: 4 — title slide + 3 content slides.)
- "How do you rearrange slides?" (Answer: Click and drag the slide thumbnails in the Slides Pane.)

Part 3: Apply a Theme (5 minutes)

7. "Go to the Design tab. Hover over a few themes to see the previews."
8. "Choose a theme you like. Click on it to apply it to the entire presentation."
9. "Click the Variants dropdown. Choose a colour variant."

Instructor Prompts:

- "What happened when you applied the theme?" (Answer: The colours, fonts and background of all slides changed.)
- "Why is it important to use a theme?" (Answer: It gives the presentation a consistent, professional appearance.)

Part 4: Save the Presentation (2 minutes)

10. "Press Ctrl+S. Navigate to the Desktop. Type 'My First Presentation' in the File name box. Click Save."

Instructor Prompts:

- "What is the file extension for PowerPoint?" (Answer: .pptx)

Expected Result: Every learner has created a 4-slide presentation with a theme and saved it to the Desktop.

Troubleshooting Guidance:

- If the learner cannot find PowerPoint: "Click the Start button and type 'PowerPoint.' It should appear in the search results."
- If the learner cannot add a slide: "Go to the Home tab and click the New Slide button."
- If the learner cannot apply a theme: "Go to the Design tab. The Themes gallery is in the middle of the Ribbon."

Completion Checklist:

- ☐ Title slide created with name and subtitle
- ☐ 3 content slides added
- ☐ Theme applied to entire presentation
- ☐ Colour variant chosen
- ☐ Presentation saved as "My First Presentation" on the Desktop

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I. INDEPENDENT CLASS EXERCISE (10 minutes)

Exercise: "PowerPoint Scavenger Hunt"

Scenario: "Your colleague asks you to find and use several tools in PowerPoint. Your task is to find each tool and use it."

Task: Without looking at your notes or your neighbour's screen, complete the following tasks in PowerPoint:

1. Open PowerPoint and create a blank presentation.
2. Add a new slide with the "Title and Content" layout.
3. Add a new slide with the "Two Content" layout.
4. Delete the second slide.
5. Rearrange the slides so the "Two Content" slide is first.
6. Apply a theme of your choice.
7. Change the colour variant.
8. Save the presentation as "Scavenger Hunt" on the Desktop.

Requirements:

- Complete all 8 tasks independently
- Do not ask the instructor for help unless you are truly stuck

Time Allowance: 10 minutes

Success Criteria:

- All 8 tasks completed correctly
- The learner can explain what each tool does

Hints Available If the Learner Gets Stuck:

- Hint 1: "To add a slide, go to the Home tab and click the New Slide button."
- Hint 2: "To delete a slide, click on the slide thumbnail and press Delete."
- Hint 3: "To apply a theme, go to the Design tab and click on a theme."

Complete Answer Key for the Instructor:

1. The learner opens PowerPoint and creates a blank presentation.
2. The learner adds a slide with the "Title and Content" layout.
3. The learner adds a slide with the "Two Content" layout.
4. The learner deletes the second slide.
5. The learner rearranges the slides.
6. The learner applies a theme.
7. The learner changes the colour variant.
8. The learner saves the presentation as "Scavenger Hunt" on the Desktop.

J. REAL-WORLD CASE STUDY (5 minutes)

Case: "Emmanuel's Business Pitch"

Situation: Emmanuel runs a small tech startup in Port Harcourt. He has developed a mobile app that helps small businesses manage their inventory. He has been invited to pitch his app to a group of investors. He needs to create a presentation.

The Problem: Emmanuel has never used PowerPoint. He creates a presentation with too many slides, too much text and no design. The investors are bored. They do not invest.

Why It Matters: A good presentation can win investors, customers and partners. A bad presentation can lose them. In the professional world, your presentation is often your only chance to make a good impression.

Relevant Concept: PowerPoint fundamentals. If Emmanuel knew how to create clean, professional slides with minimal text and good design, he would have a better chance of winning the investors.

Worked Solution: If Emmanuel had taken this course, here is what he would do:

1. Create a title slide with the app name and his name.
2. Create a slide about the problem the app solves.
3. Create a slide about how the app works.
4. Create a slide about the market opportunity.
5. Create a slide about the business model.
6. Create a slide with a call to action.
7. Apply a clean, professional theme.
8. Use minimal text — just key points.
9. Use images and screenshots to illustrate the app.
10. Deliver the presentation with confidence.

Reasoning: A good presentation tells a story. It has a beginning (the problem), a middle (the solution) and an end (the call to action). It uses visuals to support the message, not replace it. It is delivered with confidence and enthusiasm.

Possible Mistake: Emmanuel might put too much text on each slide. This makes the slides hard to read and the audience stops listening.

Correction: Use the 6x6 rule: no more than 6 words per line and no more than 6 lines per slide. If you have more to say, use speaker notes or additional slides.

Transfer Exercise: "Your friend Adaeze is presenting a school project. She has 20 slides with paragraphs of text on each one. What advice would you give her?"

Professional Connection: Every professional needs to present information at some point. Job interviews, business pitches, school presentations, church sermons — everyone. Knowing how to create and deliver a good presentation is a valuable professional skill.

K. COMMON MISTAKES AND TROUBLESHOOTING (5 minutes)

Mistake 1: Putting Too Much Text on Slides

Symptoms: The learner creates slides with paragraphs of text. **Likely Cause:** The learner does not know the 6x6 rule. **How to Investigate:** Look at the slides. How much text is on each one? **Safe Corrective Steps:** Teach the 6x6 rule: no more than 6 words per line and no more than 6 lines per slide. Use bullet points, not paragraphs. **How to Confirm the Problem Is Resolved:** The learner uses minimal text on slides. **How to Prevent Recurrence:** Make the 6x6 rule a standard requirement for every presentation exercise.

Mistake 2: Not Using a Theme

Symptoms: The learner creates a presentation with no theme. The slides look plain and unprofessional. **Likely Cause:** The learner does not know about themes. **How to Investigate:** Go to the Design tab. Is a theme applied? **Safe Corrective Steps:** Teach the learner to apply a theme from the Design tab. **How to Confirm the Problem Is Resolved:** The learner applies a theme. **How to Prevent Recurrence:** Make applying a theme a standard step in every presentation exercise.

Mistake 3: Not Saving the Presentation

Symptoms: The learner creates a presentation but does not save. The computer crashes and the work is lost. **Likely Cause:** The learner forgets to save. **How to Investigate:** Ask the learner, "Did you save your presentation?" **Safe Corrective Steps:** Teach the learner to press Ctrl+S regularly. **How to Confirm the Problem Is Resolved:** The learner saves regularly. **How to Prevent Recurrence:** Set a timer that goes off every 5 minutes. When it rings, everyone presses Ctrl+S.

Mistake 4: Using Too Many Different Layouts

Symptoms: The learner uses a different layout for every slide. The presentation looks inconsistent. **Likely Cause:** The learner does not understand the purpose of layouts. **How to Investigate:** Look at the slides. Are the layouts consistent? **Safe Corrective Steps:** Teach the learner to use the same layout for similar slides (e.g., "Title and Content" for all content slides). **How to Confirm the Problem Is Resolved:** The learner uses consistent layouts. **How to Prevent Recurrence:** Remind the learner to use the same layout for similar slides.

Mistake 5: Not Rearranging Slides

Symptoms: The learner creates slides in the wrong order and does not rearrange them. **Likely Cause:** The learner does not know how to rearrange slides. **How to Investigate:** Ask the learner, "Can you rearrange the slides?" **Safe Corrective Steps:** Teach the learner to click and drag slide thumbnails in the Slides Pane. **How to Confirm the Problem Is Resolved:** The learner rearranges slides. **How to Prevent Recurrence:** Make rearranging slides a standard part of every presentation exercise.

L. SESSION ASSIGNMENT (5 minutes)

Assignment Title: "My Introduction Presentation"

Purpose: To reinforce PowerPoint fundamentals by having the learner create a simple introduction presentation.

Realistic Scenario: "You are starting a new job. On your first day, your boss asks you to introduce yourself to the team. Create a 5-slide presentation about yourself."

Detailed Instructions:

1. Open Microsoft PowerPoint. Create a new blank presentation.
2. Create the following slides:
    - Slide 1: Title Slide — Your name and "Introduction"

- Slide 2: About You — Where you are from, your background
- Slide 3: Education — Your schools and qualifications
- Slide 4: Skills — Your key skills and strengths
- Slide 5: Thank You — "Questions?"

3. Apply a theme of your choice.
4. Save the presentation as "My Introduction" on the Desktop.
5. Show the saved file to the instructor before leaving.

Required Deliverables:

- A PowerPoint presentation named "My Introduction" saved on the Desktop
- The presentation contains: 5 slides, a theme, minimal text

Submission Format:

- The file must be saved on the Desktop of the academy machine
- The learner must show the instructor the saved file before leaving

Deadline Recommendation:

- Before leaving the session (in-class assignment)

Marking Rubric:

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Title slide	Name and title present	Mostly correct	Missing or incorrect
About You slide	Clear, concise information	Mostly correct	Vague or too much text
Education slide	Schools and qualifications listed	Mostly correct	Incomplete
Skills slide	Key skills listed	Mostly correct	Incomplete
Thank You slide	Present with "Questions?"	Present but generic	Missing
Theme applied	Professional theme chosen	Theme applied but not ideal	No theme
Minimal text	Follows 6x6 rule	Mostly follows	Too much text
File saved correctly	Saved on Desktop with correct name	Saved but wrong location or name	Not saved

Total: 40 marks

Common Mistakes to Avoid:

- Putting too much text on slides
- Not applying a theme
- Not saving the file

Instructor Reference Solution:

The final presentation should have:

- 5 slides: Title Slide, About You, Education, Skills, Thank You
- A professional theme applied
- Minimal text on each slide (following the 6x6 rule)
- Saved as "My Introduction.pptx" on the Desktop

M. KNOWLEDGE CHECK (5 minutes)

Questions

1. Multiple Choice: What is Microsoft PowerPoint? a) A word processing program b) A spreadsheet program c) A presentation program d) A web browser

Answer: c) A presentation program

2. Short Answer: What is a slide layout?

Answer: A slide layout is a pre-designed arrangement of text boxes, images and other elements. Common layouts include Title Slide, Title and Content, Two Content and Blank.

3. Multiple Choice: What is the 6x6 rule? a) No more than 6 slides and 6 themes b) No more than 6 words per line and 6 lines per slide c) No more than 6 images and 6 charts d) No more than 6 colours and 6 fonts

Answer: b) No more than 6 words per line and 6 lines per slide

4. Explain in Your Own Words: Why is it important to apply a theme to a presentation?

Answer: A theme gives a presentation a consistent, professional appearance. It ensures that all slides have the same colours, fonts and design. Without a theme, a presentation looks plain and unprofessional.

5. Practical Identification: Open PowerPoint. Identify the Ribbon, the Slides Pane, the Slide Area, the Notes Pane, and the Design tab.

Answer: The learner correctly identifies all five elements.

6. Scenario-Based: Your friend says, "I created a presentation but it looks plain. What can I do to make it look better?" What advice would you give?

Answer: Apply a theme from the Design tab. A theme will give your presentation a consistent colour scheme, font and design. You can also change the colour variant to customise the look.

7. Error-Detecting: A learner writes in their assignment: "I put paragraphs of text on each slide because I wanted to include all the information." What is wrong with this, and what should they have done instead?

Answer: Putting paragraphs of text on slides makes them hard to read and the audience stops listening. The learner should have used the 6x6 rule: no more than 6 words per line and 6 lines per slide. Additional information can go in speaker notes or additional slides.

N. REFLECTION AND RECAP (5 minutes)

Plain-English Summary

Today you learned the fundamentals of Microsoft PowerPoint. You learned what PowerPoint is, how to open it, and how to identify the main parts of the interface. You learned how to create a new presentation, add and delete slides, apply slide layouts and apply themes. You also learned the 6x6 rule for slide text. These are the foundation skills for creating professional presentations.

Five Important Takeaways

1. Microsoft PowerPoint is a presentation program used for creating slideshows.
2. A slide layout is a pre-designed arrangement of text boxes and other elements.
3. Themes give a presentation a consistent, professional appearance.
4. The 6x6 rule: no more than 6 words per line and 6 lines per slide.
5. Save regularly with Ctrl+S and use descriptive file names.

Questions That Help You Recognise What You Can Now Do

- Can you open PowerPoint and identify the main parts of the interface?
- Can you create a new presentation and add slides?
- Can you apply slide layouts?
- Can you apply a theme and change the colour variant?
- Can you save and open a presentation?

Connection to the Next Session

In the next session, you will learn how to create professional presentations with images, shapes, transitions and animations. You will also learn how to deliver a presentation aloud. These are the skills that make presentations engaging and memorable. You

now have the foundation. Next, you will learn to bring your presentations to life.

O. INSTRUCTOR EVALUATION NOTES

Observable Signs of Competence

- The learner opens PowerPoint independently.
- The learner identifies the Ribbon, Slides Pane, Slide Area and Notes Pane.
- The learner creates a new presentation and adds slides.
- The learner applies slide layouts.
- The learner applies a theme and changes the colour variant.
- The learner follows the 6x6 rule.
- The learner saves with a descriptive file name.

Common Misconceptions

- **"PowerPoint is just for school projects."** Some learners think PowerPoint is only for students. Correct this by showing them how PowerPoint is used in business, churches and conferences.
- **"More text is better."** Some learners think that putting lots of text on slides makes the presentation more informative. Correct this by teaching the 6x6 rule.
- **"Themes are only for designers."** Some learners think themes are an advanced feature. Correct this by showing them how simple it is to apply a theme.

Suggested Remedial Exercises

- For learners who put too much text on slides: Have them rewrite a text-heavy slide using the 6x6 rule.
- For learners who do not apply themes: Have them apply a theme to an existing presentation and see the difference.
- For learners who cannot add slides: Walk them through the process step by step. Let them practise multiple times.

Differentiation Guidance

Level 1 — Supported (for learners who struggle):

- Provide a labelled PowerPoint interface diagram for reference.
- Pair with a more confident learner for practical exercises.
- Allow extra time for the independent exercise.
- Focus on creating slides and applying themes before moving to advanced features.

Level 2 — Independent (standard):

- Complete all exercises as written.
- Answer all knowledge check questions.
- Complete the assignment to the standard shown in the rubric.

Level 3 — Challenge (for learners who finish early):

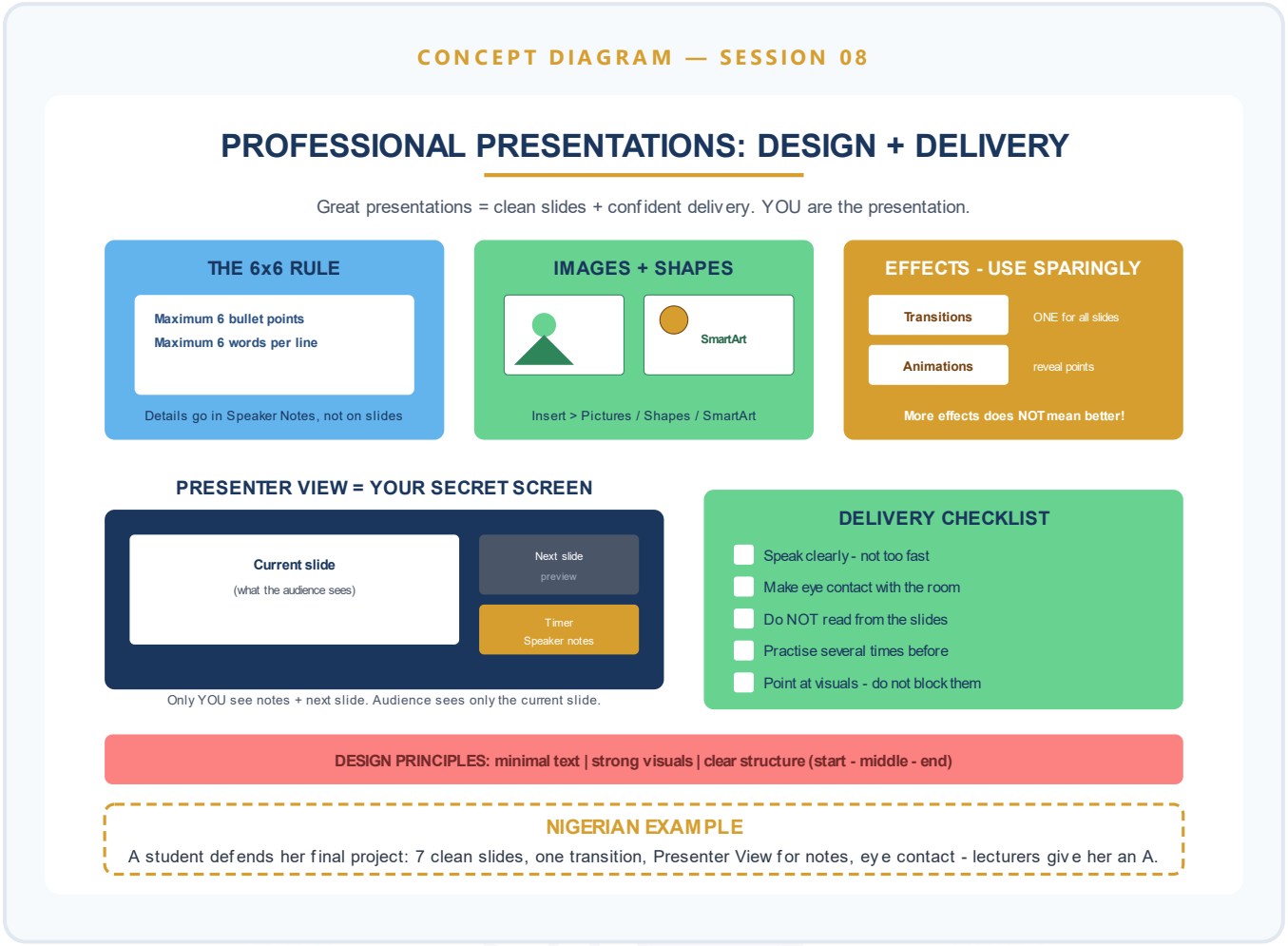
- Ask the learner to create a presentation with 10 slides on a topic of their choice.
- Ask the learner to research and write down three more PowerPoint features.
- Ask the learner to explain the difference between a theme and a variant.

End of Session 7

Next Session: Session 8 — Professional Presentations **What to Bring:** Creativity, confidence, willingness to present
Preparation: Prepare presentation delivery guidelines and feedback forms

SESSION 08 OF 9

Professional Presentations



SESSION 8: PROFESSIONAL PRESENTATIONS

Week: 3 **Module:** Microsoft PowerPoint **Duration:** 105 minutes **Difficulty:** Beginner **Prerequisite Knowledge:** Session 7 — PowerPoint fundamentals **Connection to Previous Session:** Builds on Session 7 by teaching advanced presentation design and delivery skills **Preparation Required Before Class:**

- Prepare presentation delivery guidelines and feedback forms
- Ensure Microsoft PowerPoint is installed and functional on all machines

A. SESSION IDENTIFICATION

Field	Detail
Session Number	8
Title	Professional Presentations
Week	3

Field	Detail
Module	Microsoft PowerPoint
Duration	105 minutes
Difficulty	Beginner
Prerequisite Knowledge	Session 7 — PowerPoint fundamentals
Connection to Previous Session	Builds on Session 7 by teaching advanced design and delivery
Preparation	Delivery guidelines and feedback forms prepared

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Insert and format images** in a presentation.
2. **Insert and format shapes and SmartArt** for visual communication.
3. **Add transitions** between slides for smooth flow.
4. **Add simple animations** to text and objects.
5. **Use Presenter View** to see notes and upcoming slides while presenting.
6. **Deliver a presentation aloud** with confidence and clarity.
7. **Apply the principles of good presentation design:** minimal text, strong visuals, clear structure.
8. **Give and receive constructive feedback** on presentations.

C. REQUIRED MATERIALS

Hardware:

- One desktop computer per learner
- One keyboard per learner
- One mouse per learner
- One monitor per learner
- Projector for instructor demonstration and learner presentations

Software:

- Windows 10 or Windows 11
- Microsoft PowerPoint 2019 or Microsoft 365

Printed Materials:

- "Presentation Delivery Guidelines" handout (one per learner)
- "Presentation Feedback Form" (one per learner)

Sample Files:

- Sample images for insertion practice
- A sample presentation for demonstration

Internet: Not required for this session.

D. OPENING ACTIVITY (10 minutes)

"The 2-Minute Pitch"

Instructor Setup: Deliver a short, engaging presentation on a simple topic (e.g., "Why I Love Port Harcourt") using good presentation skills — minimal text, strong visuals, confident delivery.

Instructor Script: "Good morning, everyone. Before we start the session, I am going to give you a short presentation. Watch carefully — I want you to notice what makes it engaging."

[Deliver a 2-minute presentation with 3 slides: a title slide, one image slide with a single sentence, and a closing slide. Use minimal text, a strong image and confident delivery.]

"What made that presentation engaging? [Pause for answers — minimal text, image, confidence, eye contact, enthusiasm.]

Today, you will learn how to create and deliver presentations like that. You will learn how to insert images, add transitions and animations, and deliver a presentation with confidence. By the end of this session, you will present your own short presentation to the class. Let us begin."

Expected Learner Response: Learners will identify the key elements of a good presentation: minimal text, strong visuals, confidence, enthusiasm. This sets the context for the session.

E. PREREQUISITE REVIEW (5 minutes)

Instructor Script: "Before we start, let us remember what we learned in the last session. Quick review."

Review Questions and Expected Answers:

1. "How do you open Microsoft PowerPoint?"
   - Click the Start button, type "PowerPoint," and click on Microsoft PowerPoint.
2. "What is a slide layout?"
   - A pre-designed arrangement of text boxes, images and other elements.
3. "What is a theme?"
   - A pre-designed colour scheme, font and effect that gives a presentation a consistent appearance.
4. "What is the 6x6 rule?"
   - No more than 6 words per line and 6 lines per slide.
5. "How do you add a new slide?"
   - Go to the Home tab and click the New Slide button.

How to Address Common Gaps:

- If a learner cannot remember the 6x6 rule, write it on the whiteboard.
- If a learner forgot how to apply a theme, remind them: "Go to the Design tab."

F. FULL LECTURE NOTES (35 minutes)

Topic 1: Inserting and Formatting Images (10 minutes)

Instructor Script: "Images make presentations engaging and memorable. A good image can communicate an idea more effectively than a paragraph of text.

Inserting an Image:

1. Go to the Insert tab.
2. Click the Pictures button.
3. Choose "This Device" to insert an image from your computer.
4. Navigate to the image file.
5. Click "Insert."

Formatting an Image:

1. Click on the image to select it.

2. The Picture Format tab appears on the Ribbon.
3. Use the tools to:
    - Crop the image (remove unwanted parts).
    - Adjust brightness and contrast.
    - Apply a picture style (border, shadow, reflection).
    - Resize the image (drag the corners).
    - Move the image (drag it to a new position).

Image Tips:

- Use high-quality images. Blurry or pixelated images look unprofessional.
- Use images that support your message. Do not use images just for decoration.
- Do not stretch images. Drag the corners to resize proportionally.
- Use the "Compress Pictures" tool (Picture Format tab) to reduce file size if needed."

Topic 2: Shapes and SmartArt (10 minutes)

Instructor Script: "Shapes and SmartArt are visual elements that help you communicate ideas clearly.

Inserting a Shape:

1. Go to the Insert tab.
2. Click the Shapes button.
3. Choose a shape (e.g., rectangle, oval, arrow).
4. Click and drag on the slide to draw the shape.
5. Use the Shape Format tab to change the fill colour, outline and effects.

Inserting SmartArt: SmartArt is a collection of pre-designed diagrams — processes, hierarchies, cycles, relationships. They are useful for showing:

- Processes (e.g., steps in a workflow)
- Hierarchies (e.g., organisational structure)
- Cycles (e.g., a repeating process)
- Relationships (e.g., how different elements connect)

To insert SmartArt:

1. Go to the Insert tab.
2. Click the SmartArt button.
3. Choose a category (e.g., Process, Hierarchy, Cycle).
4. Choose a diagram.
5. Click OK.
6. Type your text in the text boxes.

Shape and SmartArt Tips:

- Use shapes to highlight important information or create simple diagrams.
- Use SmartArt to show processes, hierarchies and relationships.
- Keep it simple. Do not use too many shapes or too much text in SmartArt."

Topic 3: Transitions and Animations (10 minutes)

Instructor Script: "Transitions and animations add movement and flow to a presentation. But they should be used sparingly. Too many transitions and animations are distracting.

Transitions: A transition is the effect that occurs when you move from one slide to the next. Common transitions include:

- **Fade:** The previous slide fades out and the next slide fades in.
- **Push:** The previous slide pushes the next slide into view.
- **Wipe:** The next slide wipes across the previous slide.

- **None:** No transition. The next slide simply appears.

To apply a transition:

1. Select the slide you want to apply the transition to.
2. Go to the Transitions tab.
3. Choose a transition from the gallery.
4. Click "Apply to All" if you want the same transition for all slides.

Animations: An animation is the effect that occurs when an object (text, image, shape) appears on a slide. Common animations include:

- **Appear:** The object simply appears.
- **Fade:** The object fades in.
- **Fly In:** The object flies in from a direction.
- **Wipe:** The object wipes into view.

To apply an animation:

1. Select the object you want to animate.
2. Go to the Animations tab.
3. Choose an animation from the gallery.
4. Use the Timing group to set the start option (On Click, With Previous, After Previous).

Best Practice:

- Use transitions sparingly. One transition for the entire presentation is usually enough.
- Use animations to reveal information step by step, not to entertain.
- Avoid distracting transitions like "Spinner," "Gallery" or "Conveyor Belt."
- Keep it simple and professional."

Topic 4: Presenter View and Delivery (5 minutes)

Instructor Script: "Presenter View is a special view that helps you deliver a presentation confidently. It shows:

- The current slide (large, for the audience).
- The next slide (small, so you know what is coming).
- Your speaker notes (so you do not have to memorise everything).
- A timer (so you know how long you have been speaking).

How to Use Presenter View:

1. Connect your computer to a projector or second screen.
2. Go to the Slide Show tab.
3. Click "From Beginning" or "From Current Slide."
4. PowerPoint automatically uses Presenter View if a second screen is detected.
5. If not, right-click on the slide and choose "Presenter View."

Delivery Tips:

- **Know your content.** Practise your presentation several times before delivering it.
- **Speak clearly and slowly.** Do not rush. Pause between points.
- **Make eye contact.** Look at the audience, not at the screen.
- **Use body language.** Stand up straight, use hand gestures, move around.
- **Do not read from the slides.** The slides are for the audience. Your notes are for you.
- **Handle questions confidently.** If you do not know the answer, say so and offer to follow up."

G. INSTRUCTOR DEMONSTRATION (15 minutes)

Demonstration 1: Inserting and Formatting an Image (5 minutes)

What the Instructor Prepares:

- A presentation open on the projector
- A sample image file

What the Instructor Says and Does: "I am going to insert and format an image."

1. "I go to the Insert tab. I click Pictures. I choose 'This Device.' I navigate to the image file. I click Insert."
2. "The image appears on the slide. I click on it to select it. The Picture Format tab appears."
3. "I drag the corners to resize the image proportionally."
4. "I go to the Picture Format tab. I choose a picture style — a simple border."
5. "I drag the image to the centre of the slide."

What Learners Should Observe: The instructor inserting, resizing and formatting an image.

Expected Result: Learners can describe how to insert and format an image.

Demonstration 2: Adding a Transition and Animation (5 minutes)

What the Instructor Prepares:

- The presentation from Demonstration 1

What the Instructor Says and Does: "Now I will add a transition and an animation."

1. "I select the first slide. I go to the Transitions tab. I choose 'Fade.' I click 'Apply to All.' Now all slides have the same transition."
2. "I select the title text on the first slide. I go to the Animations tab. I choose 'Appear.' The title will appear when I click."
3. "I select the image. I go to the Animations tab. I choose 'Fade.' The image will fade in after the title."

What Learners Should Observe: The instructor adding a transition and animations.

Expected Result: Learners can describe how to add transitions and animations.

Demonstration 3: Using Presenter View (5 minutes)

What the Instructor Prepares:

- The presentation from Demonstrations 1 and 2
- A projector connected

What the Instructor Says and Does: "Now I will show you how to use Presenter View."

1. "I go to the Slide Show tab. I click 'From Beginning.'"
2. "The presentation starts. The audience sees the current slide. I see Presenter View — the current slide, the next slide, my notes and a timer."
3. "I click to advance to the next slide. The transition plays. The title appears. The image fades in."
4. "I right-click and choose 'End Show' to exit."

What Learners Should Observe: The instructor using Presenter View to deliver a presentation.

Expected Result: Learners can describe how to use Presenter View.

H. GUIDED PRACTICAL EXERCISE (20 minutes)

Exercise: "My Business Pitch Presentation"

Practical Objective: The learner will create a 5-slide business pitch presentation with images, transitions, animations and speaker notes.

Starting Materials:

- PowerPoint open on each machine with a blank presentation
- Sample images provided by the instructor

Instructions:

Part 1: Create the Slides (10 minutes)

1. "Create a title slide with the name of a fictional business (e.g., 'Adaeze's Fashion Design') and your name."
2. "Add a slide with the title 'About Us' and a short description of the business."
3. "Add a slide with the title 'Our Products' and insert an image of a product."
4. "Add a slide with the title 'Pricing' and a simple price list."
5. "Add a slide with the title 'Contact Us' and contact information."

Instructor Prompts:

- "How many slides do you have?" (Answer: 5.)
- "Did you follow the 6x6 rule?" (Answer: Yes — minimal text on each slide.)

Part 2: Add Images, Transitions and Animations (5 minutes)

6. "Insert an image on the 'About Us' slide. Resize and format it."
7. "Go to the Transitions tab. Choose 'Fade.' Click 'Apply to All.'"
8. "Select the title on the first slide. Go to the Animations tab. Choose 'Appear.'"
9. "Select the image on the 'Our Products' slide. Go to the Animations tab. Choose 'Fade.'"

Instructor Prompts:

- "Why did we use 'Apply to All' for the transition?" (Answer: To keep the presentation consistent.)
- "Why did we use different animations for the title and the image?" (Answer: To reveal information step by step.)

Part 3: Add Speaker Notes and Save (5 minutes)

10. "Click on the first slide. Go to the Notes Pane at the bottom. Type a short note about what you will say."
11. "Repeat for all slides."
12. "Press Ctrl+S. Navigate to the Desktop. Type 'My Business Pitch' in the File name box. Click Save."

Instructor Prompts:

- "What is the purpose of speaker notes?" (Answer: To help you remember what to say without reading from the slides.)

Expected Result: Every learner has created a 5-slide business pitch presentation with images, transitions, animations and speaker notes.

Troubleshooting Guidance:

- If the learner cannot insert an image: "Go to the Insert tab, click Pictures, and choose 'This Device.'"
- If the learner cannot add a transition: "Go to the Transitions tab and choose a transition."
- If the learner cannot add speaker notes: "Look at the bottom of the screen for the Notes Pane. If you do not see it, go to the View tab and click 'Notes.'"

Completion Checklist:

- [ ] 5 slides created (Title, About Us, Products, Pricing, Contact)
- [ ] Image inserted and formatted on the About Us slide
- [ ] Transition applied to all slides
- [ ] Animations added to title and image
- [ ] Speaker notes added to all slides
- [ ] Presentation saved as "My Business Pitch" on the Desktop

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I. INDEPENDENT CLASS EXERCISE (10 minutes)

Exercise: "Present My Business Pitch"

Scenario: "You are pitching your fictional business to a group of investors (the class). You have 2 minutes to present your 5-slide presentation."

Task: Deliver your "My Business Pitch" presentation aloud to the class. Use the following guidelines:

1. Stand up and face the audience.
2. Speak clearly and slowly.
3. Make eye contact with the audience.
4. Do not read from the slides — use your speaker notes.
5. Use the 6x6 rule — let the slides support your message, not replace it.

Requirements:

- Present all 5 slides
- Speak for at least 1 minute but no more than 3 minutes
- Use Presenter View if possible

Time Allowance: 2–3 minutes per learner (depending on class size)

Success Criteria:

- The learner presents all 5 slides
- The learner speaks clearly and confidently
- The learner makes eye contact with the audience
- The learner does not read from the slides
- The learner finishes within the time limit

Hints Available If the Learner Gets Stuck:

- Hint 1: "Take a deep breath before you start. It is normal to be nervous."
- Hint 2: "If you forget what to say, look at your speaker notes. They are there to help you."
- Hint 3: "Remember: the slides are for the audience. Your words are the presentation."

Complete Answer Key for the Instructor:

There is no single correct answer for a presentation. The instructor should use the "Presentation Feedback Form" to evaluate each learner on:

- Content (Is the information clear and organised?)
- Delivery (Does the speaker speak clearly and confidently?)
- Eye contact (Does the speaker look at the audience?)
- Time management (Does the speaker finish within the time limit?)
- Slide design (Are the slides clean and professional?)

J. REAL-WORLD CASE STUDY (5 minutes)

Case: "Emmanuel's Investor Pitch"

Situation: Emmanuel has developed a mobile app for small businesses. He has been invited to pitch to investors. He creates a presentation with 20 slides, too much text and no design. He delivers it by reading from the slides. The investors are bored. They do not invest.

The Problem: Emmanuel's presentation was too long, too text-heavy and poorly delivered. The investors could not see the value of his app.

Why It Matters: A good presentation can win investors, customers and partners. A bad presentation can lose them. In the professional world, your presentation is often your only chance to make a good impression.

Relevant Concept: Professional presentation design and delivery. If Emmanuel knew how to create clean slides, use minimal text and deliver with confidence, he would have a better chance of winning the investors.

Worked Solution: If Emmanuel had taken this course, here is what he would do:

1. Create 7 slides: Title, Problem, Solution, How It Works, Market, Business Model, Call to Action.

2. Use minimal text — just key points.
3. Use images and screenshots to illustrate the app.
4. Apply a clean, professional theme.
5. Add a simple transition (Fade) to all slides.
6. Add animations to reveal information step by step.
7. Write speaker notes for each slide.
8. Practise the presentation several times.
9. Deliver with confidence, eye contact and enthusiasm.

Reasoning: A good presentation tells a story. It has a beginning (the problem), a middle (the solution) and an end (the call to action). It uses visuals to support the message, not replace it. It is delivered with confidence and enthusiasm.

Possible Mistake: Emmanuel might think that more slides means a more thorough presentation. In reality, fewer slides with clear, concise information are more effective.

Correction: Aim for 1 slide per minute of presentation time. A 10-minute presentation should have no more than 10–12 slides. A 2-minute pitch should have no more than 5–7 slides.

Transfer Exercise: "Your friend Adaeze is presenting a school project. She has 30 slides and reads from each one. What advice would you give her?"

Professional Connection: Every professional needs to present information at some point. Job interviews, business pitches, school presentations, church sermons — everyone. Knowing how to create and deliver a good presentation is a valuable professional skill.

K. COMMON MISTAKES AND TROUBLESHOOTING (5 minutes)

Mistake 1: Reading from the Slides

Symptoms: The learner reads the text on the slides word for word. **Likely Cause:** The learner is nervous or has not practised. **How to Investigate:** Watch the learner present. Are they looking at the screen or the audience? **Safe Corrective Steps:** Teach the learner to use speaker notes instead of reading from the slides. Encourage them to practise several times. **How to Confirm the Problem Is Resolved:** The learner does not read from the slides. **How to Prevent Recurrence:** Make speaker notes a standard requirement for every presentation.

Mistake 2: Using Too Many Transitions and Animations

Symptoms: The learner uses a different transition for every slide and multiple animations per slide. **Likely Cause:** The learner thinks more effects make the presentation more engaging. **How to Investigate:** Watch the presentation. Are the transitions and animations distracting? **Safe Corrective Steps:** Teach the learner to use one transition for the entire presentation and animations sparingly. **How to Confirm the Problem Is Resolved:** The learner uses transitions and animations sparingly. **How to Prevent Recurrence:** Make "one transition for all slides" a standard requirement.

Mistake 3: Not Making Eye Contact

Symptoms: The learner looks at the screen or the floor while presenting. **Likely Cause:** The learner is nervous or reading from the slides. **How to Investigate:** Watch the learner present. Where are they looking? **Safe Corrective Steps:** Encourage the learner to look at the audience. Teach them to use speaker notes instead of reading from the slides. **How to Confirm the Problem Is Resolved:** The learner makes eye contact with the audience. **How to Prevent Recurrence:** Make eye contact a standard part of presentation delivery practice.

Mistake 4: Speaking Too Fast

Symptoms: The learner rushes through the presentation. **Likely Cause:** The learner is nervous or has not practised. **How to Investigate:** Time the presentation. Is it much shorter than expected? **Safe Corrective Steps:** Encourage the learner to speak slowly and pause between points. Teach them to practise with a timer. **How to Confirm the Problem Is Resolved:** The learner speaks at a normal pace. **How to Prevent Recurrence:** Make practising with a timer a standard part of presentation preparation.

Mistake 5: Not Using Presenter View

Symptoms: The learner presents without Presenter View. They cannot see their notes or the next slide. **Likely Cause:** The learner does not know about Presenter View. **How to Investigate:** Ask the learner, "Did you use Presenter View?" **Safe Corrective Steps:** Teach the learner to use Presenter View. Show them how to access it. **How to Confirm the Problem Is Resolved:** The learner uses Presenter View. **How to Prevent Recurrence:** Make Presenter View a standard part of presentation delivery practice.

L. SESSION ASSIGNMENT (5 minutes)

Assignment Title: "My Business Pitch Presentation"

Purpose: To reinforce professional presentation skills by having the learner create and deliver a business pitch presentation.

Realistic Scenario: "You are an entrepreneur pitching your business idea to potential investors. Create a 5–7 slide presentation and deliver it to the class."

Detailed Instructions:

1. Open Microsoft PowerPoint. Create a new blank presentation.
2. Create a 5–7 slide presentation for a fictional business. Include:
    - Title slide (business name and your name)
    - Problem slide (what problem does your business solve?)
    - Solution slide (how does your business solve the problem?)
    - Product/Service slide (with an image)
    - Market slide (who are your customers?)
    - Business Model slide (how will you make money?)
    - Contact slide (how can people reach you?)
3. Apply a professional theme.
4. Insert at least one image.
5. Add a transition to all slides.
6. Add at least two animations.
7. Write speaker notes for each slide.
8. Save the presentation as "My Business Pitch" on the Desktop.
9. Show the saved file to the instructor before leaving.

Required Deliverables:

- A PowerPoint presentation named "My Business Pitch" saved on the Desktop
- The presentation contains: 5–7 slides, a theme, an image, a transition, animations, speaker notes

Submission Format:

- The file must be saved on the Desktop of the academy machine
- The learner must show the instructor the saved file before leaving

Deadline Recommendation:

- Before leaving the session (in-class assignment)

Marking Rubric:

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Number of slides	5–7 slides	4 or 8 slides	3 or fewer, 9 or more
Content	Clear, complete, well-organised	Mostly clear	Unclear or incomplete
Theme	Professional theme chosen	Theme applied but not ideal	No theme
Image	Relevant, high-quality image	Image present but not ideal	No image or irrelevant

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Transition	One transition for all slides	Multiple transitions	No transition
Animations	Used sparingly and purposefully	Used but excessive	Too many or none
Speaker notes	Present on all slides	Present on some slides	Missing
File saved correctly	Saved on Desktop with correct name	Saved but wrong location or name	Not saved

Total: 40 marks

Common Mistakes to Avoid:

- Putting too much text on slides
- Using too many transitions and animations
- Not writing speaker notes
- Not saving the file

Instructor Reference Solution:

The final presentation should have:

- 5–7 slides with clear, organised content
- A professional theme applied
- At least one relevant image
- One transition applied to all slides
- Animations used sparingly
- Speaker notes on all slides
- Saved as "My Business Pitch.pptx" on the Desktop

M. KNOWLEDGE CHECK (5 minutes)

Questions

1. Multiple Choice: What is the purpose of a transition in PowerPoint? a) To add text to a slide b) To create an effect when moving from one slide to the next c) To insert an image d) To apply a theme

Answer: b) To create an effect when moving from one slide to the next

2. Short Answer: What is Presenter View and what does it show?

Answer: Presenter View is a special view that shows the current slide, the next slide, speaker notes and a timer. It helps the presenter deliver the presentation confidently.

3. Multiple Choice: What is the 6x6 rule? a) No more than 6 slides and 6 themes b) No more than 6 words per line and 6 lines per slide c) No more than 6 images and 6 charts d) No more than 6 colours and 6 fonts

Answer: b) No more than 6 words per line and 6 lines per slide

4. Explain in Your Own Words: Why should you not read from the slides during a presentation?

Answer: Reading from the slides makes the presentation boring and disengaging. The slides are for the audience — they should support your message, not replace it. The presenter should speak to the audience, using the slides as visual aids.

5. Practical Identification: Open PowerPoint. Identify the Insert tab, the Transitions tab, the Animations tab, and the Slide Show tab.

Answer: The learner correctly identifies all four tabs.

6. Scenario-Based: Your friend says, "I used a different transition for every slide because I wanted the presentation to be interesting." What advice would you give?

Answer: Using too many different transitions is distracting and unprofessional. Use one transition for the entire presentation. This keeps the presentation consistent and allows the audience to focus on the content.

7. Error-Detecting: A learner writes in their assignment: "I read from the slides because I was nervous and did not want to forget what to say." What is wrong with this, and what should they have done instead?

Answer: Reading from the slides makes the presentation boring. The learner should have written speaker notes and practised the presentation several times. Speaker notes are there to help you remember what to say without reading from the slides.

N. REFLECTION AND RECAP (5 minutes)

Plain-English Summary

Today you learned how to create and deliver professional presentations in Microsoft PowerPoint. You learned how to insert and format images, add shapes and SmartArt, add transitions and animations, and use Presenter View. You also learned the principles of good presentation delivery: speak clearly, make eye contact, do not read from the slides, and practise beforehand. These are the skills that make presentations engaging and memorable.

Five Important Takeaways

1. Use images to support your message, not replace it.
2. Use one transition for the entire presentation.
3. Use animations sparingly to reveal information step by step.
4. Use Presenter View to see your notes and the next slide.
5. Practise your presentation several times before delivering it.

Questions That Help You Recognise What You Can Now Do

- Can you insert and format images?
- Can you add transitions and animations?
- Can you use Presenter View?
- Can you write speaker notes?
- Can you deliver a presentation with confidence?

Connection to the Next Session

This is the second-to-last session. You now have all the skills you need to create professional presentations. In the next session — the final session — you will combine everything you have learned in Word, Excel and PowerPoint to create a complete document package. This is your capstone project. You are ready.

O. INSTRUCTOR EVALUATION NOTES

Observable Signs of Competence

- The learner inserts and formats images.
- The learner adds transitions and animations appropriately.
- The learner uses Presenter View.
- The learner writes speaker notes.
- The learner delivers a presentation with confidence and eye contact.

Common Misconceptions

- **"More transitions and animations make a better presentation."** Some learners think that more effects make a presentation more engaging. Correct this by explaining that too many effects are distracting.
- **"Reading from the slides is fine."** Some learners think that reading from the slides is acceptable. Correct this by explaining that reading from the slides makes the presentation boring.
- **"Presenter View is only for experts."** Some learners think Presenter View is an advanced feature. Correct this by showing them how simple it is to use.

Suggested Remedial Exercises

- For learners who read from the slides: Have them practise presenting with speaker notes only (no looking at the screen).
- For learners who use too many transitions: Have them create a presentation with one transition for all slides.
- For learners who do not make eye contact: Have them practise presenting to a mirror or a friend.

Differentiation Guidance

Level 1 — Supported (for learners who struggle):

- Provide a "Presentation Delivery Guidelines" handout.
- Pair with a more confident learner for presentation practice.
- Allow extra time for the presentation.
- Focus on content and delivery before design.

Level 2 — Independent (standard):

- Complete all exercises as written.
- Answer all knowledge check questions.
- Complete the assignment to the standard shown in the rubric.

Level 3 — Challenge (for learners who finish early):

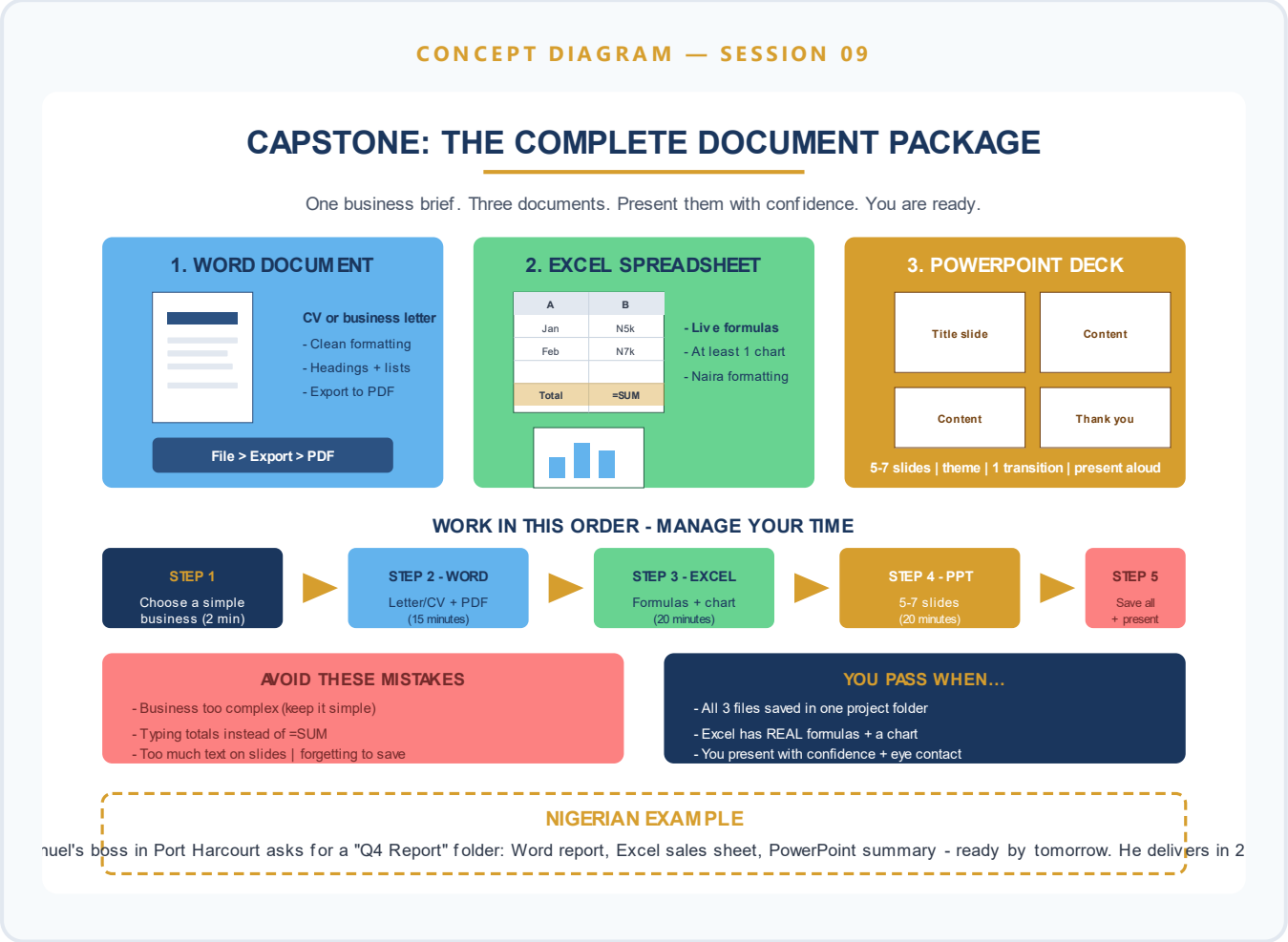
- Ask the learner to create a presentation with 10 slides on a topic of their choice.
- Ask the learner to research and write down three more presentation delivery tips.
- Ask the learner to explain the difference between a transition and an animation.

End of Session 8

Next Session: Session 9 — Final Project: Complete Document Package (Capstone) **What to Bring:** All the skills you have learned, confidence, creativity **Preparation:** Print capstone briefs, prepare assessment rubrics, ensure all software is functional

SESSION 09 OF 9

Final Project Capstone



SESSION 9: FINAL PROJECT — COMPLETE DOCUMENT PACKAGE (CAPSTONE)

Week: 3 **Module:** Integration **Duration:** 105 minutes **Difficulty:** Beginner (tests all accumulated skills) **Prerequisite Knowledge:** Sessions 1–8 — all Word, Excel and PowerPoint skills **Connection to Previous Session:** Integrates and tests everything taught in Sessions 1–8 **Preparation Required Before Class:**

- Print capstone briefs (one per learner)
- Prepare assessment rubrics
- Ensure all Microsoft Office applications are functional
- Prepare certificate sign-off sheets

A. SESSION IDENTIFICATION

Field	Detail
Session Number	9

Field	Detail
Title	Final Project — Complete Document Package (Capstone)
Week	3
Module	Integration
Duration	105 minutes
Difficulty	Beginner (tests all accumulated skills)
Prerequisite Knowledge	Sessions 1–8 — all prior knowledge
Connection to Previous Session	Integrates and tests everything taught in Sessions 1–8
Preparation	Capstone briefs, assessment rubrics, certificate sign-off sheets

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Plan and organise** a multi-part project from a brief.
2. **Create a formatted CV or business letter** in Microsoft Word, exported to PDF.
3. **Create a working spreadsheet** in Microsoft Excel with live formulas and at least one chart.
4. **Create a 5–7 slide presentation** in Microsoft PowerPoint.
5. **Present the presentation aloud** to the class with confidence.
6. **Explain their design and formatting decisions** for all three deliverables.
7. **Reflect on their learning** and identify areas for continued growth.

C. REQUIRED MATERIALS

Hardware:

- One desktop computer per learner
- One keyboard per learner
- One mouse per learner
- One monitor per learner
- Projector for presentation delivery

Software:

- Windows 10 or Windows 11
- Microsoft Word, Excel and PowerPoint 2019 or Microsoft 365
- Adobe Acrobat Reader (for PDF verification)

Printed Materials:

- Capstone brief (one per learner)
- Assessment rubric (one per learner)
- Certificate sign-off sheet (one per learner)

Sample Files:

- Sample CV template (optional)
- Sample spreadsheet with formulas (optional)
- Sample presentation (optional)

Internet: Not required for this session.

D. OPENING ACTIVITY (5 minutes)

"You Are Ready"

Instructor Script: "Good morning, everyone. This is it — the final session. Over the past three weeks, you have learned Microsoft Word, Excel and PowerPoint. Today, you are going to prove that you can use all three to create a professional document package.

I am not going to help you. I am not going to give you step-by-step instructions. I am going to give you a project brief, and you are going to complete it on your own. This is not a test of perfection. It is a test of competence. Can you sit down at a computer and produce professional work?

If you have attended the previous sessions and practised, you are ready. If you are nervous, that is normal. Take a deep breath. You know how to do this. Let us begin."

Expected Learner Response: A mix of excitement and nervousness. Some learners will be confident. Others will be anxious. The instructor's calm, encouraging tone sets the atmosphere.

E. PREREQUISITE REVIEW (5 minutes)

Instructor Script: "Before we start, let us do a quick review of everything you have learned. I will ask questions and you answer out loud."

Review Questions and Expected Answers:

1. "How do you change the font in Word?"
    - Select the text, go to the Home tab, and choose a font from the Font group.
2. "What does the SUM function do in Excel?"
    - It adds a range of cells.
3. "What is the 6x6 rule in PowerPoint?"
    - No more than 6 words per line and 6 lines per slide.
4. "How do you apply a theme in PowerPoint?"
    - Go to the Design tab and click on a theme.
5. "How do you export a Word document to PDF?"
    - Click File > Export > Create PDF/XPS Document.
6. "What is the most important rule of presentation delivery?"
    - Do not read from the slides. Speak to the audience.

How to Address Common Gaps:

- If a learner cannot answer a question, do not embarrass them. Ask another learner to help. The goal is to activate prior knowledge, not to test.

F. FULL LECTURE NOTES (10 minutes)

The Capstone Project Explained

Instructor Script: "Here is what you are going to do today. I will give you the full project brief, and then you will complete it on your own. I will be watching, but I will not help unless you are truly stuck.

The Project: You are a small business owner in Port Harcourt. You need a complete set of professional documents for your business. You must create:

1. **A formatted CV or business letter** in Microsoft Word, exported to PDF.
2. **A working spreadsheet** in Microsoft Excel with live formulas and at least one chart.

3. **A 5–7 slide presentation** in Microsoft PowerPoint.

All three documents must be related to the same business. You can choose any type of business — a fashion boutique, a restaurant, a tech startup, a provision shop, a salon, etc.

After you have created all three documents, you will present your presentation aloud to the class. This is your moment to show what you have learned.

Any questions?"

Key Point: The capstone is not a trick. It is not designed to make you fail. It is designed to prove that you can do the things you have been taught. If you have been paying attention and practising, you will pass.

G. INSTRUCTOR DEMONSTRATION (0 minutes)

There is no instructor demonstration in this session.

The entire session is dedicated to the learner's independent project work and presentation. The instructor's role is to observe, record results, and provide encouragement.

H. GUIDED PRACTICAL EXERCISE (0 minutes)

There is no guided practical in this session.

The learner completes the entire project independently. This is the whole point of the capstone.

I. INDEPENDENT CLASS EXERCISE (60 minutes)

Exercise: "Complete Document Package"

Scenario: "You are a small business owner in Port Harcourt. You need a complete set of professional documents for your business. Create all three documents as a unified package."

Task: Complete the following project without assistance:

Step 1: Choose Your Business (2 minutes) Choose a type of business. Examples:

- Adaeze's Fashion Design
- Chinedu's Provision Shop
- Funmi's Restaurant
- Tunde's Tech Startup
- Ngozi's Salon
- Or any other business you like

Step 2: Create a Formatted CV or Business Letter in Word (15 minutes) Create a professional CV for the business owner OR a business letter from the business. Include:

- Proper formatting (fonts, sizes, alignment, spacing)
- Professional appearance
- Export to PDF

Step 3: Create a Working Spreadsheet in Excel (20 minutes) Create a spreadsheet for the business. Include:

- At least 10 rows of data
- At least 3 formulas (SUM, AVERAGE, or other)
- At least one chart
- Proper formatting (currency, borders, header formatting)

Step 4: Create a 5–7 Slide Presentation in PowerPoint (20 minutes) Create a presentation about the business. Include:

- Title slide
- At least 4 content slides
- A professional theme
- At least one image
- A transition
- Animations
- Speaker notes

Step 5: Save All Documents (3 minutes) Save all three documents on the Desktop in a folder called "CEA Final Project — [Your Name]."

Requirements:

- Complete all 5 steps independently
- Do not ask the instructor for step-by-step guidance
- All three documents must be related to the same business
- The spreadsheet must use formulas (not manual calculations)
- The presentation must follow the 6x6 rule

Time Allowance: 60 minutes

Success Criteria:

- All three documents created to a satisfactory standard
- The spreadsheet uses formulas and has a chart
- The presentation has 5–7 slides with a theme, image, transition and animations
- All documents saved in the correct folder

Hints Available If the Learner Gets Stuck:

- The instructor may provide a single hint if a learner is truly stuck for more than 5 minutes. The hint should be a question, not an answer.
- The instructor should NOT provide the full answer. The learner should figure it out themselves.

Complete Answer Key for the Instructor:

The instructor observes each learner and records their progress on the assessment rubric. There is no single correct answer — the learner's business choice and document designs will vary. The instructor evaluates based on the assessment rubric criteria.

J. REAL-WORLD CASE STUDY (5 minutes)

Case: "Emmanuel's Business Package"

Situation: Emmanuel completed the Microsoft Office course at CEA. He passed the capstone assessment. Two weeks later, he starts a new job as an office assistant at a company in Port Harcourt. On his first day, his boss says: "Create a folder on your desktop called 'Q4 Report.' Create a formatted report, a spreadsheet with our sales data and a presentation summarising the key points. Have it ready by tomorrow."

Emmanuel sits down at the computer. He takes a deep breath. He has done this before. He creates the folder, opens Word, types and formats the report, opens Excel, enters the data and creates formulas and a chart, opens PowerPoint, creates a clean presentation, and saves everything. Total time: 2 hours.

His boss is impressed. "You are the fastest new starter we have had," she says.

The Problem: Emmanuel's colleague, Chinedu, also started on the same day. Chinedu did not take the CEA course. When his boss gives him the same task, he stares at the screen. He does not know how to format a document, create a formula or design a presentation. He has to ask for help. His boss is not impressed.

Why It Matters: Emmanuel and Chinedu have the same job. But Emmanuel can work independently. Chinedu cannot. In a workplace, independence is valuable. Employers do not have time to hold your hand. They expect you to be able to sit down and get to work.

Relevant Concept: All the skills taught in Sessions 1–8. Emmanuel's success is the result of three weeks of focused, practical learning.

Worked Solution: Emmanuel succeeded because he had practised every step of the workflow multiple times. He had created documents, spreadsheets and presentations. By the time he did it in a real workplace, it was second nature.

Reasoning: The capstone assessment is not just a test. It is a rehearsal for real life. When you can create a complete document package independently in class, you can create it independently in a workplace.

Possible Mistake: Emmanuel might have been nervous on his first day. Nervousness can make you forget things. But because he had practised so much, his hands remembered what to do even when his brain was stressed.

Correction: Practice until the skills become automatic. Do not just practise until you can do it. Practise until you cannot fail.

Transfer Exercise: "Think about a task you might be asked to do in a real job. Write down the steps you would take. Then practise those steps until they become automatic."

Professional Connection: Every office job requires the skills taught in this course. Administrators, sales officers, teachers, accountants, managers — everyone. The ability to create professional documents, working spreadsheets and polished presentations is the foundation of all professional office work.

K. COMMON MISTAKES AND TROUBLESHOOTING (5 minutes)

Mistake 1: Choosing a Business That Is Too Complex

Symptoms: The learner chooses a business that is too complex (e.g., a multinational corporation) and cannot complete the project in time. **Likely Cause:** The learner does not understand the scope of the project. **How to Investigate:** Ask the learner, "What business did you choose? How complex is it?" **Safe Corrective Steps:** Encourage the learner to choose a simple business (e.g., a small shop, a salon, a restaurant). The goal is to demonstrate skills, not to create a realistic business plan. **How to Confirm the Problem Is Resolved:** The learner chooses a simple business. **How to Prevent Recurrence:** Provide a list of simple business examples in the project brief.

Mistake 2: Not Managing Time

Symptoms: The learner spends too much time on one document and does not have time for the others. **Likely Cause:** The learner does not plan their time. **How to Investigate:** Ask the learner, "How much time have you spent on each document?" **Safe Corrective Steps:** Teach the learner to allocate time for each document (e.g., 15 minutes for Word, 20 minutes for Excel, 20 minutes for PowerPoint, 5 minutes for saving). **How to Confirm the Problem Is Resolved:** The learner manages their time effectively. **How to Prevent Recurrence:** Provide a time allocation guide in the project brief.

Mistake 3: Not Using Formulas in Excel

Symptoms: The learner types calculated values manually instead of using formulas. **Likely Cause:** The learner forgets to use formulas or does not know how. **How to Investigate:** Ask the learner, "Did you use formulas in your spreadsheet?" **Safe Corrective Steps:** Remind the learner to use formulas for calculations. Show them how to use SUM, AVERAGE and other functions. **How to Confirm the Problem Is Resolved:** The learner uses formulas. **How to Prevent Recurrence:** Make formulas a standard requirement for every spreadsheet exercise.

Mistake 4: Not Following the 6x6 Rule in PowerPoint

Symptoms: The learner puts too much text on slides. **Likely Cause:** The learner forgets the 6x6 rule. **How to Investigate:** Look at the slides. How much text is on each one? **Safe Corrective Steps:** Remind the learner of the 6x6 rule. Encourage them to use bullet points and speaker notes. **How to Confirm the Problem Is Resolved:** The learner follows the 6x6 rule. **How to Prevent Recurrence:** Make the 6x6 rule a standard requirement for every presentation exercise.

Mistake 5: Not Saving Documents

Symptoms: The learner creates documents but does not save them. **Likely Cause:** The learner forgets to save or runs out of time. **How to Investigate:** Ask the learner, "Did you save your documents?" **Safe Corrective Steps:** Remind the learner to save early and save often. Provide a "Save All Documents" step at the end of the project. **How to Confirm the Problem Is Resolved:** The learner saves all documents. **How to Prevent Recurrence:** Make saving a standard step in every project.

L. SESSION ASSIGNMENT (5 minutes)

Assignment Title: "My Learning Reflection"

Purpose: To encourage the learner to reflect on their learning and identify areas for continued growth.

Realistic Scenario: "You have completed the Microsoft Office course. Your instructor asks you to write a short reflection on what you have learned, what you found challenging, and what you want to learn next."

Detailed Instructions:

1. Open Microsoft Word.
2. Write a reflection that includes:
    - Your name and the date
    - Three things you learned in this course
    - Two things you found challenging
    - One thing you want to learn next
    - A sentence about how you feel now compared to how you felt in Session 1
3. Save the file as "My Learning Reflection" on the Desktop.
4. Show the saved file to the instructor before leaving.

Required Deliverables:

- A document named "My Learning Reflection" saved on the Desktop
- The document contains: three things learned, two challenges, one next step, one feeling comparison

Submission Format:

- The file must be saved on the Desktop of the academy machine
- The learner must show the instructor the saved file before leaving

Deadline Recommendation:

- Before leaving the session (in-class assignment)

Marking Rubric:

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Three things learned	Three specific, accurate points	Two specific points	One or vague points
Two challenges	Two specific, honest challenges	One specific challenge	Vague or missing
One next step	Clear, specific goal	Present but vague	Missing
Feeling comparison	Insightful, honest reflection	Present but generic	Missing
File saved correctly	Saved on Desktop with correct name	Saved but wrong location or name	Not saved

Total: 25 marks

Common Mistakes to Avoid:

- Writing vague statements (e.g., "I learned a lot") without specifics
- Not saving the file
- Copying from another learner

Instructor Reference Solution:

Three Things I Learned:

1. I learned how to create and format professional documents in Microsoft Word.
2. I learned how to use formulas and create charts in Microsoft Excel.
3. I learned how to design and deliver professional presentations in Microsoft PowerPoint.

Two Things I Found Challenging:

1. I found it hard to remember all the keyboard shortcuts.
2. I found it challenging to design presentations that are visually appealing.

One Thing I Want to Learn Next: I want to learn how to use Microsoft Excel for more advanced data analysis, such as pivot tables and VLOOKUP.

How I Feel Now: In Session 1, I was nervous and did not think I could use Microsoft Office. Now I feel confident and ready to use these skills in a real workplace.

M. KNOWLEDGE CHECK (15 minutes)

Quiz

Instructions: Answer all 20 questions.

Section A: Word (5 questions)

1. How do you change the font of selected text in Word?
2. What does the Format Painter do?
3. How do you insert a table in Word?
4. What is the correct way to shut down a computer? (Trick question — this is from Session 1!)
5. How do you export a Word document to PDF?

Section B: Excel (5 questions)

6. What must every Excel formula start with?
7. What does the SUM function do?
8. What is the difference between a workbook and a worksheet?
9. How do you format numbers as currency in Excel?
10. What type of chart is best for comparing values across categories?

Section C: PowerPoint (5 questions)

11. What is the 6x6 rule?
12. How do you apply a theme in PowerPoint?
13. What is the difference between a transition and an animation?
14. What does Presenter View show?
15. Why should you not read from the slides during a presentation?

Section D: Integration (5 questions)

16. What are the three applications in Microsoft Office that you learned in this course?
17. Which application would you use to create a budget? Why?
18. Which application would you use to create a CV? Why?
19. Which application would you use to create a presentation? Why?
20. What is the most important skill you learned in this course? Why?

Answer Key

Section A: Word

1. Select the text, go to the Home tab, and choose a font from the Font group.
2. The Format Painter copies formatting from one section of text to another.
3. Go to the Insert tab and click the Table button.
4. Click Start, click Power, click Shut down. (Trick question — always check your work!)
5. Click File > Export > Create PDF/XPS Document.

Section B: Excel

6. An equals sign (=).

7. It adds a range of cells.
8. A workbook is a collection of worksheets. A worksheet is a single page.
9. Select the numbers, go to the Home tab, click the Number Format dropdown, and choose Currency.
10. A bar chart.

Section C: PowerPoint

11. No more than 6 words per line and 6 lines per slide.
12. Go to the Design tab and click on a theme.
13. A transition is the effect when moving from one slide to the next. An animation is the effect when an object appears on a slide.
14. Presenter View shows the current slide, the next slide, speaker notes and a timer.
15. Reading from the slides makes the presentation boring. The slides are for the audience. Your words are the presentation.

Section D: Integration

16. Microsoft Word, Microsoft Excel and Microsoft PowerPoint.
17. Excel — because it is designed for calculations and working with numbers.
18. Word — because it is designed for creating text documents.
19. PowerPoint — because it is designed for creating presentations.
20. (Open-ended — any reasonable answer is acceptable.)

N. REFLECTION AND RECAP (5 minutes)

Plain-English Summary

Today you demonstrated that you can use Microsoft Office to create professional documents, working spreadsheets and polished presentations. You completed the capstone project — a complete document package for a fictional business. You have completed the Microsoft Office course. You should be proud of yourself.

Five Important Takeaways from the Entire Course

1. Word is for documents. Excel is for spreadsheets. PowerPoint is for presentations.
2. Good formatting makes documents professional. Keep it clean and consistent.
3. Formulas automate calculations in Excel. Use them instead of manual calculations.
4. The 6x6 rule keeps presentations clean. Use speaker notes for your talking points.
5. You are now ready to use Microsoft Office in a real workplace.

Questions That Help You Recognise What You Can Now Do

- Can you create and format a professional document in Word?
- Can you use formulas and create charts in Excel?
- Can you design and deliver a professional presentation in PowerPoint?
- Can you explain your design and formatting decisions?
- Can you work independently without step-by-step guidance?

What Comes Next

You have completed the Microsoft Office course. You are now ready for the next level. At Cyber Elias Academy, you can continue with:

- **Graphic Design** — Learn to create professional designs.
- **Web Design** — Learn to build websites.
- **Data Entry** — Learn to handle data accurately.
- **Computer Repairs** — Learn to diagnose and fix computer problems.
- **Digital Marketing** — Learn to promote businesses online.
- **Content Creation** — Learn to create photos, videos and written content.

The skills you learned today are the foundation for all of these courses. You are ready.

O. INSTRUCTOR EVALUATION NOTES

Observable Signs of Competence

- The learner creates a formatted document in Word and exports to PDF.
- The learner creates a working spreadsheet with formulas and a chart in Excel.
- The learner creates a 5–7 slide presentation in PowerPoint with a theme, image, transition and animations.
- The learner presents the presentation with confidence and eye contact.
- The learner explains their design and formatting decisions.
- The learner works independently without step-by-step guidance.

Common Misconceptions

- **"I can just learn this on my own."** Some learners think they do not need a course. They may be right — some people can figure things out on their own. But most people benefit from structured, guided learning.
- **"I will never need to use Office."** Some learners think they will avoid Office in their daily life. In 2026, this is almost impossible. Office skills are required in almost every job.
- **"I am too old to learn this."** Some older learners feel they cannot learn new software. This is not true. Age is not a barrier to learning computer skills.

Suggested Remedial Exercises

- For learners who do not pass the capstone: Allow them to practise on academy machines during opening hours. Provide a step-by-step guide they can follow. Let them retake the assessment when they are ready.
- For learners who cannot use formulas: Provide additional formula practice exercises. Walk them through the process step by step.
- For learners who cannot design presentations: Provide sample presentations and ask them to recreate the design.

Differentiation Guidance

Level 1 — Supported (for learners who struggle):

- Allow extra time for the capstone project.
- Provide templates and examples for reference.
- Offer one-on-one support during the project.
- Allow them to retake the assessment after additional practice.

Level 2 — Independent (standard):

- Complete the capstone project as written.
- Answer all knowledge check questions.
- Complete the reflection assignment.

Level 3 — Challenge (for learners who finish early):

- Ask the learner to create a more complex project (e.g., a 10-slide presentation, a spreadsheet with multiple charts).
- Ask the learner to research and write down three more Office skills they want to learn.
- Ask the learner to help a struggling peer.

CERTIFICATE SIGN-OFF

Learner Name: _____

Date: _____

Capstone Project Result: Pass / Not Yet Competent

Knowledge Check Score: _____ / 100

Deliverables Completed:

- ☐ Formatted document in Word (exported to PDF)
- ☐ Working spreadsheet in Excel (with formulas and chart)
- ☐ 5–7 slide presentation in PowerPoint
- ☐ Presentation delivered aloud

Instructor Signature: _____

Certificate Awarded: Yes / No

Comments:

End of Session 9 — End of Course

Congratulations! You have completed the Microsoft Office course at Cyber Elias Academy.

Next Step: Continue your learning with Graphic Design, Web Design, or another CEA course.